MachineSense

Machinery fault diagnosis that can flag unfamiliar faults and explain the evidence behind each result

PROJECT REPORT

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List of Abbreviations

Table 1. Abbreviations used in this report.

Abbreviation	Meaning
AE	Autoencoder
AE-6	The healthy-only autoencoder on the six vibration channels
AUROC	Area under the receiver operating characteristic curve
B1s	Speed-aware healthy baseline on physics features
BPFI / BPFO	Ball-pass frequency of the inner / outer race (bearing defect rates)
BSF / FTF	Ball-spin frequency / fundamental train (cage) frequency
CNN	Convolutional neural network
CNN-6	The project's six-channel convolutional classifier
CPU / GPU / MPS	Central / graphics processing unit / Apple Metal Performance Shaders
E1	Gated condition-specialist autoencoder bank (research stage R5)
F1 (stage)	The one-time final evaluation on the reserved tests
ISSM	Intelligent Sensing and Sense Making (course)
kNN	k-nearest neighbours
LOCO	Leave-one-class-out
LR	Logistic regression
MAFAULDA	Machinery Fault Database (Federal University of Rio de Janeiro)
MSP	Maximum softmax probability (classifier confidence)
OSCR	Open-set classification rate
PCA	Principal component analysis
PRMLS	Pattern Recognition and Machine Learning Systems (course)
PSUPR	Problem Solving Using Pattern Recognition (course)
Q1 / Q2 / Q3	The three diagnostic questions: healthy? which known fault? unfamiliar fault?
RMS	Root mean square (overall vibration level)
R0–R10	The development-data research stages
STFT	Short-time Fourier transform
SVM	Support vector machine
1×, 2×	Vibration once and twice per shaft revolution (shaft orders)

Acknowledgements and Use of AI Tools

I thank the NUS-ISS Pattern Recognition Systems teaching team for its guidance during the practice module. I also thank the Signal, Multimedia and Telecommunications Laboratory at the Federal University of Rio de Janeiro for making MAFAULDA publicly available [1].

Use of AI tools

AI tools helped with planning, implementation and drafting. I reviewed the design, evaluation protocol and final report, and I remain responsible for the work and its claims.

- **Planning and review:** Claude helped plan research stages, review proposed protocols and compare coding-agent reports with the repository. ChatGPT provided a second review of selected prompts.
- **Implementation:** Claude Code wrote code for stages described in the project record. I reviewed the changes and ran the test suites.
- **Writing:** Claude helped prepare an initial draft from the saved results. I then checked the reported findings against the result files and revised the wording and limitations.
- **Inside the application:** a locally hosted Gemma model writes optional explanations and answers case questions. Code checks its output, and the model cannot change a diagnostic verdict. The scores and evaluation statistics come from executable code.

Table of Contents

List of Abbreviations 2
Acknowledgements and Use of AI Tools 3
Table of Contents 4
Executive Summary 7
1. Business Case 8
1.1 Why machinery fault diagnosis matters 8
1.2 The problem: three questions an engineer asks 8
1.3 The solution: MachineSense Web application 8
1.4 Why honesty matters more than a headline score 9
1.5 Intended users 9
2. Related Work 10
2.1 Condition-monitoring tools today 10
2.2 Prior work on MAFAULDA 10
2.3 Unknown-input detection 10
2.4 The gaps this project targets 11
3. Objectives and Success Measures 12
3.1 Functional objectives 12
3.2 Non-functional objectives 12
3.3 Success metrics and declared bars 12
3.4 Out of scope 13
4. Data 14
4.1 MAFAULDA: the rig and the recordings 14
4.2 Data audit 15
4.3 Splits and leakage control 17
4.4 Learning from imbalanced data 19
4.5 Data pipeline: from raw files to model inputs 19
5. System Design and Models 24
5.1 System architecture 24
5.2 Signal preparation 26
5.3 Representations 26
5.4 Q2: recognising known faults 28
5.5 Q1: the healthy check 29
5.6 Q3: unfamiliar faults 30
5.7 The decision rule 32
5.8 Evidence, explanation and guidance 33
5.9 Storage and API 34

5.10 The investigation assistant: agent architecture 34
5.11 Why these models: choices and evidence 37
6. Tools and Techniques Used 39
6.1 The practice-module aspects 39
6.2 Course topics by course 39
6.3 Techniques considered and not used 39
7. Implementation 40
7.1 Tech stack and rationale 40
7.2 Code structure 40
7.3 The application 40
7.4 Reproducibility and testing 42
7.5 Demonstration walkthrough 43
8. Evaluation Protocol 49
8.1 Development data and reserved tests 49
8.2 Rules that protect the results 49
8.3 Metrics and uncertainty 49
8.4 Why these metrics 49
9. Evaluation Results 52
9.1 Q2: recognising known faults 52
9.2 Q1: the healthy check 54
9.3 Q3: unfamiliar faults 56
9.4 The investigation assistant pilot (D7) 57
9.5 The conversation assistant evaluation (App-9) 58
9.6 Latency and system checks 59
10. Findings and Discussion 60
10.1 The improvement stages 60
10.2 Each experiment: why it was tried and what it taught 61
10.3 Why distance methods fail and reconstruction works 64
10.4 Pretrained encoders 66
10.5 What the results mean for a maintenance engineer 67
10.6 What surprised me 67
11. Limitations and Future Work 68
11.1 Limitations 68
11.2 Changes from the proposal 68
11.3 Future work 68
12. Conclusion 70
Appendix A. Project Proposal 71
Appendix B. Course Topics Mapped to the Project 72

B.1 PSUPR: Problem Solving Using Pattern Recognition 72
B.2 PRMLS: Pattern Recognition and Machine Learning Systems 73
B.3 ISSM: Intelligent Sensing and Sense Making 74
B.4 Taught but not used 76
Appendix C. Installation and User Guide 78
Appendix D. Detailed Results 79
D.1 Q2 per-class F1 on the reserved tests 79
D.2 Q2 confusion on the closed split 79
D.3 Q3 known-side rejection by partition 79
Appendix E. Research-Stage Protocols 81
Appendix F. D7 Pilot Protocol 82
Appendix G. References 83

Executive Summary

MachineSense analyses a five-second recording of a rotating $_{test}$ rig. It uses eight channels: a tachometer, two three-axis accelerometers and a microphone. The system asks three separate questions:

Q1: Does the recording look healthy?

Q2: Which of ten known conditions does it resemble?

Q3: Could it contain a fault the models have not seen?

The web app presents the result and its supporting evidence at two reading levels, Mechanic and Developer. A case-based AI agent assistant can answer follow-up questions.

The study uses the public MAFAULDA dataset: 1,951 recordings from one test rig. Its main contribution is a careful evaluation, rather than a new network. I split recordings before cutting them into windows, grouped development folds by source folder, and opened the reserved tests once, after the models, thresholds and decision rule were fixed.

Headline reserved-test results. Q2: ten-class macro-F1 was 0.916 (95% interval 0.845–0.972) on 290 recordings, below the 0.983–0.994 development range. Q3: across nine withheld fault classes, macro AUROC was 0.879 for Mahalanobis familiarity and 0.834 for classifier confidence. The rule rejected 925 of 1,123 unfamiliar-fault recordings (82.4%) and 765 of 2,327 known-side scorings (32.9%); the latter are repeated scores across withheld-class runs, not distinct recordings. Q1: no fault was called healthy in the reserved tests, but 5 of 7 and 19 of 20 healthy recordings in the closed and speed-block tests were also not called plainly healthy. These results describe the frozen system on one rig.

The system rejected almost every unfamiliar bearing fault, but often missed unfamiliar imbalance and misalignment. In the classifier's learned feature space, those shaft faults overlap known classes, so a distance score alone struggles to separate them. I tested improvements on development data. One method met the full predeclared bar: a bank of fault-specific autoencoders (E1) increased unfamiliar-fault recall by an average of 10.1 percentage points across three seeds, without increasing the number called healthy. A fine-tuned ResNet-18 helped the difficult shaft classes in a smaller exploratory study. A frozen pretrained encoder did not. These research models are not in the app: E1 was tested after the reserved evaluation, and ResNet-18 was weaker than CNN-6 on known faults.

In a 20-recording local benchmark, the median diagnosis-path time was 885 ms on a CPU with the recording already in memory. A language-model assistant answer took much longer (median: 21.5–30.6 s). The main limits are the 49 healthy recordings from one source folder, one test rig and a small assistant pilot scored by its author. The report gives those limits alongside the results.

1. Business Case

1.1 Why machinery fault diagnosis matters

Changes in vibration or sound can be early signs of bearing damage, imbalance or shaft misalignment in motors, pumps and other rotating machines. Condition-based maintenance uses those signals to decide when an inspection may be useful, instead of relying only on a calendar or waiting for a breakdown.

A diagnosis can help a maintenance team decide where to look first. It can also mislead them: a false alarm costs inspection time and may reduce trust. This project measures missed faults and false alarms. It does not claim to reduce downtime or maintenance cost.

1.2 The problem: three questions an engineer asks

A new recording raises three different questions. A model that names a familiar fault does not automatically answer whether the machine is healthy or whether the fault is unfamiliar.

Table 2. The three diagnostic questions and why each needs its own method.

Question	What it asks	Why it is hard
Q1: Is it healthy?	Does the recording depart from healthy behaviour?	Only 49 healthy recordings are available, all in one folder; acquisition-session IDs are unavailable and healthy behaviour changes with speed.
Q2: Which known fault?	Which of ten labelled conditions best matches?	Published scores are often inflated by windows of one recording appearing in both training and test sets.
Q3: Is it unfamiliar?	Is this a fault the system was never trained on?	A closed-set classifier must pick a known label; it can name an unseen fault wrongly, with high confidence.

A recording can look unusual because the fault is new, the speed or operating setting is new, or a sensor is unreliable. I evaluate these cases separately.

1.3 The solution: MachineSense Web application

MachineSense takes one eight-channel recording and returns a finding, the evidence behind it and suggested checks. It can report healthy, name a known fault as a model prediction, or ask for further inspection when the checks do not support a familiar match.

- **Signal preparation:** validate the file, estimate shaft speed from the tachometer, remove known interference and cut the recording into nine overlapping one-second windows.
- **Models:** CNN-6 predicts a known condition. A familiarity score checks whether its pattern resembles training data. Two models trained only on healthy recordings check for departures from healthy behaviour, while logistic regression provides a second opinion.
- **Decision:** a fixed rule combines the checks. The healthy models can add a caution, but they cannot overturn a confident, familiar result.
- **Evidence and explanation:** the app stores measurements, scores and warnings for each case. A local language model can turn these into plain language.
- **Application:** the dashboard has a plain-language Mechanic view and a more detailed Developer view. The app also includes a pipeline page and a case-based AI Agent assistant.

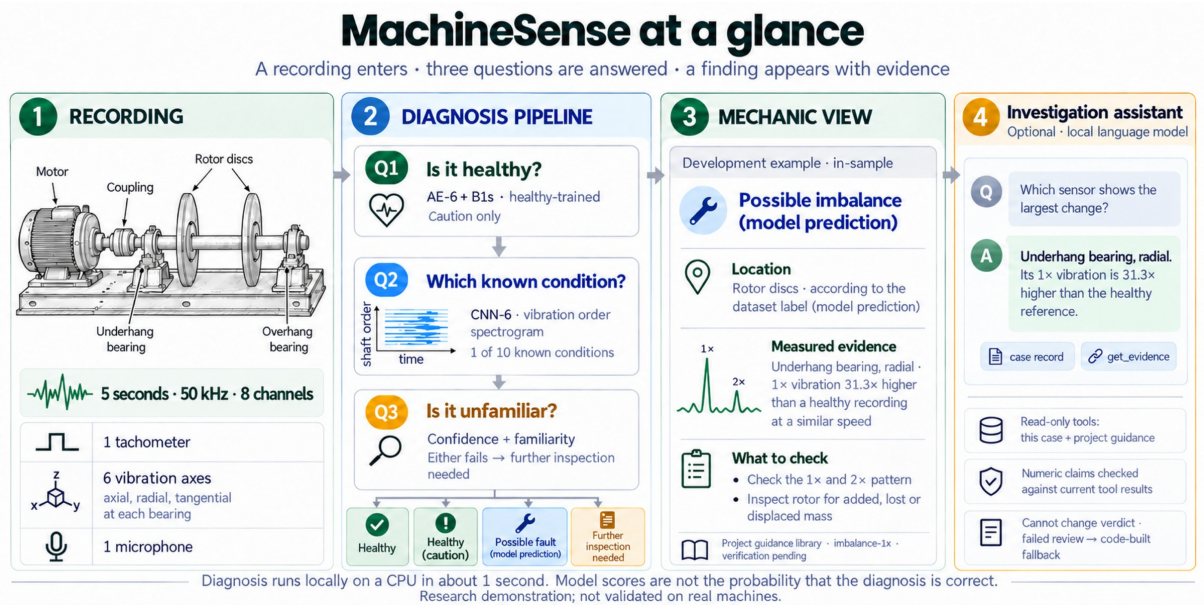


Figure 1. MachineSense at a glance: a recording enters, three questions are answered, and the result is shown with evidence.

1.4 Why honesty matters more than a headline score

A maintenance engineer may act on a diagnosis, so the app must be clear about uncertainty. It labels fault names as model predictions, identifies recordings used to build the model, and does not present model scores as probabilities of correctness. It also states that the system is a research demonstration, not a tool validated on operating machinery. Automated tests protect these statements.

1.5 Intended users

- **Maintenance engineer or mechanic** (Mechanic view): needs a readable finding, a place to start looking and concrete checks.
- **Reliability or data engineer** (Developer view): needs scores, thresholds, processing steps and the reason behind the decision.
- **Examiner or reviewer:** needs enough detail to judge whether the reported results are credible. Section 8 describes the evaluation protocol.

2. Related Work

2.1 Condition-monitoring tools today

Commercial condition-monitoring systems already help teams monitor machines. Augury combines sensors, cloud analytics and expert-reviewed alerts [11]; Siemens Senseye uses machine and maintenance data to prioritise work across a fleet [12]. Their public descriptions do not provide the detail needed to compare unfamiliar-fault evaluation directly with this study. MachineSense is smaller in scope: it is a local, inspectable research system with a documented protocol.

2.2 Prior work on MAFAULDA

MAFAULDA is widely used for vibration-based fault diagnosis [1]. Ribeiro et al. studied similarity-based models [2]; Souza et al. used a one-dimensional CNN on frequency-domain inputs [5]; T4PdM adapted a transformer to a subset of accelerometer channels [6]. Ignjatovska et al. combined data-driven and physics-informed features and also tested an independent rig [7]. Their class sets and evaluation splits differ, so headline accuracy is not directly comparable. Unfamiliar-fault rejection is not the main question in these papers.

Two choices can make a score look better than it is. Windows from one recording are highly alike; putting them on both sides of a train–test split leaks information [3], [4]. Here, a random window split scored 1.000 macro-F1, while a recording-grouped split on the same training pool scored 0.991 (Section 4.3). A closed-set test also cannot show what happens when the fault type was absent from training.

Table 3. Prior work on MAFAULDA, as described in the reviewed texts.

Work	Approach	Split (as reported)	Unknown faults
Ribeiro et al. [2]	Similarity-based models on features	Random split of feature samples	Not addressed
Souza et al. [5]	1-D CNN on frequency-domain vibration	Closed-set; grouping by recording not reported	Not addressed
T4PdM [6]	Transformer, 3 accelerometer channels, overhang data excluded	Closed-set; grouping not reported	Not addressed
Ignjatovska et al. [7]	Data-driven + physics features, supervised ML	Closed-set; grouping not reported	Not addressed
MachineSense	Physics features + CNN-6 + healthy-only detectors + three-way rule	Recording-level; folder-grouped development; sealed reserved tests	Primary question (Q3)

2.3 Unknown-input detection

Unfamiliar-input detection has a larger literature beyond machinery diagnosis. A low maximum softmax probability is a simple warning sign [8]. Class-conditional Mahalanobis distance tests whether a network feature is near a known class [9], while Relative Mahalanobis targets difficult near-unknown inputs [13]. Reconstruction error provides another anomaly signal, and open-set surveys compare these approaches [10]. This project adapts simplified versions to MAFAULDA; it does not claim to reproduce every original method.

2.4 The gaps this project targets

1. **Healthy-only vibration detection.** The reviewed MAFAULDA literature offers little direct evidence for fault detection trained only on healthy vibration data.
2. **Unknown-fault rejection.** The reviewed MAFAULDA papers focus on known labels rather than systematically withholding a fault class.
3. **Leakage control.** This study splits by recording, groups development folds by folder and opens reserved tests once.
4. **Useful evidence.** The app connects each finding to measured sensor changes, shaft-order and bearing-rate lines, and labelled inspection guidance.

3. Objectives and Success Measures

3.1 Functional objectives

Table 4. Functional objectives and their status.

#	Objective	Status
F1	Q1: detect departure from healthy behaviour using healthy-only detectors	Delivered (AE-6, B1s)
F2	Q2: classify ten known conditions, with a six-group view	Delivered (CNN-6; LR second opinion)
F3	Q3: withhold a label and request inspection for unfamiliar faults	Delivered (MSP + Mahalanobis rule)
F4	Store an evidence record per case: measurements, scores, warnings, verdict, reason	Delivered
F5	Plain-language explanation checked by code, with a fixed-template fallback	Delivered
F6	Local web application with Mechanic and Developer views	Delivered
F7	Optional tool-using AI investigation assistant	Delivered

3.2 Non-functional objectives

- **Runs locally on a CPU.** A cloud service is not required. Without the optional local language model, fixed-template explanations remain available.
- **Responsive diagnosis.** The measured median was 885 ms for a local, in-memory recording benchmark. Language-model answers are timed separately (Section 9.6).
- **Reproducible.** The app checks model files against stored hashes, and the data split has a recorded identifier.
- **Careful wording.** Tests protect key statements about uncertainty and model predictions (Section 7.4).

3.3 Success metrics and declared bars

Each question has its own main metric. I report counts with rates, especially where the test group is small.

Table 5. Primary metrics per question.

Question	Primary metric	Also reported
Q1	Fault detection and healthy false-alarm rate at the frozen thresholds; AUROC	Detection at the mildest settings; per-class detection
Q2	Ten-class macro-F1 and balanced accuracy at recording level	Six-group macro-F1; per-class F1; confusion matrix
Q3	Macro AUROC over withheld classes; unknown recall; false-unknown rate on known faults	OSCR; withheld faults called healthy; per-class outcomes

Every improvement stage set its acceptance rule before running. For unfamiliar-fault work, a method had to improve recall by at least five percentage points at the same known-side rejection count, without increasing the number of withheld faults called healthy, on all three seeds. A stage that missed that bar is reported as unsuccessful rather than tuned after the result.

3.4 Out of scope

- Predicting remaining useful life, continuous monitoring, or natural fault progression over time.
- Confirming a physical root cause, prescribing repairs, or controlling a machine.
- Performance on machines other than the MAFAULDA rig.

4. Data

4.1 MAFAULDA: the rig and the recordings

MAFAULDA comes from a bench-top fault simulator at the Federal University of Rio de Janeiro [1]. A motor drives a shaft with two rotor discs. One bearing sits between motor and rotor (underhang); the other sits beyond the rotor (overhang). The database contains deliberately induced faults at speeds of roughly 12–61 Hz.

The dataset contains 1,951 five-second recordings sampled at 50 kHz. Each recording has 250,000 samples on eight synchronised channels: one tachometer, three accelerometer axes at each bearing, and one microphone.

Table 6. Conditions in MAFAULDA and the operating settings that vary within each.

Condition (class)	Recordings	Operating setting
Normal	49	Speed only
Imbalance	333	Added mass 6–35 g
Horizontal misalignment	197	Offset 0.5–2.0 mm
Vertical misalignment	301	Offset 0.51–1.90 mm
Underhang bearing: ball / cage / outer race	186 / 188 / 184	Added imbalance mass 0, 6, 20, 35 g
Overhang bearing: ball / cage / outer race	137 / 188 / 188	Added imbalance mass 0, 6, 20, 35 g
Total	**1,951**	

Two data constraints shape the study. First, all 49 normal recordings are in one folder; the dataset has no session identifiers, so they may share acquisition conditions. Second, the bearing subfolders vary added imbalance mass, not bearing-defect size. I therefore call those variations operating settings, not fault severity.

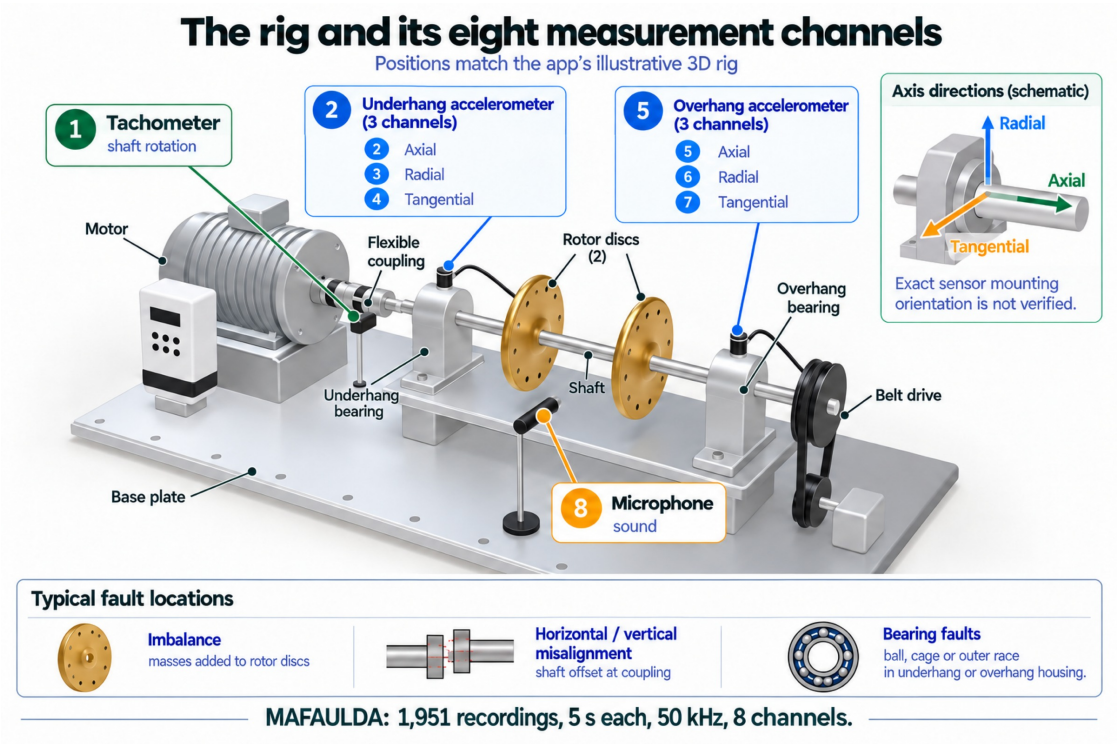


Figure 2. The MAFAULDA rig and the positions of its eight measurement channels.

Before any model, I checked how the recordings spread over shaft speed. Figure 3 is the plot from the indexing notebook.

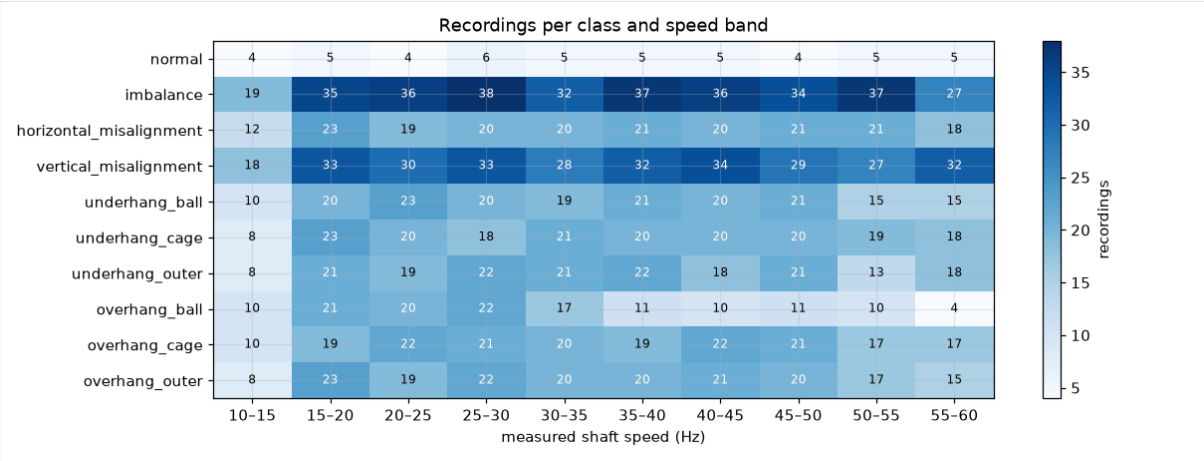


Figure 3. Recordings per class and measured speed band (all 1,951 recordings, from the index). Real output of notebook 01 (01_index_audit_splits.ipynb), inserted unedited.

- **Why I plotted it:** Each class must cover the full speed range. If a class appeared only at some speeds, a classifier could learn the speed instead of the fault.
- **How to read it:** Rows are the ten classes. Columns are 5 Hz bands of measured shaft speed. Each number is a count of recordings, and darker cells hold more recordings. The plot uses file metadata only; no signal is loaded.
- **What it shows:** Every class covers 10 to 60 Hz, so speed alone cannot identify a class. The normal row is pale: only 4 to 6 recordings per band, 49 in total. Imbalance has 19 to 38 per band. Overhang ball thins above 35 Hz (4 recordings at 55 to 60 Hz).
- **Decision:** I kept every class and handled the imbalance with class weights and macro-F1 (Sections 4.4 and 8.4). The thin healthy row is the first reason I treat the healthy result as the weakest evidence in this report.

4.2 Data audit

I indexed and audited the files before training. The audit found issues that could affect both validity and model behaviour.

Table 7. Data audit findings and how each was handled.

Finding	Scale	Handling
Filename speed differs from measured speed	Measured speed is about 2.3 % lower (median ratio 0.977)	Speed estimated from the tachometer; three methods cross-checked (median agreement 0.010 %)
Tachometer glitches (displaced pulses)	228 files; 94 severe	Speed from clean intervals only; tachometer FFT used for the 94 severe files
Microphone electrical bursts, linked to classes	168 files, almost all imbalance 20–35 g or underhang cage	Bursts interpolated (threshold from healthy training files); microphone peak statistics banned as features
Interference line near 22 kHz	All accelerometer channels	Fixed band-stop filter, 21.87–22.20 kHz
Accelerometer saturation	9 overhang-ball files at 47–55 Hz	Kept; flagged as a quality warning

Finding	Scale	Handling
DC offsets on overhang accelerometers	44 files above 0.2 V	DC removed per window
Bearing sub-folders are added mass, not defect size	All bearing classes	Called "operating settings" throughout

The microphone bursts were a real shortcut risk. A simple classifier using only microphone peak statistics identified "imbalance or underhang cage" with **99.3 % precision** whenever a burst was present. A model could have learned the recording session instead of the fault. The bursts were removed and the features that could detect them were banned before any classifier was trained.

The audit flagged spikes in the microphone channel. Figures 4 and 5 are the two notebook plots that decided how to handle them.

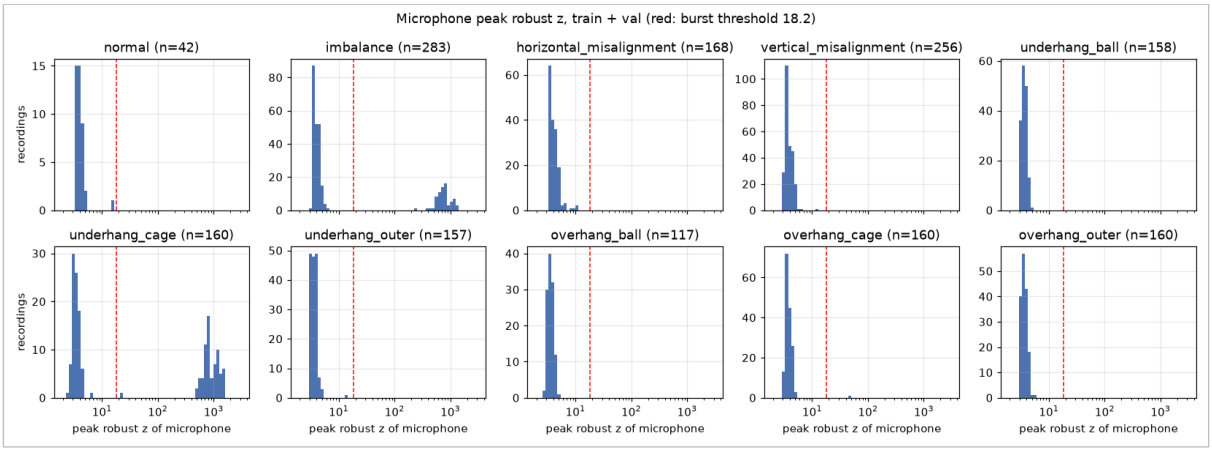


Figure 4. Largest microphone spike score per recording, one panel per class (closed-split train and validation recordings).
Real output of notebook 02 (02_signal_exploration.ipynb), inserted unedited.

- **Why I plotted it:** I needed to know which classes have the spikes, and where to put a threshold that catches bursts but not ordinary loud moments.
- **How to read it:** The x-axis is the largest robust z-score of the microphone in one recording, on a log scale. A clean microphone stays in single digits. The y-axis counts recordings. The red dashed line is the burst threshold.
- **What it shows:** Imbalance and underhang cage have a second group of recordings far to the right, with scores in the hundreds or above 1,000. The other classes stay left of the line. The threshold rule is 1.25 × the largest score among training normal recordings (14.57), which gives 18.2. It falls in a clear gap: no recording scores between 14.57 and 21.18. It catches 168 files: 83 imbalance, 84 underhang cage and 1 overhang cage.
- **Decision:** The value 18.2 became the fixed burst rule in the pipeline (Section 4.5, Step 6). Because the bursts identify two classes, I also banned the microphone peak statistics as features.

A spike could be a real sound from the machine or an electrical glitch. Figure 5 looks at three bursts sample by sample.

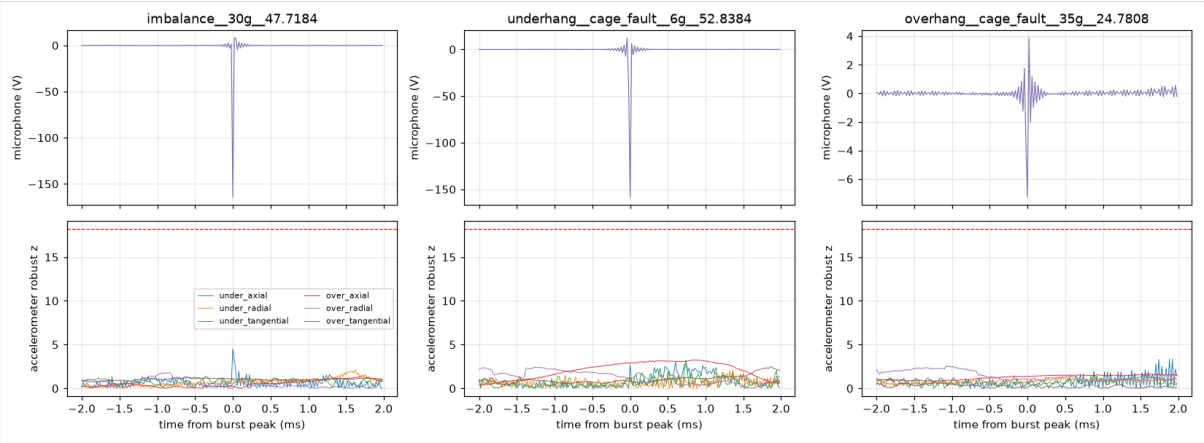


Figure 5. Three microphone bursts, 2 ms either side of the peak, with the six accelerometers below. Real output of notebook 02 (02_signal_exploration.ipynb), inserted unedited.

- **Why I plotted it:** A real acoustic event rings for a while, and it usually shakes the frame, so the accelerometers would react. An electrical glitch does neither.
- **How to read it:** Each column is one recording. The top row is the microphone in volts. The bottom row shows the six accelerometers as robust z-scores over the same 4 ms; the red dashed line is the same threshold of 18.2.
- **What it shows:** Each burst is a spike a few samples wide with no long ring-down, and the accelerometers stay low. Over all 143 train and validation burst files, the median accelerometer score at a burst was 2.51, against 2.46 at random times. The median ring-down ratio was 1.00.
- **Decision:** I treated the bursts as electrical artefacts, not machine signal. The pipeline replaces the burst samples by interpolation before any feature is computed. A later check in the same notebook found residual filter ringing, so I widened the patch from 5 to 15 samples.

4.3 Splits and leakage control

I assigned each recording to a split before creating windows and saved the assignment in metadata.csv under a split identifier. Windows from one recording always stay in the same partition.

Table 8. How the data were partitioned.

Partition	Purpose	Size
Closed split (train / val / test)	Q2 on unseen recordings of known conditions	1,367 / 294 / 290
Speed block	Familiar faults at unseen speeds (measured ≥ 40 Hz held out)	776 test recordings
Severity	Familiar faults at the largest, unseen setting of each family	370 test recordings
Development folds (K = 3)	All model development; folds grouped by source folder	804 recordings
Leave-one-class-out (LOCO)	Q3: each of 9 fault classes withheld in turn; normal never withheld	9 classes × 3 folds × 3 seeds = 81 runs

Figure 6a shows the same partition as a flow. The closed split gives about 70 % training, 15 % validation and 15 % test recordings (1,367 / 294 / 290). The speed-block and largest-setting recordings are then removed, and what remains is the development pool of 804 recordings.

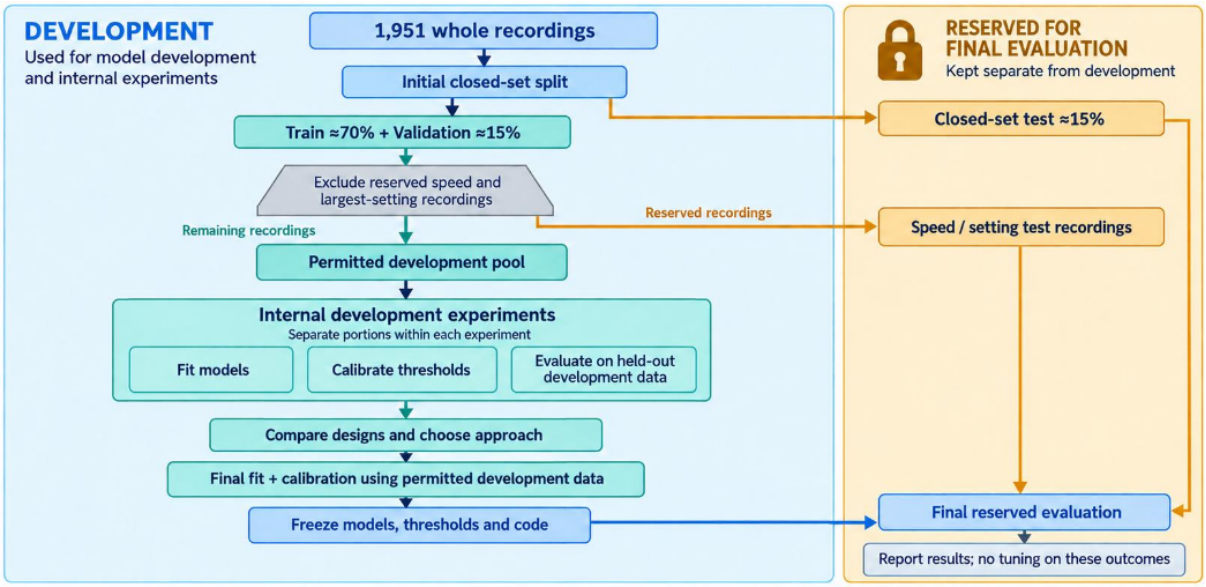


Figure 6a. How the recordings move from the split to the one final evaluation. Left: development data, used for every fit, calibration and design choice. Right: reserved recordings, kept apart until the models, thresholds and code were frozen.

Reserved tests. The closed, speed-block and severity partitions remained sealed until the one-time final evaluation (F1), after the model, thresholds and decision rule were fixed. I disclose overlaps between test partitions rather than treating their results as independent samples.

Folder grouping. MAFAULDA does not provide session IDs. I use each class-and-setting source folder as a proxy for a session, and hold out whole folders in development folds. The normal recordings occupy one folder, so I divide them into speed blocks instead. This difference limits how strongly the healthy result can be interpreted.

Leakage check. On the same training pool and physics features, a random window split scored 1.000 macro-F1. Every test window then had another window from its recording in training. A recording-grouped split scored 0.991 ± 0.005. The small gap does not make the random split valid; it hides the dependence between windows.

Figure 6b repeats the leakage check with a third, stricter protocol and with matched training sizes.

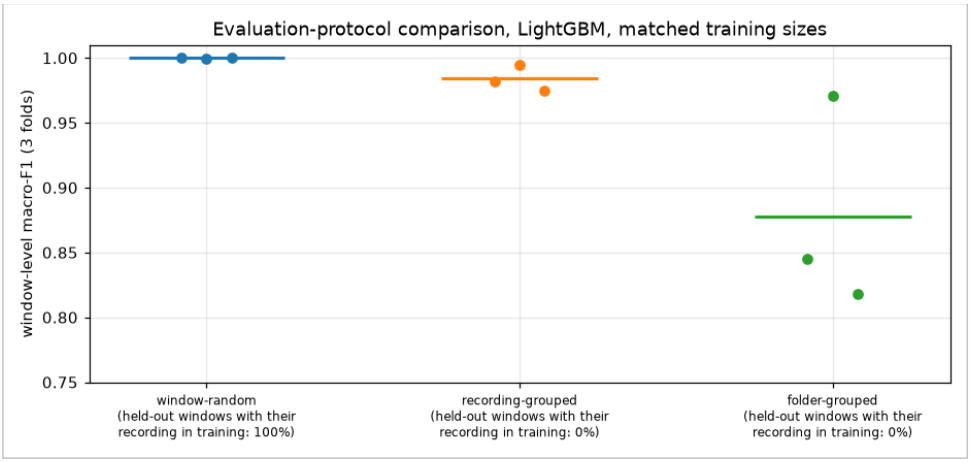


Figure 6b. The same LightGBM model under three ways of holding out data (development pool, three folds). Real output of notebook 05b (05b_dev_folder_grouped.ipynb), inserted unedited.

- **Why I plotted it:** I wanted to show how much the score depends on what is held out, with the model and the training size fixed. Only the choice of held-out data changes.
- **How to read it:** Each column is one protocol. Each dot is the window-level macro-F1 of one fold, and the line is the mean. The label under each column gives the share of held-out windows whose recording also had windows in training.
- **What it shows:** Window-random scores 1.000, and 100 % of its held-out windows have a sibling window in training. Holding out whole recordings gives 0.984. Holding out whole source folders gives 0.878. All three train on exactly the same number of windows per fold (4,842, 4,707 and 4,923). The drop from 0.984 to 0.878 is the similarity between recordings of the same folder, which a recording-level split cannot see.
- **Decision:** All model development used folder-grouped folds, the strictest protocol. I never report a window-random score as a result.

4.4 Learning from imbalanced data

The dataset has 49 normal recordings but 333 imbalance recordings. For supervised models, balanced class weights reduce this imbalance, while macro-F1 and balanced accuracy give every class equal weight in evaluation. The Q1 detectors use healthy data only; their main problem is the limited healthy coverage, which class weighting cannot fix. I did not use SMOTE because synthetic samples from the same folder would not add new operating conditions.

4.5 Data pipeline: from raw files to model inputs

This section follows one recording from the downloaded archive to the two inputs the models use. Every step runs in code in the `ml/` package, with fixed settings, and the same steps run in the app for a new upload.

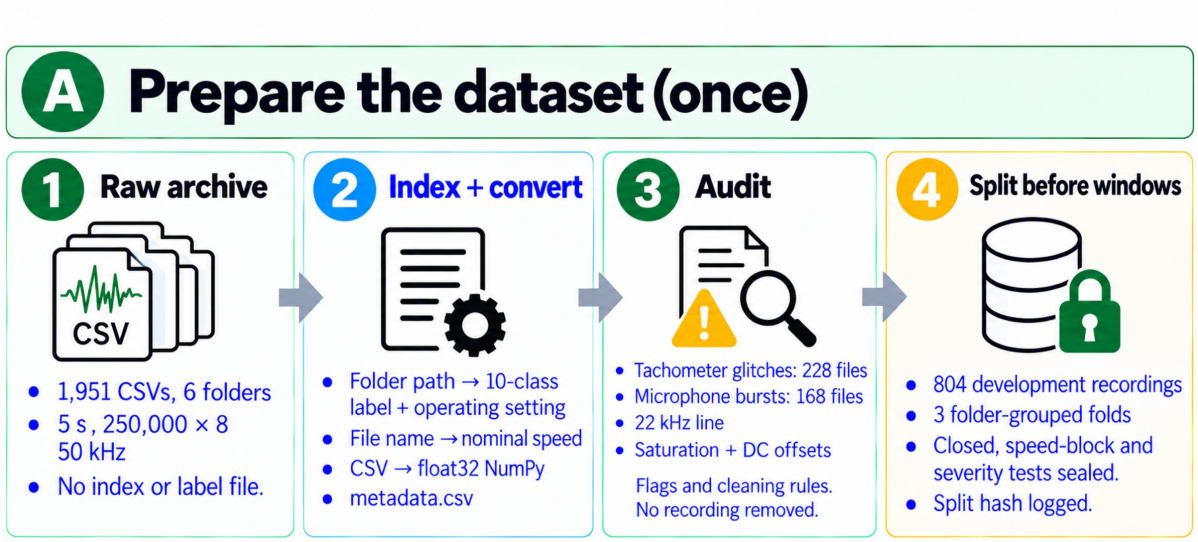


Figure 7a. The data pipeline, part A: preparing the dataset once (steps 1 to 4).

Step 1: the raw archive

- The download is one archive with six top-level folders and 1,951 CSV files. It has no index and no label file.
- Each file is one 5-second recording: 250,000 rows at 50 kHz and 8 columns (tachometer, three accelerometer axes at each bearing, microphone).

Step 2: index and convert

- I built an index from the folder path and the file name: the condition (one of ten classes), the operating setting (for example the added mass or the offset) and the nominal speed in the file name.
- Each CSV was converted once to a float32 NumPy array, so later steps read binary data quickly and identically.
- The index became `metadata.csv`, the single table that later also holds the measured speed and every split assignment.

Step 3: audit before any model

- I checked every file for shape, missing values, clipping, DC offsets, tachometer glitches and microphone bursts (Table 7).
- Problems became **flags and cleaning rules**, not deletions. No recording was removed, so the class counts stay as published.
- The audit found the microphone shortcut (99.3 % precision for two classes) before any classifier was trained.

Step 4: split by recording, before any window exists

- Each recording was assigned to its partitions (Table 8) before windowing, and the assignment was saved with a split hash. Windows from one recording therefore can never be split between training and testing.
- The three reserved test partitions were sealed. Every later step, including learned preprocessing, saw development data only.
- Inside the development pool, three folds hold out whole source folders, and a separate calibration set is kept apart from model fitting.

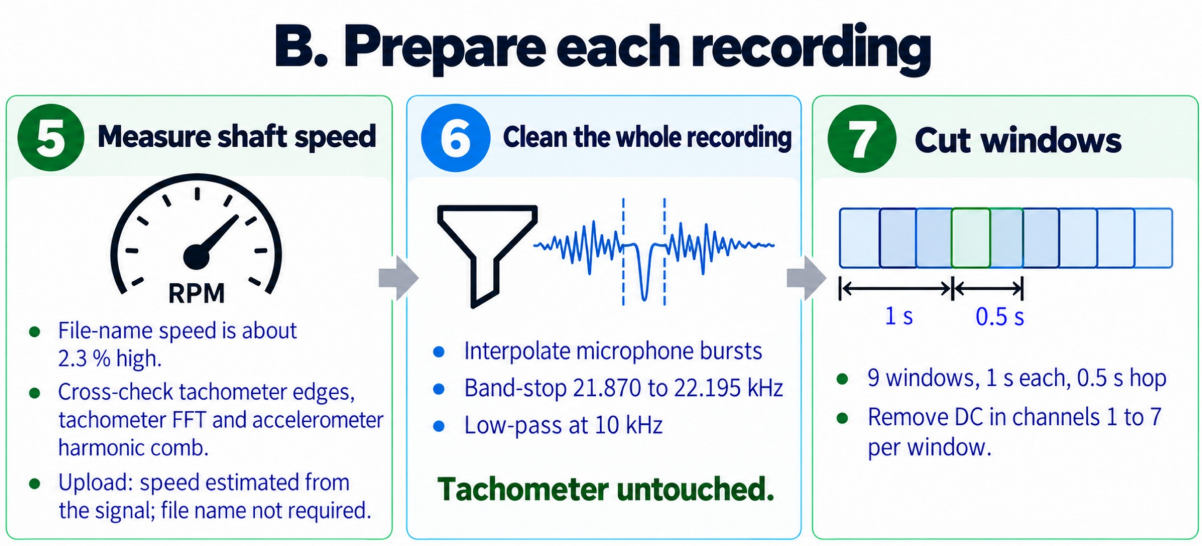


Figure 7b. The data pipeline, part B: preparing each recording (steps 5 to 7).

Step 5: measure the shaft speed

- File-name speeds are about 2.3 % higher than the measured speed, so the speed is always measured from the signal.
- Three estimates are compared: tachometer pulse edges, a tachometer FFT and a harmonic comb on the accelerometers. The edges win when another method confirms them; otherwise the FFT, if the comb confirms it; otherwise the median.
- Each window also gets its own speed from at least three clean tachometer intervals; otherwise it uses the recording speed.

Step 6: clean the signal

- **Microphone bursts:** samples with a robust z-score above 18.2 (median absolute deviation) are widened by 15 samples and interpolated.
- **Interference:** a Kaiser band-stop filter removes the 21.87–22.20 kHz line on the seven sensor channels.
- **Low-pass:** pass band to 9.8 kHz, stop band from 10.2 kHz, so the analysis uses 0–10 kHz.
- The tachometer is never filtered, because the speed depends on its exact pulse timing.

Step 7: cut into windows

- Each recording becomes **nine 1-second windows** with a 0.5-second hop. DC is removed per window (not on the tachometer).
- Models score each window; the nine window scores are averaged into one recording-level score.

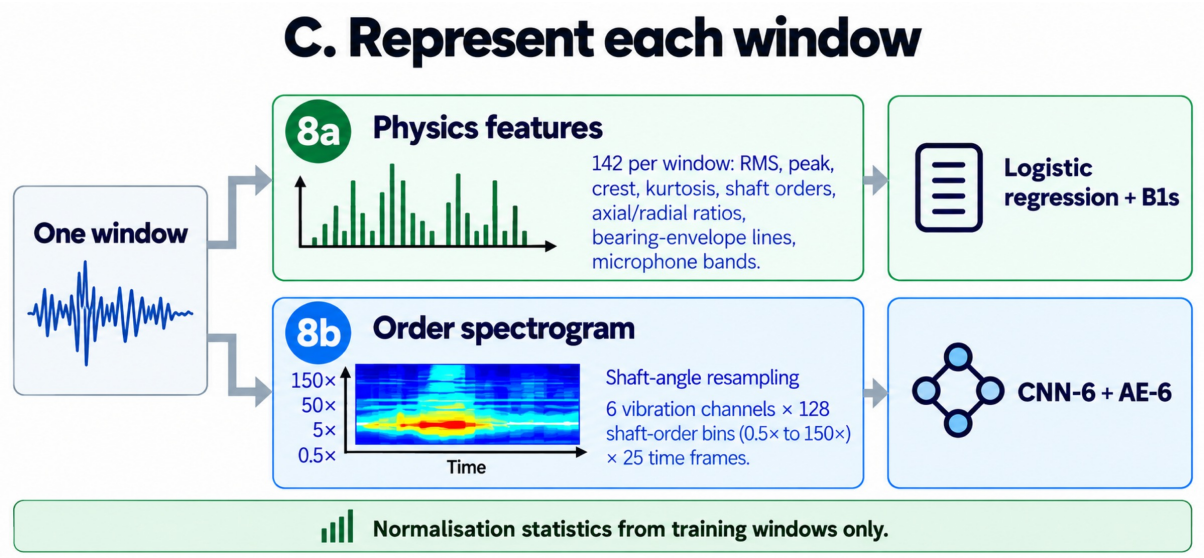


Figure 7c. The data pipeline, part C: representing each window (step 8).

Step 8: build the two representations

- **8a, physics features:** 142 numbers per window, each defined in a feature dictionary with its unit (Section 5.3): per-axis RMS, peak, crest factor, kurtosis and shaft-order amplitudes; axial/radial ratios; bearing envelope lines; microphone bands. Used by logistic regression and B1s.
- **8b, order spectrogram:** each vibration channel is resampled to the shaft angle (512 samples per revolution, 8 revolutions per frame) and mapped to 128 log-spaced orders from 0.5× to 150×, over 25 frames. The six vibration channels give a 6 × 128 × 25 input for CNN-6 and AE-6. The spectrogram also has an aligned microphone channel, but no deployed model uses that channel.
- **Normalisation:** statistics come from the training windows of each model only, never from calibration or test data.

Leakage controls in one place: splits fixed per recording before windowing; reserved tests sealed until one final run; normalisation and thresholds fitted on separate development subsets; class-linked microphone artefacts removed and their features banned; and a test that a random window split scores a misleading 1.000 (Section 4.3).

Before any model used the physics features, I checked them against known machine physics (Figure 8). A bug in feature code would show here first.

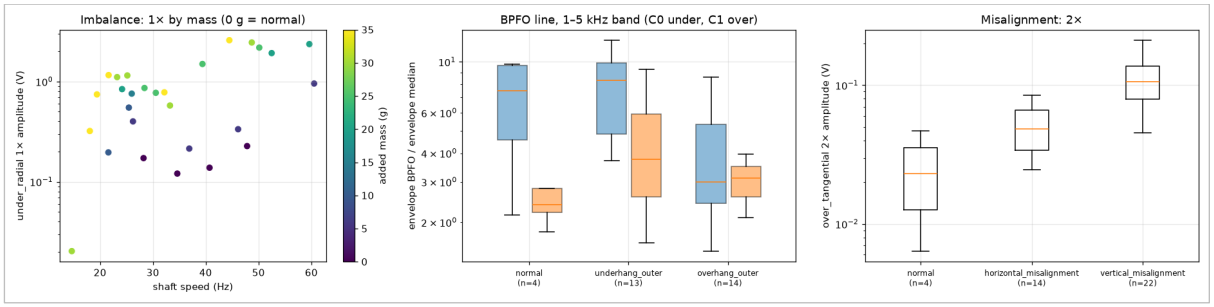


Figure 8. Three physics checks on the feature table (a sample of recordings; group sizes under each box). Real output of notebook 03 (03_windows_physics_features.ipynb), inserted unedited.

- **Why I plotted it:** Each feature has a physical meaning. If imbalance did not raise the 1× amplitude, or misalignment did not raise 2×, the feature code would be wrong.

- **How to read it:** Left: the 1× amplitude on the underhang radial channel against shaft speed; colour is the added imbalance mass (0 g is normal). Middle: the envelope line at the outer-race fault frequency (BPFO), blue for the underhang radial channel and orange for the overhang radial channel. Right: the 2× amplitude on the overhang tangential channel. All y-axes are logarithmic.
- **What it shows:** The 1× amplitude rises with mass: median 0.16 V for normal, 0.37 V at 6 g and 1.9 V at 20 g. Above 20 g the trend is not monotonic, because each mass has only about 3 recordings and the spread of speeds dominates. The 2× amplitude doubles for horizontal misalignment (0.048 V against 0.023 V) and more than quadruples for vertical misalignment (0.106 V). The BPFO line is weak: on the overhang channel the median rises only from 2.4 to 3.8 (underhang outer race) and 3.1 (overhang outer race).
- **Decision:** I kept the features unchanged. I recorded the weak BPFO line as a known limit of the hand-made bearing features. The notebook also flagged one imbalance recording with an unusually low 1× value on one channel; I checked the channel and kept the recording.

5. System Design and Models

5.1 System architecture

MachineSense has four layers. The scoring layer alone runs the diagnostic models; the other layers handle input, storage and presentation.

Table 9. The four layers of MachineSense.

Layer	What it does	Main code
Web application	Mechanic and Developer views, pipeline page, unseen-fault wizard, assistant panel	React 18 + Vite (`frontend/src`)
API and case services	Checks the input, runs a diagnosis, stores the case, builds evidence, report, trace and explanation, runs the assistant	FastAPI (`backend/app.py`, `backend/engine.py`)
Scoring	Cleans the signal, builds features and spectrograms, runs every model and the decision rule; the only entry point is `ml.inference.diagnose()`	Python package `ml/`, CPU, float32
Storage and models	Hash-verified model bundle; SQLite case store; guidance cards	`artifacts/bundle`, `backend/data/cases.db`, `knowledge/`

- **Model bundle:** twelve packaged files hold the trained models, thresholds, decision rule, normalisation, feature dictionary and calibration scores. A manifest lists their SHA-256 hashes; the engine checks them at startup and refuses a mismatch.
- **Optional local model:** LM Studio serves Gemma for explanations and assistant answers. In the documented local configuration, requests stay on the machine. The diagnosis and fixed-template report still work without the language model.
- **Saved views:** the report, pipeline and dashboard read the stored case record. Opening a case does not score it again.

A diagnosis moves through six stages. The pipeline page displays the stored result and timing of each stage.

Table 10. The six stages of a diagnosis.

Stage	Question it answers	Main components
1. Recording	What was measured?	8-channel CSV upload or a stored development recording
2. Preparation and quality	Is the recording usable?	Input checks, reserved-recording guard, speed estimation, cleaning, windowing, signal-quality warnings
3. Features	What does the signal contain?	142 physics features per window; order spectrogram
4. Models	What do the models see?	CNN-6, familiarity (Mahalanobis), AE-6, B1s, LR second opinion
5. Decision	Do the checks support a verdict?	Fixed three-way rule: healthy / known fault / further inspection
6. Outputs	What is saved and shown?	Evidence record, explanation, report, dashboard, assistant

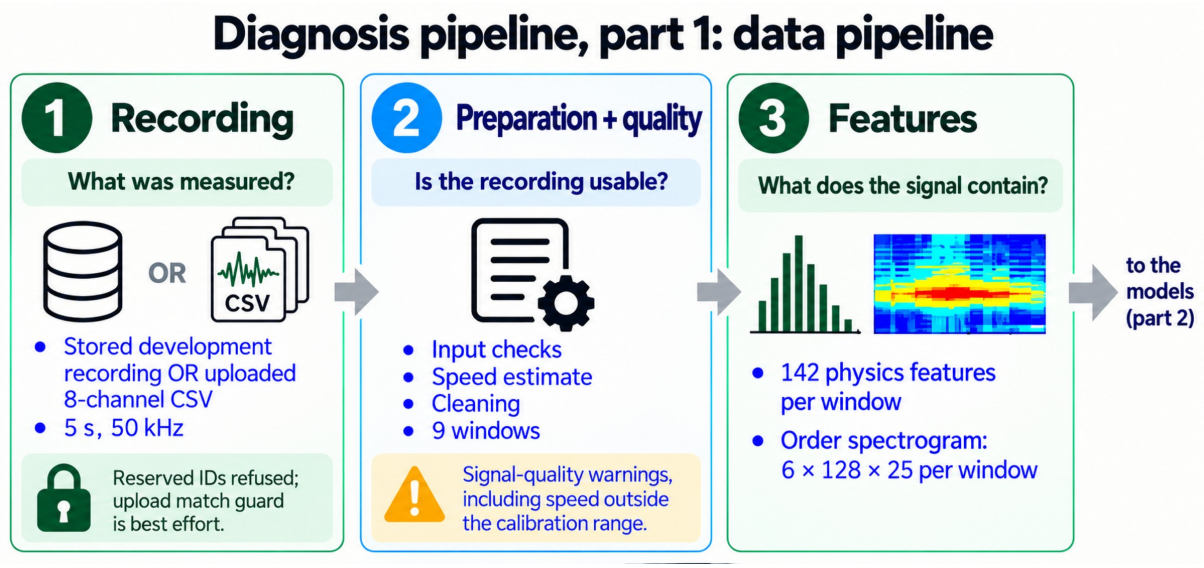


Figure 9a. The diagnosis pipeline, part 1: the data pipeline (stages 1 to 3).

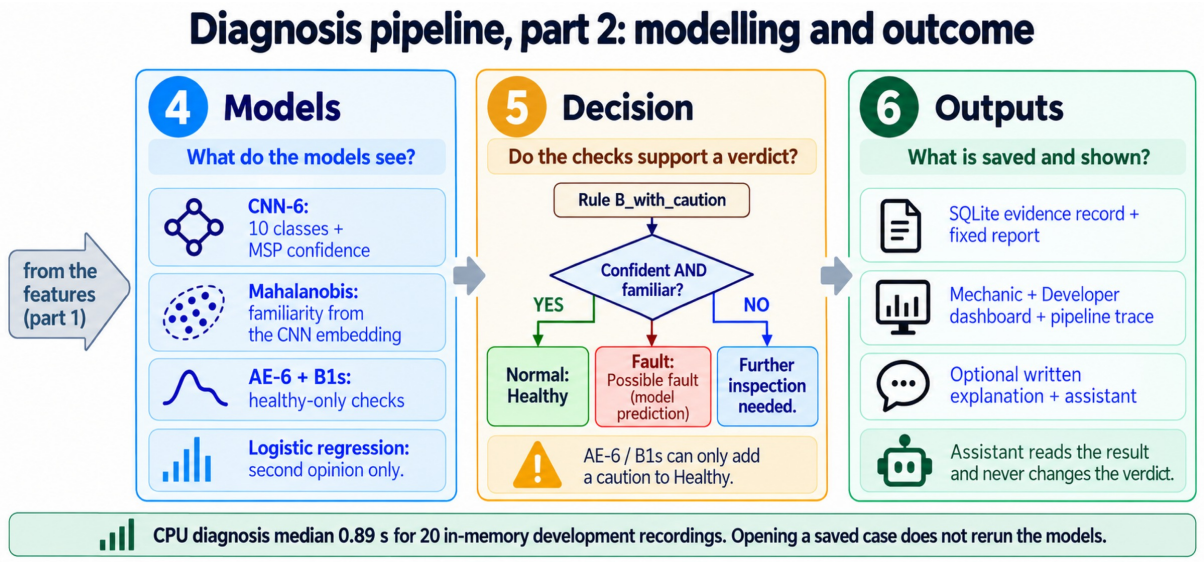


Figure 9b. The diagnosis pipeline, part 2: modelling and outcome (stages 4 to 6).

Input checks and the reserved-recording guard

- **Stored recordings:** the selector exposes development recordings only. The API refuses a reserved test recording (HTTP 403).
- **Uploads:** the app accepts a headerless CSV with exactly eight columns, 250,000 rows and finite values. It explains why any other shape is refused.
- **Reserved-upload guard:** the app compares uploaded samples with reserved recordings using a hash and a second rounded fingerprint checked with channel statistics. A match is refused. This guard reduces accidental exposure but cannot guarantee that every transformed copy is caught.

What happens when a user starts an investigation

1. The user chooses a development recording or uploads a CSV in the browser.
2. The frontend sends `POST /api/diagnose`. The engine checks the input and the reserved-recording guard.

3. `ml.inference.diagnose()` measures the speed, cleans and windows the signal, builds both representations, runs all five models and applies the decision rule. This takes under 1 second on a CPU.
4. The result is validated against the evidence-record schema and saved in SQLite with its timing and provenance.
5. The dashboard loads the record, the report, the measured channel evidence, the signal views and the pipeline trace. All of these are built from the saved record.
6. In the background, the app asks the local language model for a written explanation (about 15–20 seconds). Until it arrives, or if it fails a check, the page shows a fixed template.
7. The assistant panel answers questions about this case on demand (Section 5.10).

Design principles

- **One scoring path.** The app, the threshold calibration and the final evaluation all call the same scoring function on the CPU. A GPU-versus-CPU mismatch found during development was fixed before the final evaluation, so the thresholds match what the app computes.
- **The saved record is the single source of truth.** Every page reads the stored evidence record. Nothing is re-scored when a case is opened, so two people see the same result.
- **Kinds of fact stay apart.** Measured values, model outputs, guidance and missing information are tagged differently in the report, the dashboard and the assistant.
- **The language model is optional and never decides.** It writes words around the result; code checks every number it uses. With it switched off, every page still works.
- **The app sees development data only.** Reserved recordings cannot be listed, chosen or uploaded.

5.2 Signal preparation

- **File checks:** the system checks shape, channel count, missing values and clipping. Quality problems appear as warnings in the evidence record; they do not become model features.
- **Speed:** the pipeline compares tachometer pulse edges, a tachometer FFT and an accelerometer harmonic comb. It prefers a confirmed edge estimate, then a confirmed FFT estimate, and otherwise uses the median. A window uses its own speed when at least three clean pulse intervals are available; otherwise it uses the recording estimate.
- **Tachometer glitches:** hysteresis-edge checks remove double triggers and dropouts. The count becomes a quality flag, not a model input.
- **Cleaning:** large microphone bursts are detected with a robust z-score, widened by 15 samples and interpolated. A Kaiser band-stop removes the 21.9–22.2 kHz interference, followed by a low-pass filter. The tachometer is left unfiltered.
- Quality warnings cover out-of-range or unstable speed, window-speed fallbacks, tachometer glitches, removed microphone bursts, missing values and clipped channels.
- **Windowing:** the five-second signal becomes nine one-second windows at half-second intervals. DC is removed within each window, and the nine scores are combined into one recording-level decision.

5.3 Representations

Physics features (R1)

Each window produces 142 physics features with documented units. They include:

- Per channel: RMS, peak, crest factor and kurtosis.

- Per accelerometer: amplitudes from 0.5× to 10× shaft order. A strong 1× component can indicate imbalance; 2× and axial vibration can support a misalignment interpretation, but neither pattern proves a cause.
- On one radial channel at each bearing, an envelope spectrum measures how clearly the bearing-defect lines rise above the surrounding background. The rig-specific rates are BPFO 2.998×, BPFI 5.002×, BSF 1.871× and FTF 0.375× shaft speed [14].
- Scale-free ratios, such as 2×/1× and axial/radial 1×, which change less with speed than raw amplitudes.
- Microphone band energies, spectral centroid and roll-off (peak statistics excluded, Section 4.2).

Order spectrogram

A vibration peak moves in hertz as the shaft speed changes, but stays near the same shaft order. The pipeline therefore uses tachometer-based order tracking: 512 samples per revolution, eight revolutions per frame and 25 frames per window. It maps each frame to 128 log-spaced orders from 0.5× to 150×. The six vibration channels form a 6 × 128 × 25 input, normalised using training data only.

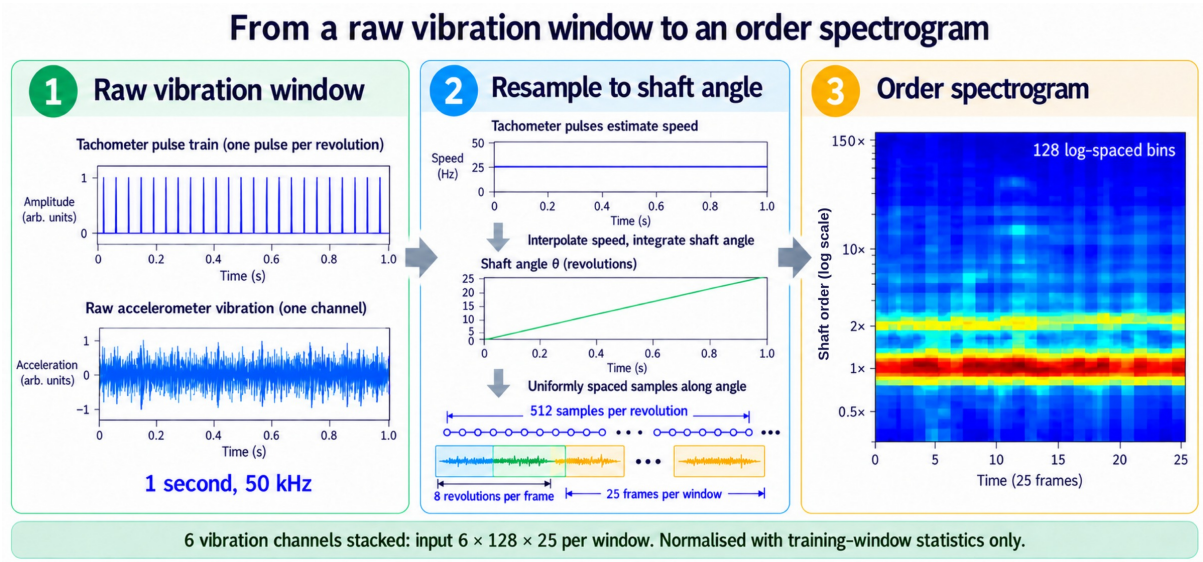


Figure 10a. From a raw vibration window to an order spectrogram.

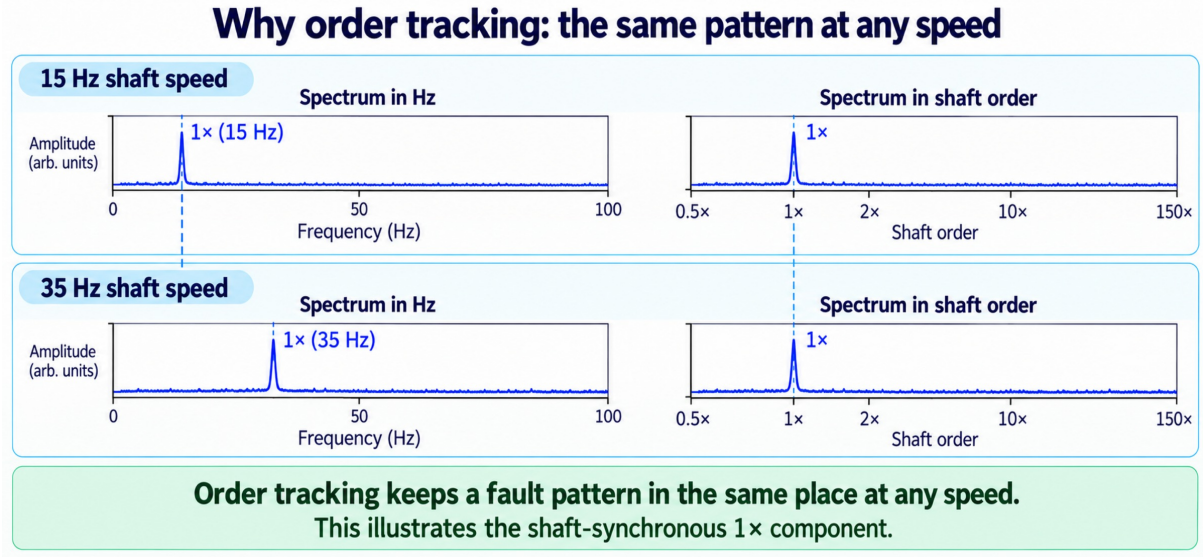


Figure 10b. Why order tracking: the same pattern at any shaft speed.

Figure 11a shows the real model input, read back from the feature cache, for one recording of each class.

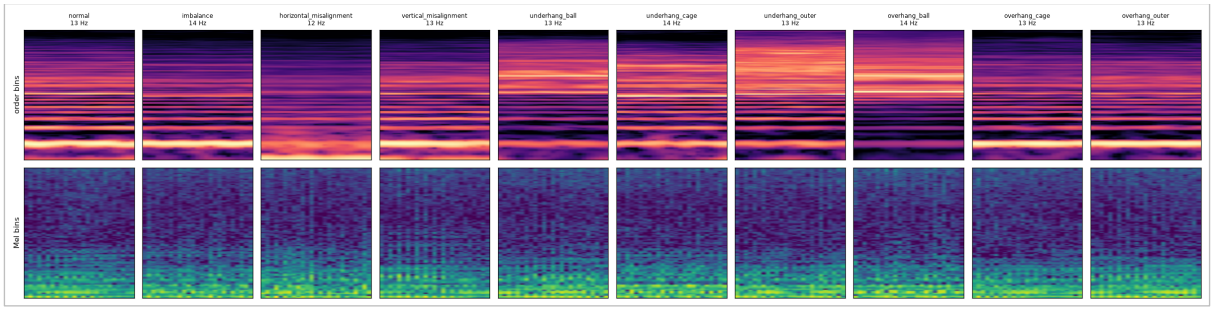


Figure 11a. Cached inputs for one training recording per class, middle window: order spectrogram (top) and microphone log-Mel (bottom). Real output of notebook 04 (04_representations.ipynb), inserted unedited.

- **Why I plotted it:** I wanted to see what the CNN actually receives, and to check that the cache holds a sensible picture for every class before training.
- **How to read it:** Each column is one class, with its measured speed. In the top row the y-axis is shaft order (log scale, 0.5× to 150×), the x-axis is time over 25 frames, and brighter means more vibration; the image is the mean of the six accelerometer maps. The bottom row is the microphone log-Mel for the same window.
- **What it shows:** The horizontal lines are shaft orders. They stay straight across the frames because the axis is in orders, not hertz. The shaft-fault classes keep most energy at low orders, while the bearing classes show brighter bands at higher orders. The microphone images look much alike across classes.
- **Decision:** The cached order spectrogram became the input of CNN-6 and AE-6. The microphone channel stays in the cache, but no deployed model uses it.

5.4 Q2: recognising known faults

CNN-6, a compact two-dimensional convolutional network, is the main known-fault classifier. It learns from the six-channel order spectrogram.

- **CNN-6 has 601,674 parameters.** Four convolution blocks use 32, 64, 128 and 128 filters, followed by global average pooling, a 128-dimensional embedding and a dropout-regularised ten-class head.
- **Training:** class-weighted cross-entropy (Section 4.4), AdamW with weight decay, a warm-up then cosine learning-rate schedule and a fixed epoch budget.
- **Augmentation (training windows only):** ±1 dB gain jitter per channel, time shift and time masking.
- **Per recording:** the mean log-probability over the nine windows; MSP is the top probability.

Figure 11b shows the network. One network answers two questions: its top class is the Q2 label, and its confidence and embedding feed the two Q3 checks.

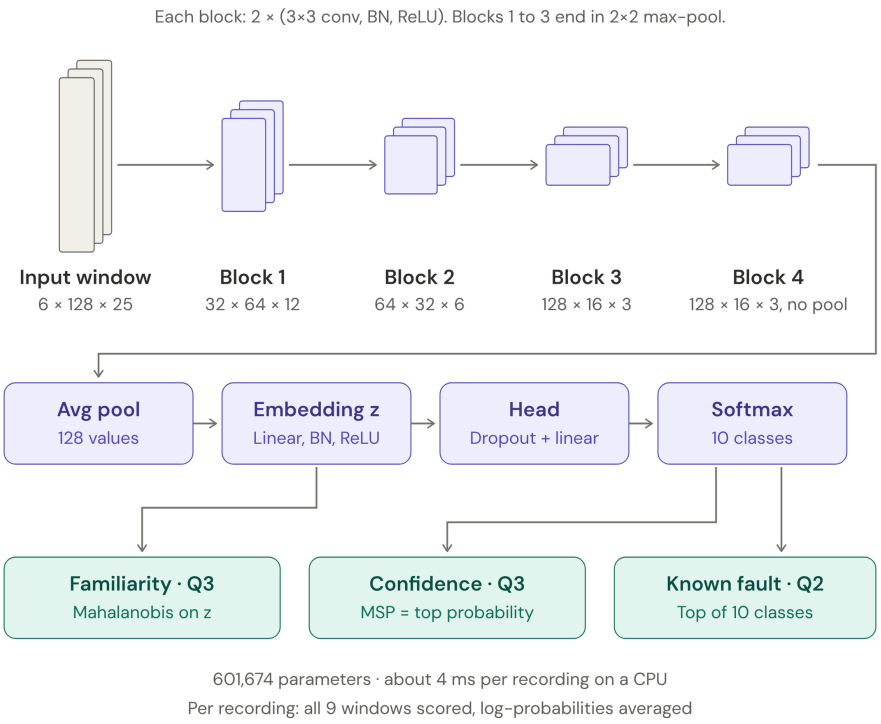


Figure 11b. CNN-6 architecture: one window through four convolution blocks to the embedding and the ten-class head. The embedding feeds familiarity (Q3); the softmax gives MSP confidence (Q3) and the known-fault label (Q2). Layer sizes from ml/models/cnn.py.

I also trained classical models on the 142 physics features:

- The comparison included multinomial logistic regression, k-nearest neighbours, an RBF support vector machine, Random Forests and LightGBM. The SVM was calibrated on recording-grouped folds.
- **Logistic regression** remains in the app as a second opinion. Its feature contributions provide context, but it cannot change the verdict.

The transfer-learning study (ML-5) adapted an ImageNet-pretrained ResNet-18 [16] to the six-channel spectrogram with a learned 1 × 1 adapter. It compared a frozen backbone with a new head against full fine-tuning at a lower backbone learning rate (Section 10.2).

5.5 Q1: the healthy check

Two models trained only on healthy recordings check whether the new recording resembles the healthy examples:

- AE-6 reconstructs the six vibration-channel order maps. A high reconstruction error means the recording differs from the healthy examples. Errors are averaged across channels and windows.
- B1s models each healthy vibration feature as a function of log shaft speed. It scores how far the recording lies from the value expected at that speed.
- **Thresholds:** τ_e and τ_b are the maximum score over the healthy calibration recordings.

- I also compared nine classical healthy-only detectors on development data, including simple rules, Mahalanobis distance, Gaussian mixtures, PCA reconstruction, one-class SVM, isolation forest and local outlier factor (Section 9.2).

5.6 Q3: unfamiliar faults

Two checks ask whether the pattern looks familiar to the known-fault model:

- **MSP:** the classifier's top softmax probability [8]. A low value means the classifier is unsure.
- Familiarity is the distance from the recording's CNN-6 embedding to the nearest known class centre. The score uses a shared Ledoit–Wolf covariance estimate [9], [17], computed per window and then averaged.
- **Thresholds:** each is set on calibration recordings to accept 97.5 % of known recordings.
- For Q3, I retrained CNN-6 across 81 leave-one-class-out combinations: nine withheld fault classes, three development folds and three seeds. The withheld class was excluded from fitting, preprocessing and calibration, then tested as an unfamiliar fault.

How I trained models with a hidden fault (leave-one-class-out)

A model cannot be tested on an unfamiliar fault if it has already seen that fault. To test Q3 honestly, I trained separate CNN-6 models, each with one fault class **completely hidden**. The hidden class then plays the part of a new, unknown fault.

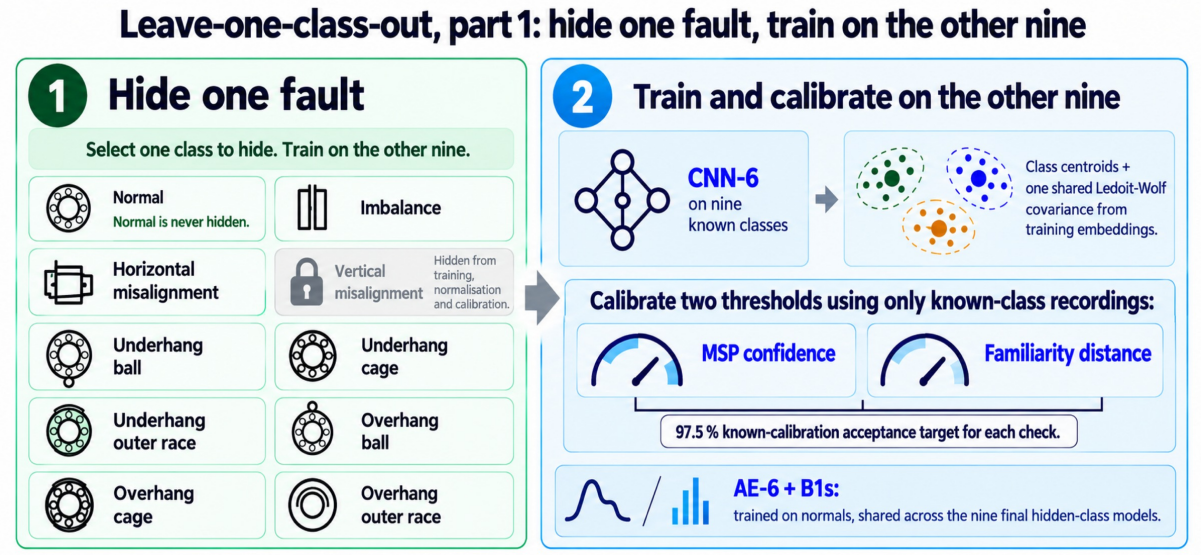


Figure 12a. Leave-one-class-out, part 1: hide one fault and train on the other nine.

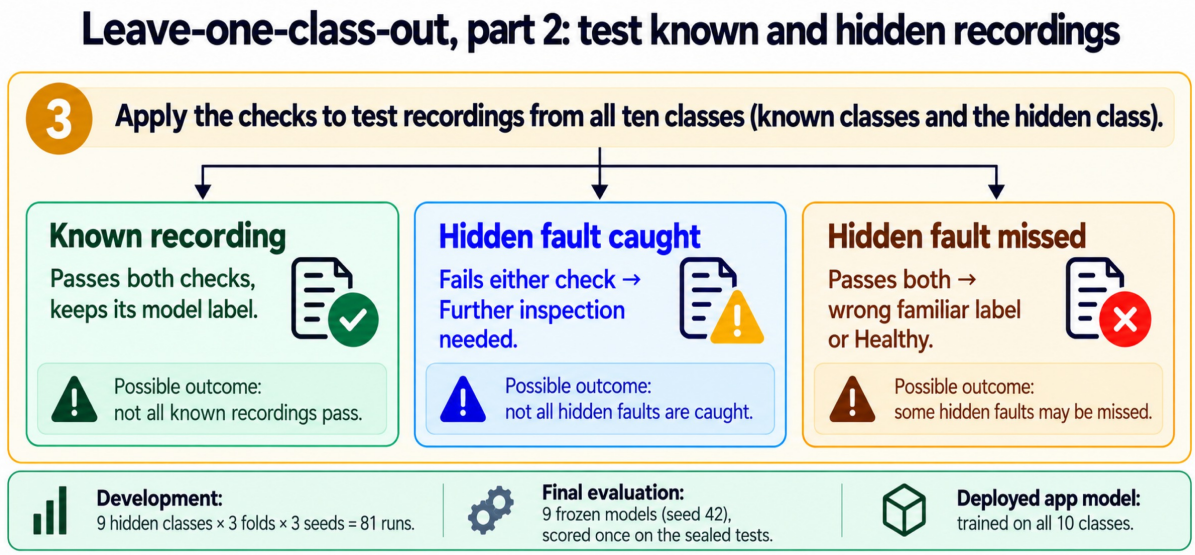


Figure 12b. Leave-one-class-out, part 2: test known and hidden recordings.

For each hidden class c, one run does the following:

1. **Remove class c everywhere it could leak.** The fit set and the calibration set contain only the other classes, and class c is also left out of the normalisation statistics. In the final runs, these are the development training and validation recordings minus class c.
2. **Train CNN-6 on the remaining nine classes** (normal plus eight faults), with the same architecture and settings as the deployed model.
3. **Build the familiarity model** from the fit-window embeddings: one centroid per known class and one shared Ledoit–Wolf covariance.
4. **Set the thresholds** on the calibration recordings of the known classes only: τ_p and τ_d accept 97.5 % of them. Nothing is chosen by looking at the hidden class.
5. **Score the held-out recordings.** Known-class recordings give the known side; recordings of class c give the unknown side. A good model passes known recordings and sends class c to "further inspection needed".

- **Healthy checks:** normal is never hidden, so AE-6 and B1s do not depend on the hidden class. In the final runs they were trained once and shared by all nine models.
- **Development runs:** 9 hidden classes × 3 development folds × 3 seeds = **81 runs**. Each development recording is scored once, by the model of its own outer fold. The research stages R4, R5 and R7–R10 reuse these saved runs.
- **Final runs:** before the final evaluation, **9 final models** (one per hidden class, seed 42) were trained on development data only and frozen. In F1 each was scored once on the sealed recordings: the known side is the reserved known-class recordings, and the unknown side is the reserved recordings of its hidden class (Section 9.3).
- **Measures:** AUROC and the open-set classification rate (threshold-free), and, at the frozen thresholds, unknown recall, the false-unknown rate on known faults and the number of hidden-fault recordings called healthy. Methods are compared at the same number of known recordings rejected.

How the app answers Q3 for a real recording

- The deployed model is trained on **all ten classes**. It has no "unknown" class to predict.

- Q3 is answered by the two checks: a recording that makes the classifier unsure (low MSP) **or** lies far from every known class (high distance) gets "further inspection needed" instead of a fault name.
- The leave-one-class-out models are not used for diagnosis. They exist to measure how well these checks work, and the "Explore an unseen fault" page uses the seed-42 models to show it.
- **What the tests showed:** unseen bearing faults are almost always rejected, while unseen imbalance and misalignment often land inside another shaft-fault cluster and get a wrong familiar label (Sections 9.3 and 10.3).

5.7 The decision rule

The fixed rule B_with_caution combines the checks into a finding. It was frozen before the final evaluation. The app displays it as the rule in use.

1. If the top-class score passes its confidence threshold and the feature distance passes its familiarity threshold, accept the top class: normal becomes healthy; a fault class becomes a known-fault prediction.
2. If either check fails, ask for further inspection.
3. AE-6 and B1s can add a caution to a healthy result when they disagree; they do not overturn a confident, familiar prediction.

The Developer view also calculates four rule variants for comparison. Only the rule in use determines the saved finding.

- **A, classifier first with an autoencoder override:** a non-confident result whose top class is normal becomes healthy if the autoencoder agrees.
- **B, classifier first (the rule in use, plus the caution):** as described above.
- **C, strict:** healthy also needs the autoencoder check to pass.
- **D, strict on speed:** healthy also needs the speed-aware check (B1s) to pass; otherwise further inspection.
- The fixed thresholds are τ_p = 0.998895 for MSP, τ_d = 30.04 for familiarity, τ_e = 0.243 for AE-6 and τ_b = 0.872 for B1s. They come from 144 calibration recordings; the two healthy thresholds use only five healthy calibration recordings. A value exactly on a threshold passes under the declared tie rule.

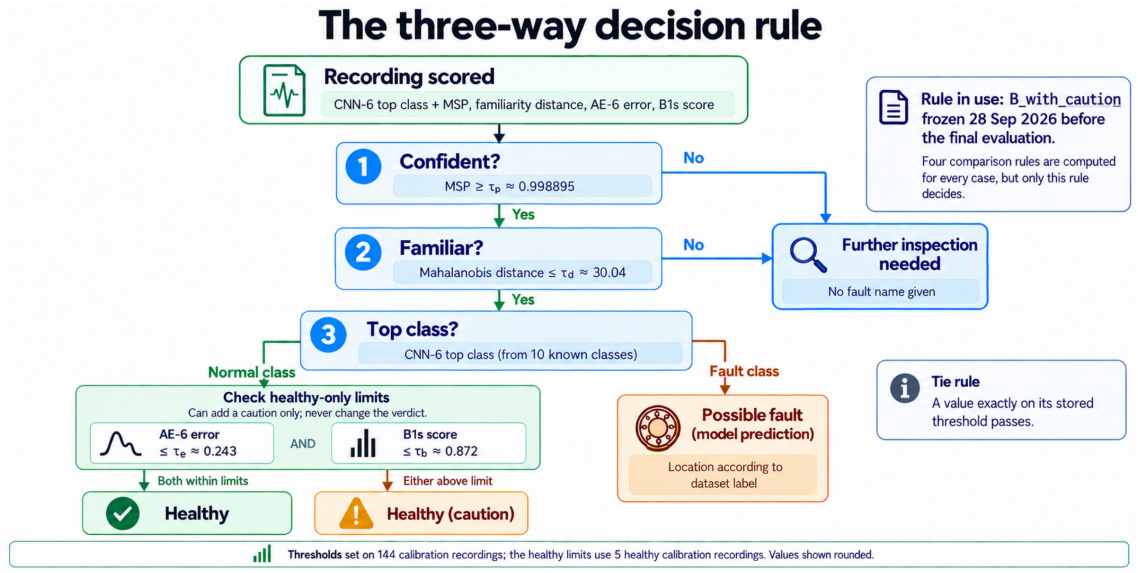


Figure 13. The three-way decision rule.

5.8 Evidence, explanation and guidance

Evidence record

- Each case is stored as a schema-validated record. It contains the input identity, signal quality, model scores and thresholds, decision variants, physical features, limitations, provenance and timing.
- The report, dashboard, pipeline and assistant read that saved record. They do not run the models again.

Measured evidence: what changed, compared with a healthy machine

- **Reference:** the healthy **training** recording nearest in measured speed (never the case itself).
- For each channel, the display averages RMS, kurtosis and 1×/2× amplitudes over the nine windows. Radial channels add four bearing-envelope lines; the microphone contributes RMS.
- The change factor uses whichever is larger, a ratio or its inverse, so a decrease counts as well as an increase. Kurtosis is not expressed as a ratio, and amplitudes below 0.005 V are left out of this comparison.
- The healthy band is the 95th percentile of change factors between healthy training recordings within ±3 Hz of each other. A channel is marked “changed” only when its factor exceeds that band.
- The comparison is descriptive, not a calibrated anomaly test. It shows where a signal changed, not where the physical fault is.

CNN attribution (Developer view)

- The Developer view can show an input × gradient map for CNN-6: which channels and order bands pushed the model towards its top class.
- The app labels it "CNN attribution · exploratory". It is not evidence of where the fault is.

Figure 14 is the notebook check of these maps on one recording per class.

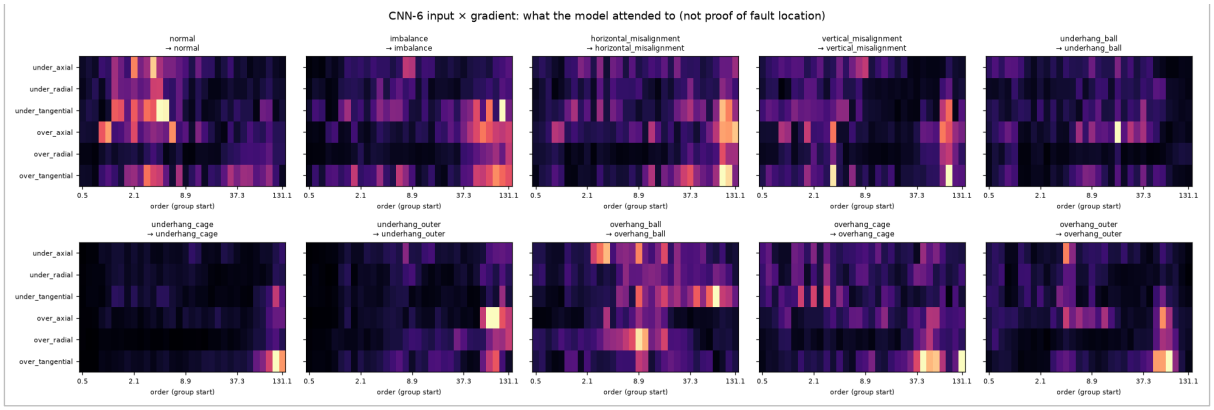


Figure 14. CNN-6 input × gradient maps for one calibration recording of each class (frozen bundle). Real output of notebook 10 (10_xai_ablations_selection.ipynb), inserted unedited.

- **Why I plotted it:** I wanted to see the maps on real cases before showing them in the app, and to confirm that they differ between classes instead of showing one fixed pattern.
- **How to read it:** Each panel is one recording; its title gives the true class, then the CNN top class after the arrow. Rows are the six vibration channels. Columns are 32 groups of shaft order on a log axis from 0.5× to 150×. Brighter cells received more attribution; each map is scaled from 0 to 1.

- **What it shows:** All ten recordings are classified correctly. The bright cells differ by class and often sit on the overhang channels and at high orders. The maps say where the model looked, not why the machine vibrates.
- **Decision:** I kept attribution in the app, but only in the Developer view and with the "exploratory" label. The Mechanic view uses measured evidence instead.

Written explanation

- **The language model receives a compact summary:** the finding, checks in plain words, top classes, warnings, measured changes and a short list of allowed numbers. It does not receive raw scores or thresholds.
- It returns a structured draft with a finding, reasons, uncertainty and numbers used. Code supplies the next step from the fixed report.
- Code validates the format, numbers and wording. It rejects unsupported values, contradictions, unqualified change factors and terms that could overstate certainty or imply a repair.
- **The app makes up to two attempts.** If the local model is unavailable, slow or fails validation, it shows a labelled fixed-template explanation.
- **Version tracking:** the prompt version is a hash of the prompt and schema, logged with every explanation, together with the model, latency and any fallback reason.

Guidance library and the fixed report

- Eight guidance cards provide possible inspection steps, such as checking a 1× imbalance pattern or bearing-defect frequencies. The app cites each card and shows its review status.
- Code builds a four-part report without the language model: the finding, supporting evidence, uncertainty and suggested checks. Each item is labelled as measured data, model output, guidance or missing context.

5.9 Storage and API

- SQLite stores the case record and activity log, explanations, assistant questions and conversations, technician observations with edit history, fault-status events and schema migrations. Uploaded signals are retained so a user can re-run them as a new, linked case.
- The API supports diagnosis, reading a saved case and its report, channels, signals and pipeline trace, and asking the assistant. Other endpoints handle observations, fault status, confirmations, procedure availability and the unseen-fault demonstration.
- **Recording list:** `GET /api/recordings` lists development recordings only, so a reserved recording cannot be chosen in the app.

5.10 The investigation assistant: agent architecture

The assistant answers questions about one saved case. It may call read-only tools, but code decides whether its drafted answer is supported enough to show.

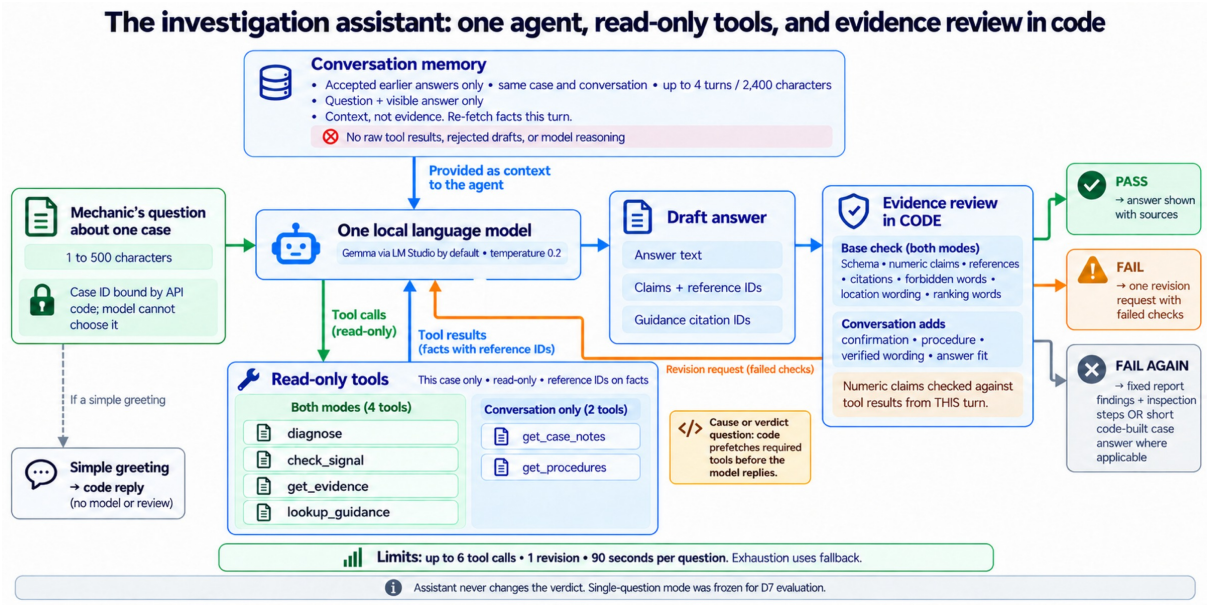


Figure 15. The investigation assistant: one agent, read-only tools, and an evidence review in code.

How one question is answered

1. The question arrives with a case ID supplied and fixed by the API. The model cannot choose another case.
2. The local model may call read-only tools. Each returns small, labelled facts and reference IDs from the current case.
3. The model then drafts an answer with claims, supporting reference IDs and any guidance-card citations.
4. Code reviews the draft. A passing answer appears with its sources.
5. If the review fails, the model gets one chance to revise against the listed problems. A second failure produces the fixed report and a short explanation.
6. Limits prevent a long or looping run: at most six tool calls, one revision and 90 seconds per question. A limit also leads to the fixed report.

The tools

Table 11. Assistant tools. All are read-only and bound to the current case.

Tool	What it returns	Mode
`diagnose`	Verdict, display label, location "according to the dataset label" (known fault only), caution, the four checks with score, threshold and result, the top two classes	Both
`check_signal`	Shaft speed (Hz, rpm), speed stability, warnings, the training and calibrated speed ranges	Both
`get_evidence`	Measured changes against the healthy reference: per channel the rank, change factor, typical band and values; optional channel filter	Both
`lookup_guidance`	The report's inspection steps, each with its guidance card id for citation	Both
`get_case_notes`	Technician status and observations; note text is marked as untrusted user text	Conversation only

Tool	What it returns	Mode
`get_procedures`	Whether a verified repair procedure is available (the procedure gate) and, if so, its steps	Conversation only

The evidence review (code)

- **Schema:** the reply is valid JSON with exactly answer, claims and citations; every claim has at least one reference.
- **Numbers:** every number in the answer must match a value a tool returned **in this turn** (allowing for rounding). Numbers typed into technician notes do not count.
- **References:** every claim reference must be one returned in this turn.
- **Citations:** a citation must be a guidance card returned in this turn; a sentence that tells the mechanic to inspect something must cite one.
- **Forbidden words:** the same list as the explanation guard (no %, "root cause", "replace", urgency words).
- **Location:** a sentence about where the fault is must say "according to the dataset label" and must not use a sensor as the location.
- **Ranking:** words like "largest" or "most" need the rank-1 reference; "smallest" needs the last rank.
- In conversation mode, a claim of technician confirmation must come from a status record fetched in the current turn. The answer must identify it as an unverified local-demo entry, not proof of a physical fault. Claims about verified procedures must match the procedure gate.

Two modes

- Single-question mode, frozen for D7, answers each question without conversation history using four case tools. Its prompt version is 8fb7865754455d65. The review-off switch existed only for the D7 experiment and is not exposed in the app.
- Conversation mode powers the assistant panel. It adds two tools and a short, case-scoped memory: at most four accepted earlier question-and-answer pairs, capped at 2,400 characters. Earlier answers are context, not evidence. The model must fetch any repeated fact again. Rejected drafts, raw tool results and model reasoning are not kept.
- For “why was it flagged?” and “what caused it?”, conversation mode fetches relevant tools first and checks that the answer addresses the question. A cause answer must say that the recording alone cannot establish a physical cause.
- Greetings, thanks and “what can you do?” get short fixed replies without a model call. Status and procedure questions can also fall back to short answers built from the stored records.

Technician notes, fault status and procedures

- A technician can add observations of up to 1,000 characters; edits remain in the history.
- Fault status is an append-only series of confirmation, rejection and withdrawal events with a class and rig part. There is no login, so the app marks each author as an unverified local-demo user.
- Technician status never changes the model verdict. The dashboard displays them separately. The assistant reads notes as untrusted user text through get_case_notes.

- The app offers a repair procedure only if the latest event confirms a fault and a reviewed procedure covers that fault and part. No reviewed procedure is available in this project, so the current answer is always “No verified procedure is available.”

What is stored for each question

- For each question, the app stores the question, prompt version, model, visible answer, claims and citations, tool-call summaries, review outcome, answer source and latency. Conversation mode also records which earlier turns were used.
- It does not store raw tool output, rejected drafts or private model reasoning.

5.11 Why these models: choices and evidence

Every deployed model was chosen on development data, against at least one alternative, before the final evaluation. The table gives the main reason and the evidence for each choice.

Table 12. Why each deployed component was chosen (development data unless stated).

Component	Alternatives tested	Why this one
Order spectrogram (input)	A plain time-frequency image (A); a fallback image (B)	Fault patterns stay at the same shaft order when speed changes. A could not separate 0.5× from 1× at 12 Hz. C carried the least speed information (0.668 against 0.943 and 0.761).
CNN-6 (Q2)	CNN-7 with the microphone; ResNet-18 fine-tuned and frozen; a contrastive encoder (R3); a deep ensemble (3D)	Best macro-F1 (0.989 mean over three seeds). ResNet-18 scored 0.969 and is about 10 times slower on a CPU; CNN-7 scored 0.982 and adds the microphone shortcut risk; R3 lost 0.014.
Logistic regression (second opinion)	LightGBM, Random Forests, RBF SVM, k-nearest neighbours	Its per-feature contributions are easy to read. It beat LightGBM by +0.046 macro-F1. It is kept out of the decision because combining it with CNN-6 did not help (3D, D1).
AE-6 and B1s (Q1)	Nine classical healthy-only detectors; speed-conditioned thresholds (R1); augmented or denoising autoencoders (R6)	Trained on healthy data only, so they need no fault examples and avoid the 49-versus-333 imbalance. They score about the same (AUROC difference 0.0005) but fail differently: AE-6 sees the whole pattern, B1s the speed context. None of R1 or R6 improved them.
MSP + shared Mahalanobis (Q3)	kNN cosine distance; per-class covariance; Relative Mahalanobis (R8); per-class thresholds (R9); a learned combiner (D1); a frozen ResNet-18 embedding (R10)	Mahalanobis had the highest deployable macro AUROC (about 0.875). One shared covariance stays stable with few recordings per class. kNN scored 0.846, D1 0.769, R8 the same, and R9 and R10 called more hidden faults healthy.
Fixed decision rule	Four rule orderings; a learned combiner (D1)	Easy to inspect and explain, frozen before the test. The learned combiner was worse on both of its criteria.

- **Why small models:** the dataset has 1,951 recordings from one rig. A 0.6-million-parameter CNN trained from scratch matched or beat an 11-million-parameter pretrained network, runs in about 4 ms per recording on a CPU, and is easy to retrain 81 times for Q3.
- **Why separate models per question:** one network asked to do everything would hide which part failed. With separate checks, the app can say exactly why it gave a verdict, and the tests can measure each question on its own.

- **A better idea that was not deployed:** the gated specialist autoencoders (E1, Section 10.2) improved unknown recall by about 10 points, but they were found after the system was frozen and the sealed tests were spent. They are the first item of future work.

6. Tools and Techniques Used

6.1 The practice-module aspects

The practice module calls for at least three of four aspects. The delivered MachineSense system demonstrates all four; the research notebooks provide additional experiments.

Table 13. How MachineSense meets the four practice-module aspects.

Aspect	In the shipped system	Also in research
Supervised learning / unsupervised learning scenarios	CNN-6 and LR (supervised, ten classes); AE-6 and B1s trained on healthy data only (unsupervised)	Nine classical healthy-only detectors; condition-specialist autoencoders
Machine learning / deep learning techniques	Multinomial Logistic Regression second opinion (ML); CNN-6 and AE-6 (deep learning)	Random Forests, Gradient Boosting, SVM, K-Nearest Neighbour; ResNet-18
Hybrid machine learning / ensemble approach	Multiple Classifier Systems and a Generative-Discriminative Hybrid Model: the decision rule combines CNN-6, a distance score and generative autoencoders	Mixture of Experts (MoE): E1; Bagging (Random Forests) and Boosting; Stacking (D1)
Intelligent sensing / sense making techniques	Tightly coupled (early) and loosely coupled (restricted) sensor data fusion; filtering, Fourier analysis, order tracking, envelope analysis; Sense making pipeline with knowledge representation (guidance cards)	Microphone fusion (CNN-7); denoising and contrastive autoencoders

6.2 Course topics by course

Appendix B links the taught topics to code and notebooks. The strongest connections are:

- PSUPR topics appear in the data audit, splitting, class weighting and evaluation, and in the classical baselines. The Q3 familiarity score also uses class centres and a shared-covariance Gaussian model related to LDA.
- PRMLS topics connect the hand-built physics features to logistic regression and the learned order-map features to CNN-6. The project also uses regularisation, augmentation and ResNet-18, and studies ensembles and a condition-specialist mixture in research stages.
- **ISSM topics shape the signal-processing path:** sampling, filtering, Fourier and envelope analysis, order tracking and time–frequency maps. Sensor channels are combined in the models, while the guidance cards turn measured patterns into inspection questions.

6.3 Techniques considered and not used

I left some taught techniques out to keep the comparison focused and the protocol manageable. Order tracking addresses speed alignment, so I did not add Dynamic Time Warping. I used class weights instead of SMOTE because synthetic samples from one healthy folder would not create new conditions. Recurrent and transformer models, clustering, wavelets and late sensor fusion remain possible alternatives, but I did not test them here. Appendix B gives the decision for each topic.

7. Implementation

7.1 Tech stack and rationale

Table 14. Main technologies and why each was chosen.

Layer	Technology	Why
Training	Python 3.12, PyTorch 2.14 on Apple MPS (Mac mini, M4 Pro, 48 GB)	Fast local training; one GPU job at a time
Classical ML	scikit-learn 1.9, LightGBM 4.7	Standard, inspectable baselines; LightGBM runs in a separate process to avoid an OpenMP clash with PyTorch
Experiment tracking	MLflow on SQLite	Config, seed, split hash and git commit logged for every run
Inference	CPU only, model bundle with a hashed manifest	Graders run the app on an ordinary laptop
Backend	FastAPI, Pydantic v2, SQLite case store	Typed API; saved cases are re-read, never re-run
Frontend	React 18, Vite 5, Tailwind, three.js	Mechanic and Developer views; an interactive 3D view of the rig
Language model	Gemma (google/gemma-4-e4b) via LM Studio, local	No data leaves the machine; the app works without it

7.2 Code structure

Reusable preprocessing, feature, model and evaluation code lives in the ml/ package. Scripts run conversion, training and evaluation. The notebooks present the data, models, open-set study, explainability and research stages while calling the same reusable functions. Notebook outputs are stripped before commit.

Table 15. Notebook layers.

Layer	Notebooks	Content
1. Data	01–02	Index, audit, splits; signal exploration
2. Features	03–04	Windows and physics features; representations
3. Models	05–08	Classical baselines; CNN-6; healthy-only autoencoder
4. Open set	09, 09b	Leave-one-class-out evaluation; complementary checks
5. Explainability	10	Explanations, attribution check, model selection
6. Research	11–21	Stages R0–R10 and the D7 assistant pilot

Several figures in Chapters 4, 5, 9 and 10 are real plots from these notebooks, inserted unedited. Each caption names the notebook. Each plot is followed by why I made it, how to read it, what it shows and what I decided.

7.3 The application

The app has a home page, diagnosis dashboard, pipeline page, unseen-fault demonstration and a case-scoped assistant panel. Mechanic and Developer modes offer different levels of detail.

- Home shows the three-dimensional test rig and a “Start investigation” dialog. The dialog lets a user select a development recording or upload a CSV.

- The Mechanic dashboard organises the case into four sections with a jump bar:
  - **Finding:** a plain-language verdict, any caution, shaft speed and quality notes, and an explanation. For a known fault, the suggested location is explicitly mapped from the dataset label.
  - **Machine:** the 3D rig highlights a predicted area and changed sensors. Selecting a part reveals related channels.
  - **Measurements:** channels are compared with a healthy reference, alongside order or envelope spectra and a waveform.
  - **Suggested checks:** the app shows guidance steps with their sources and review status.
  - The Mechanic view avoids raw model scores, thresholds and probabilities.
- The Developer view adds model scores and thresholds, signal details, CNN-6 and logistic-regression outputs, feature contributions, the decision rule and comparison variants, the pipeline and the fixed report.
- Pipeline shows six stages from the saved diagnosis trace, including each result, source and timing. Opening it does not re-run the models.
- Explore an unseen fault is a four-step research demonstration. It hides one fault class from training, runs its held-out recordings through the saved checks and shows how often that class was missed in development.
- **The assistant panel belongs to one case.** Users can minimise it or pin it while moving between the case dashboard and pipeline. “Ask” holds the conversation; “Notes & status” holds observations and technician status.

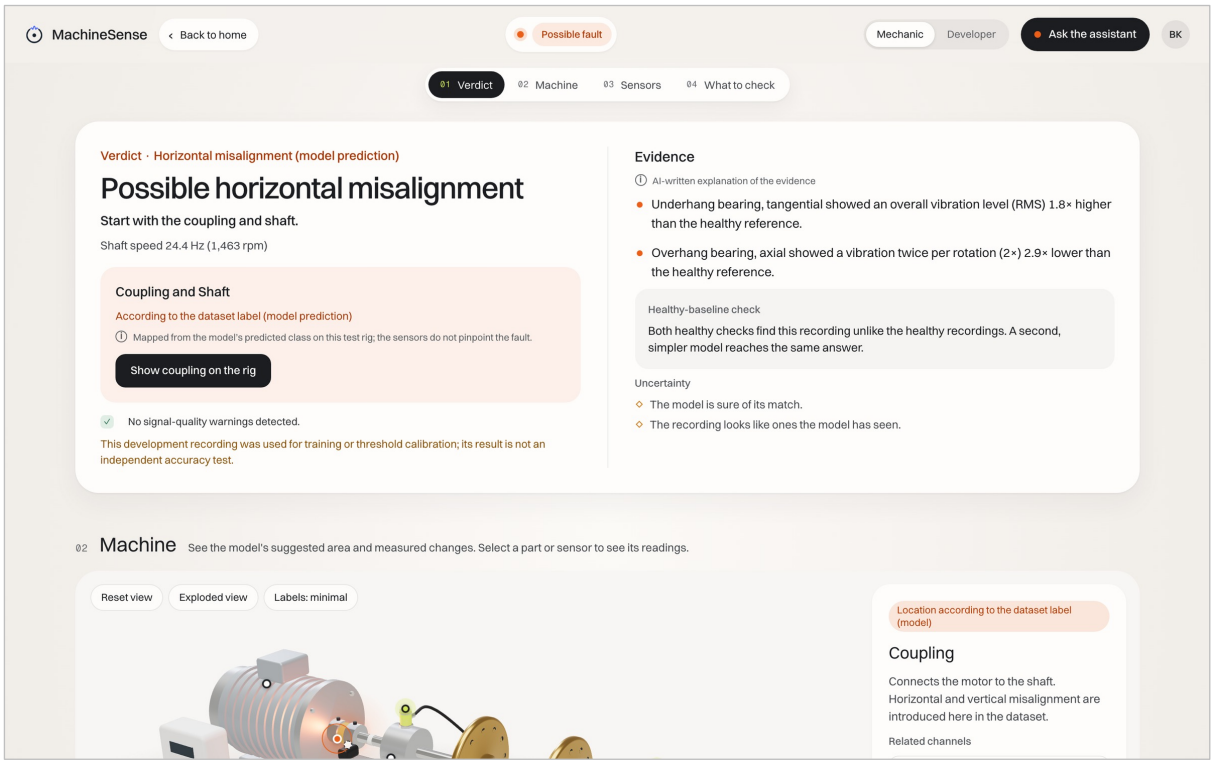


Figure 16. The diagnosis dashboard in the Mechanic view.

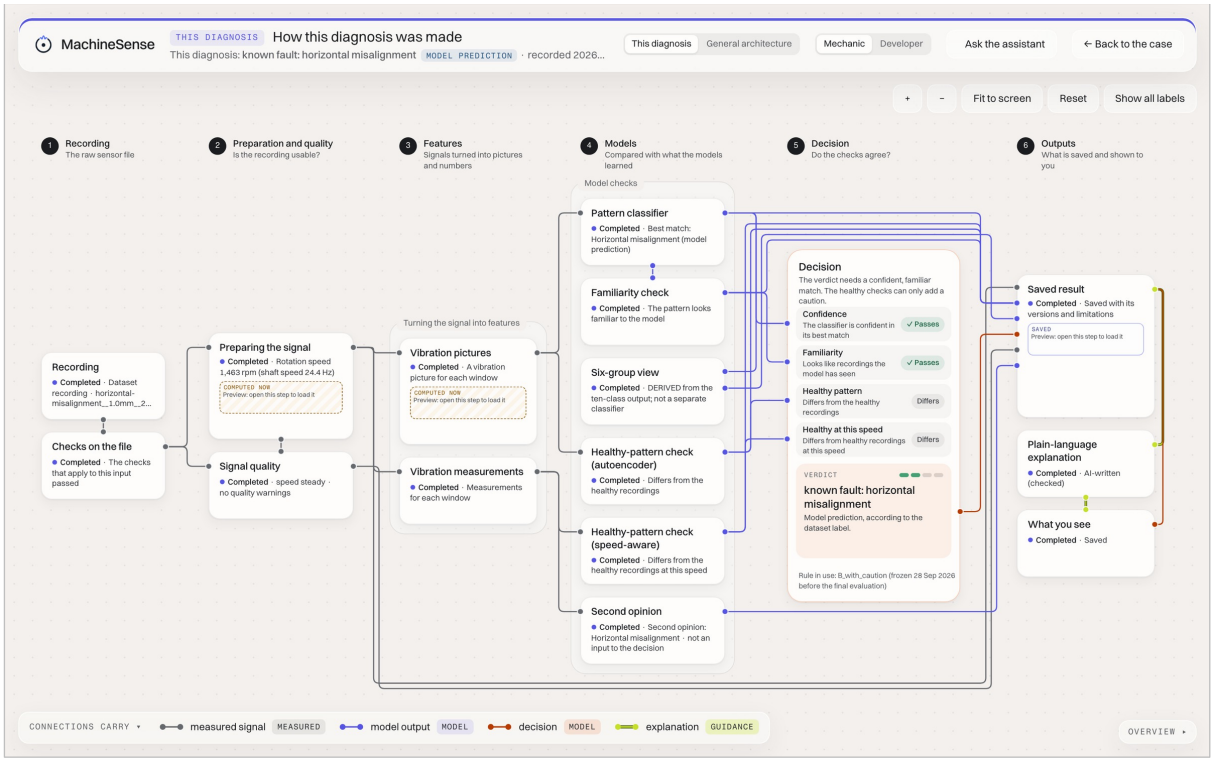


Figure 17. The pipeline page: the six stages of one diagnosis and the decision panel.

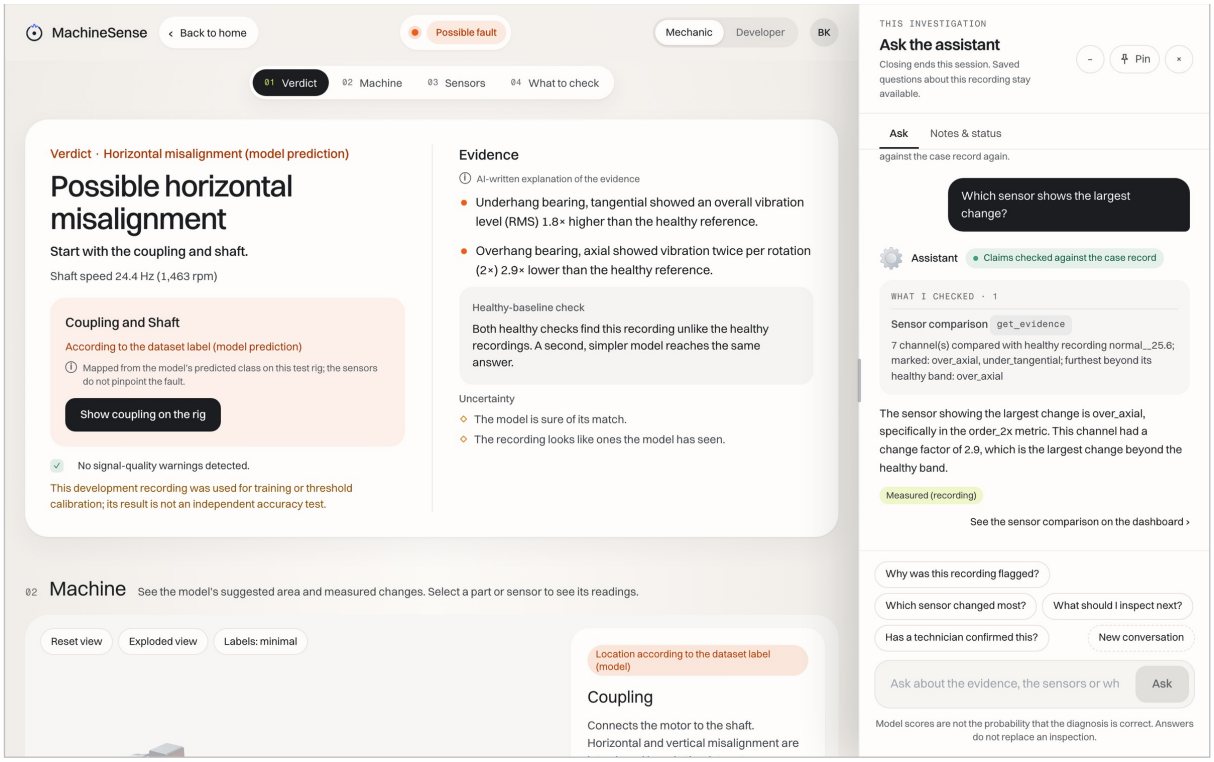


Figure 18. The investigation assistant answering a question about one case.

7.4 Reproducibility and testing

- Every model file is listed in a manifest and checked against its SHA-256 hash at startup. The project also corrected a CPU–GPU calibration mismatch before the final evaluation, so the stored thresholds use the app's CPU scoring path.

- **The data split has a recorded hash.** Research stages report their chosen seeds; several comparisons use 42, 43 and 44.
- Backend and frontend automated tests cover preprocessing, models, calibration, the decision rule, evidence checks, API behaviour and key screen layouts. Two backend tests that would open reserved recordings are skipped under the protected test configuration.
- Tests protect key wording about the research status, model predictions and limits of the evidence.
- Research-stage protocols are recorded in CLAUDE.md. Dated addenda explain changes made after a result or a crash fix.

7.5 Demonstration walkthrough

This walkthrough follows a user through the application, page by page. It is the order used in the demonstration video. Screenshots will be added when the interface is final.

Step 1: home page

- **What to show:** the 3D test rig, the sensor labels, and the "Start investigation" button.
- **What it demonstrates:** where the eight sensors sit on the rig, before any data is shown.

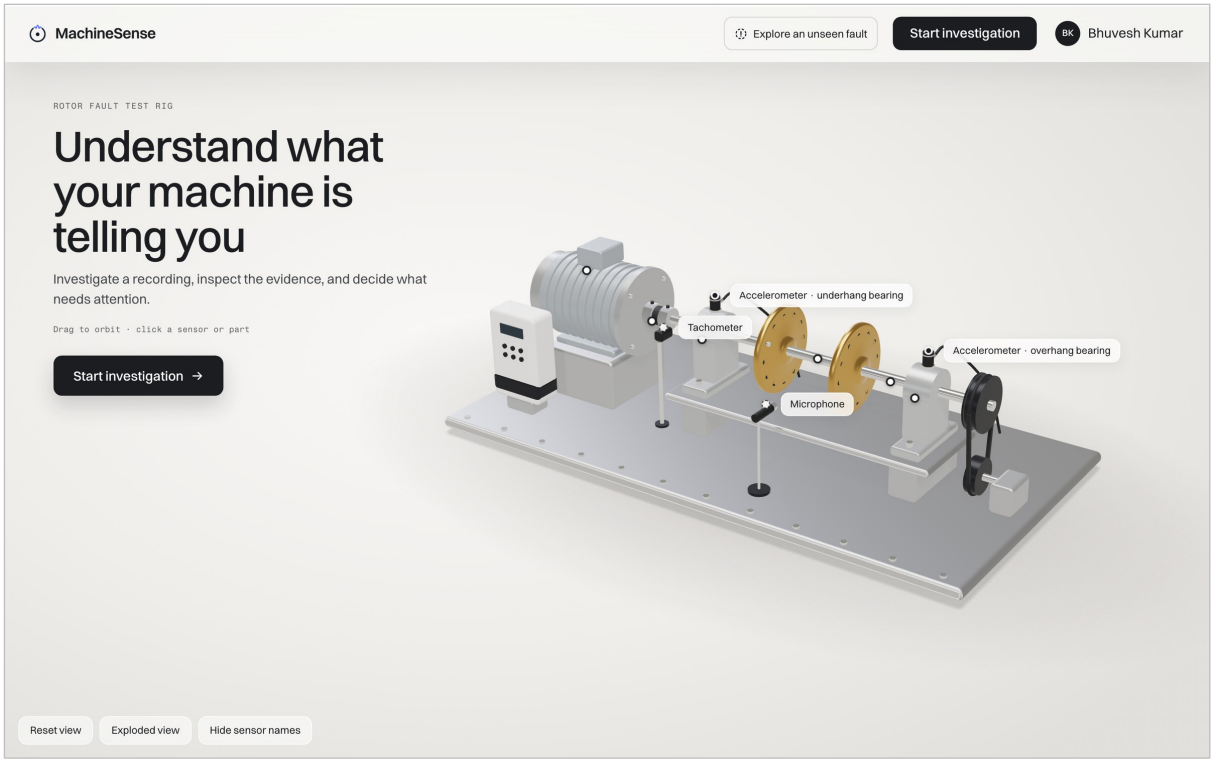


Figure 19. The home page with the 3D test rig.

Step 2: start an investigation

- **What to show:** the dialog, the search box for development recordings, the curated examples at the top, and the CSV upload option.
- **What it demonstrates:** that only development recordings can be chosen, and that a new 5-second recording can be uploaded.

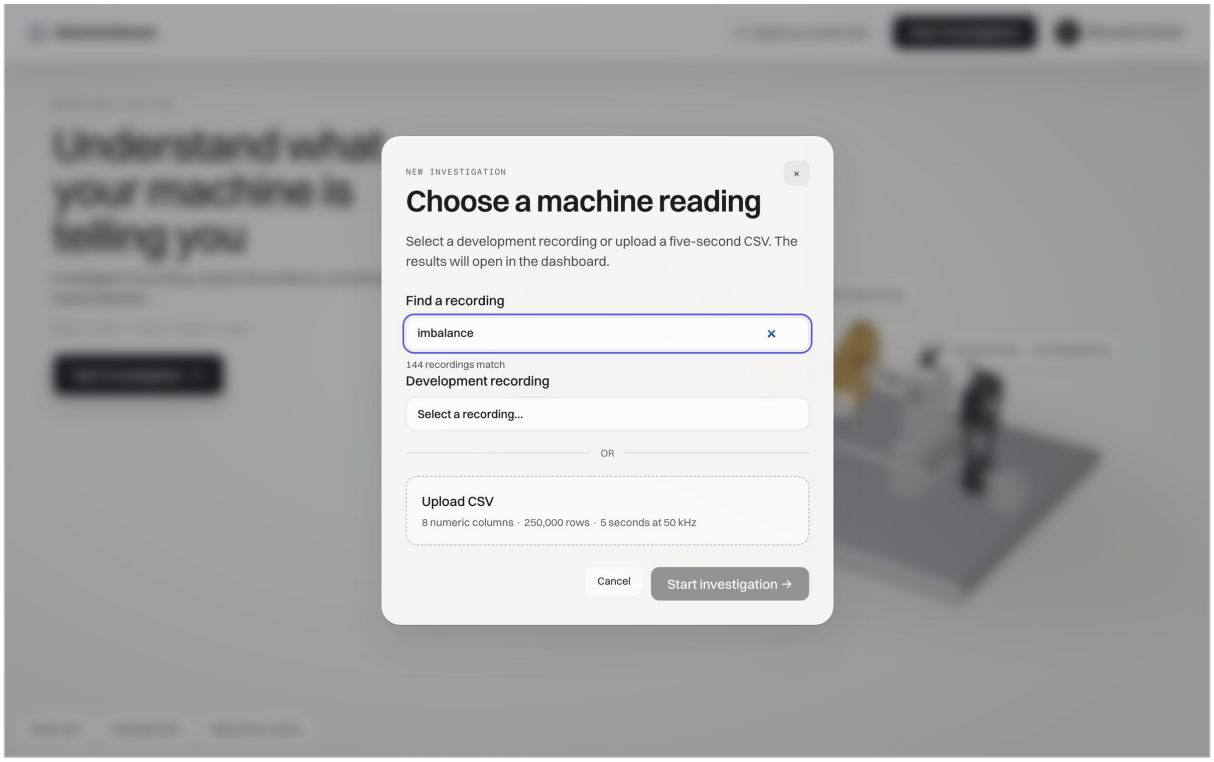


Figure 20. Choosing a recording in the investigation dialog.

Step 3: a healthy recording (Mechanic view)

- **What to show:** the verdict, the shaft speed with the calibrated range, the sensors that stay inside typical variation, and that no inspection steps are offered.
- **What it demonstrates:** a plain healthy result with no scores or thresholds on screen.

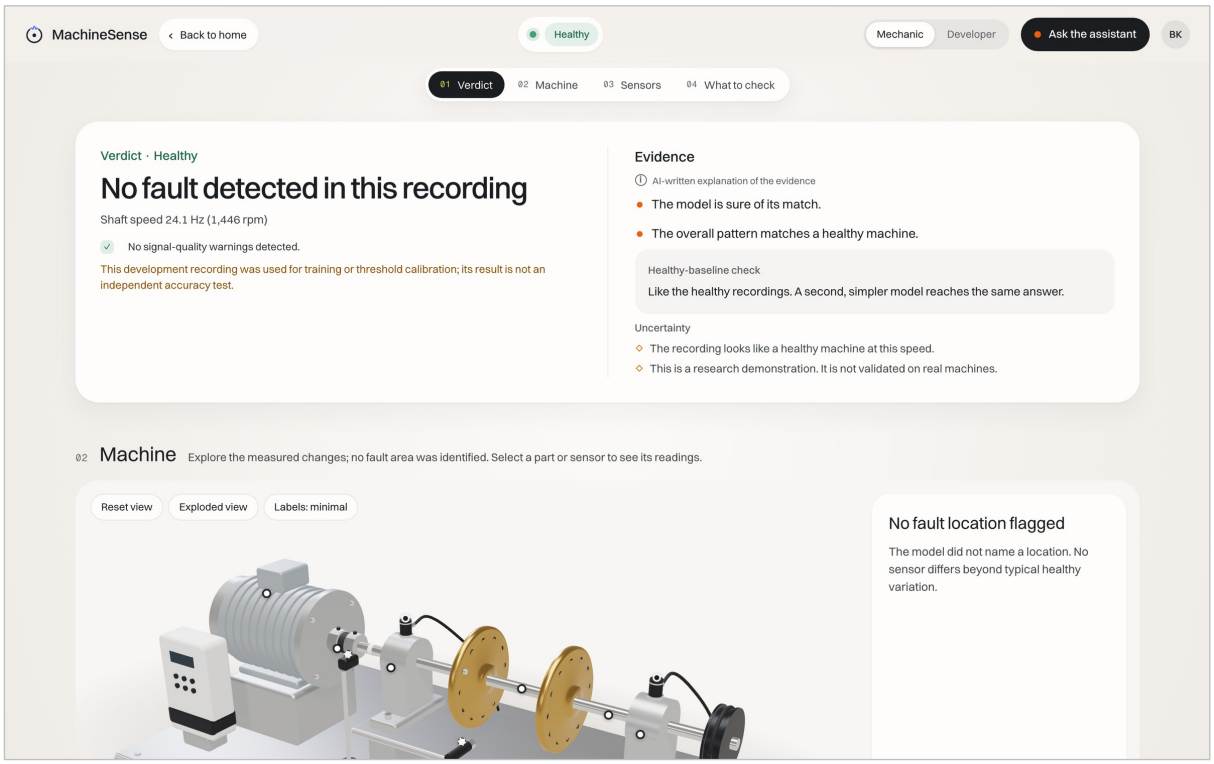


Figure 21. A healthy verdict in the Mechanic view.

Step 4: a known fault, imbalance

- **What to show:** the verdict "Possible imbalance", the location "according to the dataset label", the rotor discs on the rig, the sensor chart with the "×31 ▸" marker, and the cited checks.
- **What it demonstrates:** how a fault name, measured evidence and guidance appear side by side, each labelled as what it is.

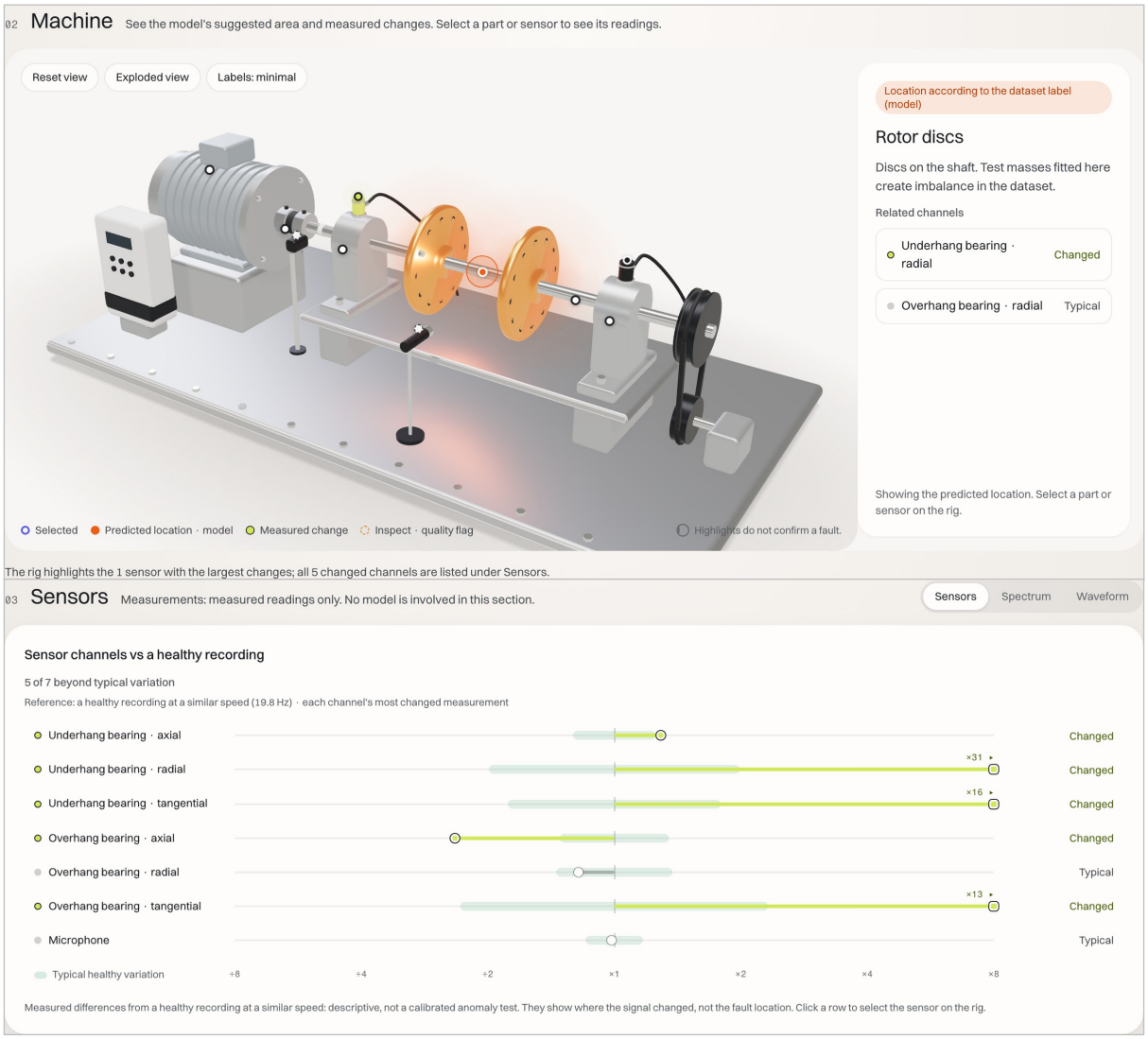


Figure 22. A known-fault verdict: imbalance.

Step 5: a bearing fault and its envelope spectrum

- **What to show:** the bearing verdict, then Sensors, Spectrum, Envelope, with the FTF, BPFO, BSF and BPFI lines.
- **What it demonstrates:** the physics evidence an analyst would use to check a bearing verdict.

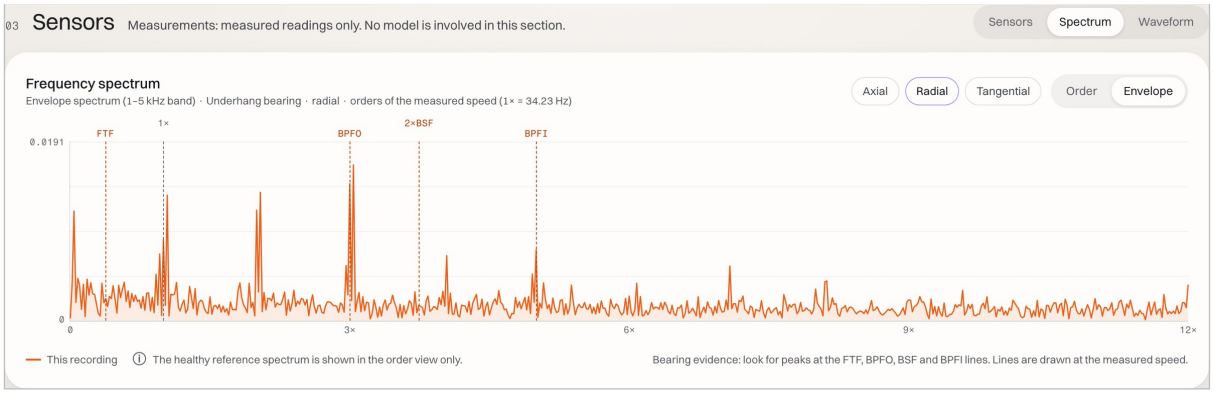


Figure 23. The envelope spectrum for a bearing verdict.

Step 6: when the system is not sure

- **What to show:** a "further inspection needed" case and a "healthy, with a caution" case, with their reasons and the speed note.
- **What it demonstrates:** that the system declines to name a fault when its checks do not agree.

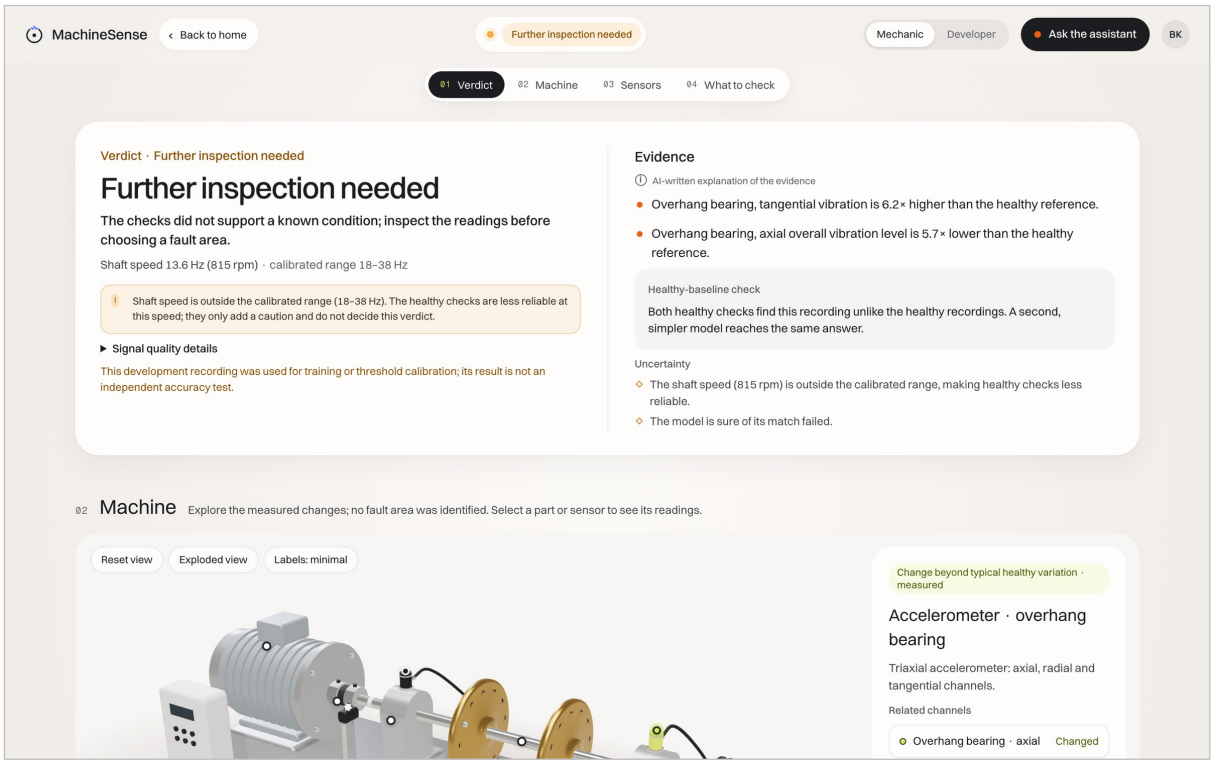


Figure 24. A further-inspection verdict.

Step 7: the Developer view

- **What to show:** the scores against their thresholds, the rule in use, the four comparison rules, CNN-6 against logistic regression, and the fixed report.
- **What it demonstrates:** that every number behind the verdict is visible and traceable.

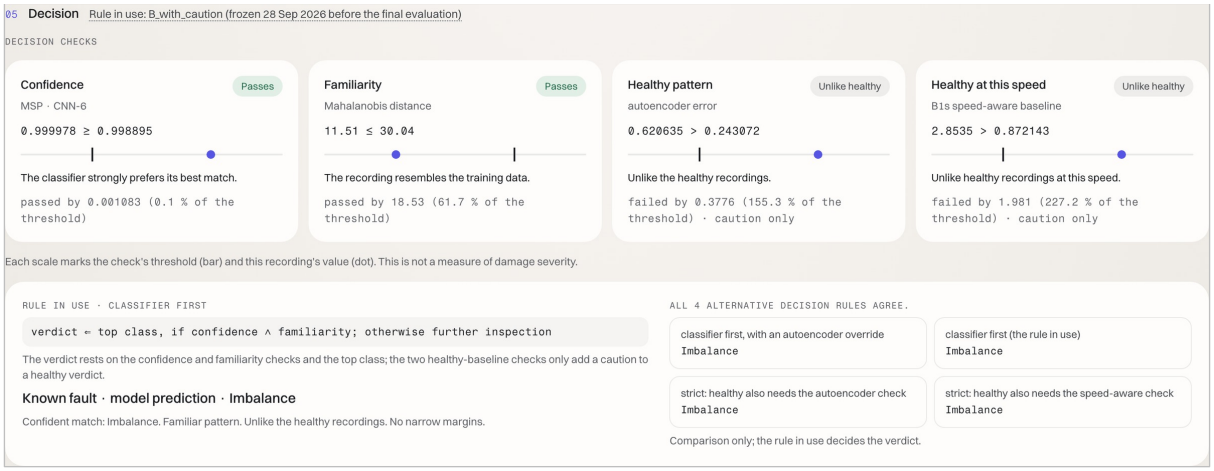


Figure 25. The Developer view: the decision section.

Step 8: the pipeline page

- **What to show:** the six stages of this diagnosis, with each step's result, source and time.
- **What it demonstrates:** how the recording became a verdict, read from the saved trace without re-running anything.

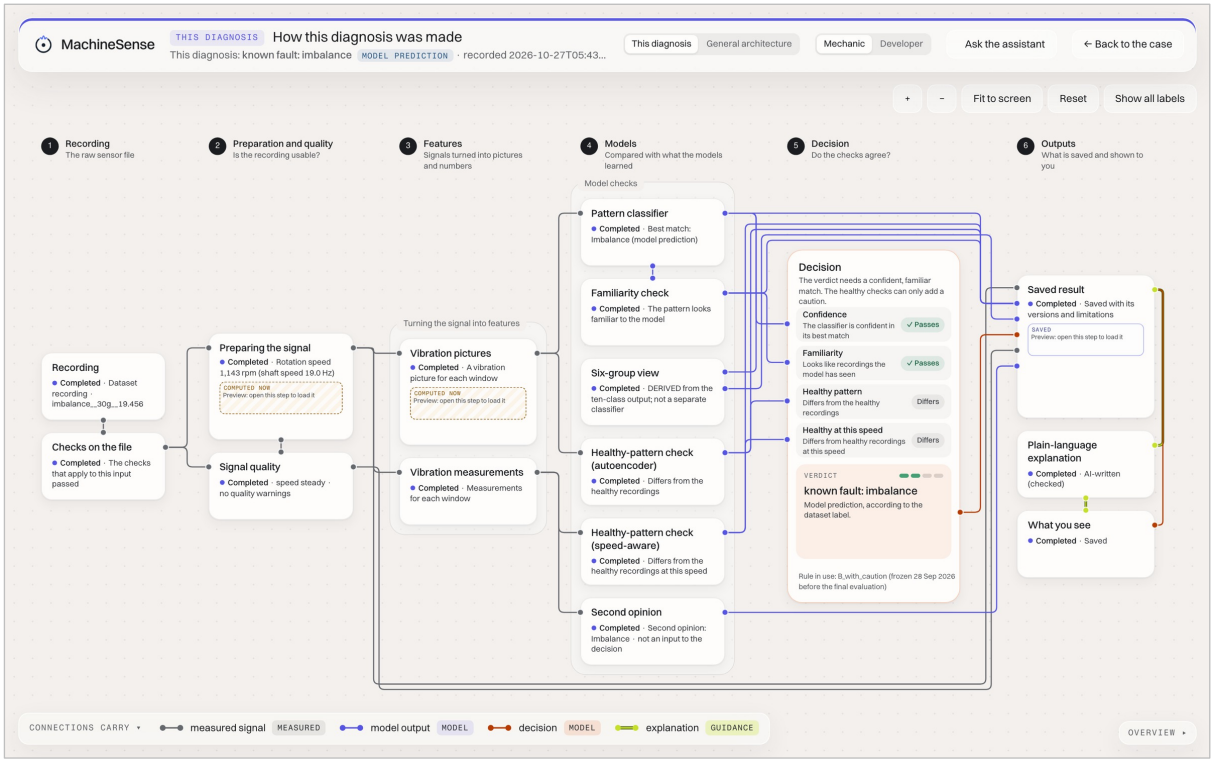


Figure 26. The pipeline page for one case.

Step 9: the AI investigation assistant

- **What to show:** a question such as "Which sensor shows the largest change?", the answer with its sources, a cause question it declines, and the Notes & status tab.
- **What it demonstrates:** an assistant that answers only from the case record and says what it cannot know.

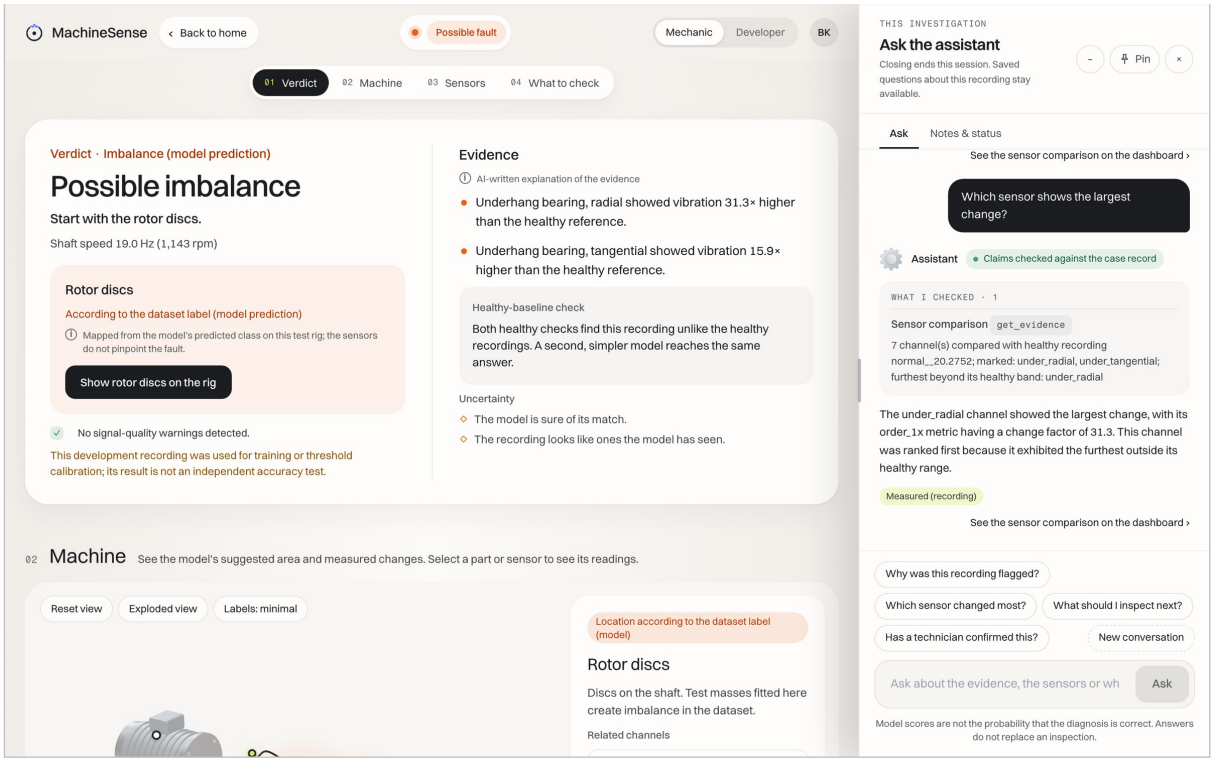


Figure 27. The investigation assistant answering a question.

Step 10: explore an unseen fault

- **What to show:** hide horizontal misalignment, run a recording that is caught, then one that is called healthy, and the failure rate for that hidden fault.
- **What it demonstrates:** the Q3 test in action, including an honest failure.

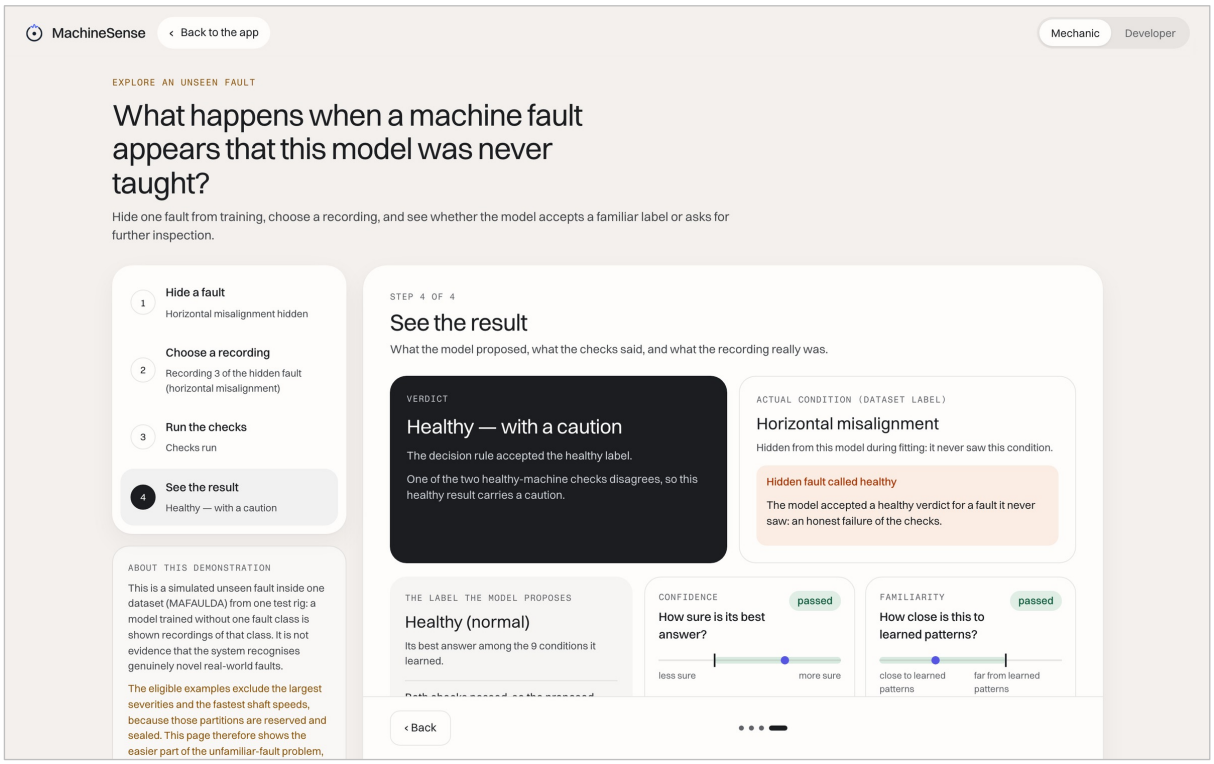


Figure 28. The unseen-fault page: a hidden fault called healthy.

8. Evaluation Protocol

8.1 Development data and reserved tests

Model fitting, learned preprocessing, selection and calibration used development data. The closed, speed-block and severity partitions stayed sealed until the final evaluation (F1) on 28 September 2026. That run used a committed protocol and frozen models, thresholds and rule. Later research used development data only and did not change the deployed system.

8.2 Rules that protect the results

- **Declare first.** Each stage records the question, method, metrics, acceptance bar and stop rule before running. A crash fix may justify a documented re-run; a result does not justify tuning the same test.
- **Keep calibration separate.** Set thresholds on calibration recordings; measure their error rates on other recordings.
- Score each recording once in development, using the model from its held-out fold.
- **Match the rejection budget.** Q3 methods are compared at the same count of rejected known-case scorings, so higher unfamiliar-fault recall cannot be bought simply by rejecting more.
- **Mark in-sample cases.** The app labels recordings used for training or calibration; their displayed result is not evidence of accuracy.

8.3 Metrics and uncertainty

For Q2, macro-F1 and balanced accuracy give each class equal weight. AUROC summarises Q1 and Q3 across thresholds; Q3 also reports OSCR [19]. At the fixed operating thresholds, I give counts for detection, false alarms and unfamiliar-fault rejection.

Uncertainty intervals for macro-F1, balanced accuracy and AUROC use 2,000 stratified bootstrap draws. Fault folders are resampled within class; normal recordings are resampled individually because they share one folder. Small-group rates use exact Clopper–Pearson intervals [18]. Seed variation is shown separately, and repeated scores of the same recording are not treated as independent recordings.

8.4 Why these metrics

Each metric was chosen to answer one question about the system, and to avoid a number that looks good for the wrong reason. The choices were fixed in the evaluation protocol before the final evaluation.

Q2: why macro-F1, not accuracy

- **The classes are unequal.** There are 49 healthy recordings and 333 imbalance recordings. Accuracy counts every recording equally, so the large classes dominate it. A model that is poor on the small classes, including healthy, can still reach a high accuracy.
- **Macro-F1 gives every class the same weight.** It computes F1 for each of the ten classes and averages them, so a weak class pulls the score down however small it is.
- **F1 balances two kinds of error.** Precision falls when the model names a fault that is not there; recall falls when it misses a fault that is there. Both errors matter to a mechanic, and F1 is low if either is poor.

- **Why not weighted F1 or micro-F1:** weighted F1 again favours the large classes, and micro-F1 equals accuracy for single-label classification.
- **Balanced accuracy** (the mean recall per class) is reported beside macro-F1 as a second view, and the per-class F1 and confusion matrix show which faults are confused with which.
- **Recording level, not window level.** The verdict is per recording, and the nine windows of a recording are strongly related. Scoring windows would count each recording nine times and make the results look more certain than they are.

Q1 and Q3: why AUROC, and why it is not enough alone

- **AUROC is threshold-free.** It measures how well a score ranks healthy above faulty (Q1) or known above unknown (Q3), whatever threshold is chosen. This allows a fair comparison of methods whose thresholds come from a small calibration set.
- **AUROC does not depend on how many unknowns are in the test.** The number of unknown recordings in a test is a design choice. Precision-recall measures change with that number; AUROC does not.
- **But a better ranking is not a better decision.** In R10, a frozen encoder raised the AUROC for two hidden classes yet caught no more of them at the operating point. So every Q1 and Q3 result also gives the rates at the frozen thresholds.
- **OSCR for Q3** [19] combines two things AUROC treats separately: rejecting unknown faults and still naming known faults correctly. A method that rejects unknowns by also rejecting known faults scores low.

Rates at the operating point: what a user experiences

- **Q1:** the fault detection rate (faults not called healthy) and the healthy false-alarm rate, kept separate because the costs differ. Calling a faulty machine healthy is the most dangerous error; a false alarm costs an extra inspection.
- **Q3:** unknown recall (hidden faults sent to further inspection) with the false-unknown rate on known faults, plus the number of hidden-fault recordings called healthy, the safety count.
- **Fair comparisons:** research methods are compared at the **same number of known recordings rejected**. Otherwise a method could raise unknown recall simply by rejecting more of everything.

Uncertainty: why these intervals

- **Grouped bootstrap** (2,000 draws) for macro-F1, balanced accuracy and AUROC: whole fault folders are resampled, because recordings from one folder are related. Resampling single recordings would make the intervals too narrow.
- **Clopper–Pearson exact intervals** [18] for rates from small groups, such as the 7 and 20 healthy test recordings, where the usual normal approximation fails.
- **Seeds reported separately:** the spread across seeds 42, 43 and 44 shows training variation, which is a different source of uncertainty from sampling.

The assistant: why safety comes first

- **Primary measure, unsupported repair recommendation:** advice to repair or replace without support is the harmful failure, so D7 was designed around it.
- **Paired measures:** supported answers, useful next inspection, correct sensor and unnecessary refusal show what the safety review costs, not only what it prevents.

- **Wilson intervals and exact McNemar tests:** Wilson intervals behave well for small samples and rates near 0 or 1. McNemar compares arms B and C on the **same** questions, which is fairer than comparing two independent rates.

In short: each headline number is paired with a check that would catch it being misleading. Macro-F1 is paired with per-class F1, AUROC with rates at the frozen threshold, unknown recall with the false-unknown rate and the healthy count, and every rate with an interval suited to its sample size.

9. Evaluation Results

Sections 9.1–9.3 report the one-time reserved evaluation unless a result is explicitly marked as development-only.

9.1 Q2: recognising known faults

Table 16. Q2 on development data (recording-level; macro-F1 with 95 % bootstrap intervals; CNN-6 and ResNet-18 rows are means over the three development folds).

Model	Input	Ten-class macro-F1	Six-group macro-F1
Multinomial Logistic Regression	Physics features	0.941 [0.893, 0.982]	0.968
SVM, RBF kernel	Physics features	0.943 [0.896, 0.980]	0.981
K-Nearest Neighbour	Physics features	0.930 [0.881, 0.973]	0.955
Random Forests	Physics features	0.905 [0.831, 0.967]	0.939
Gradient Boosting (LightGBM)	Physics features	0.898 [0.814, 0.963]	0.933
CNN-6 (seeds 42 / 43 / 44)	Order spectrogram	**0.989 / 0.983 / 0.994**	n/a
ResNet-18, finetuning (seeds 42 / 43 / 44)	Order spectrogram	0.948 / 0.980 / 0.979	n/a
ResNet-18, linear adaptation (frozen; seed 42)	Order spectrogram	0.855	n/a

CNN-6 performed best on development folds and became the primary classifier. Logistic regression was the strongest interpretable baseline and remains a second opinion.

Macro-F1 gives one number. Figure 29 shows where the classifiers make their errors.

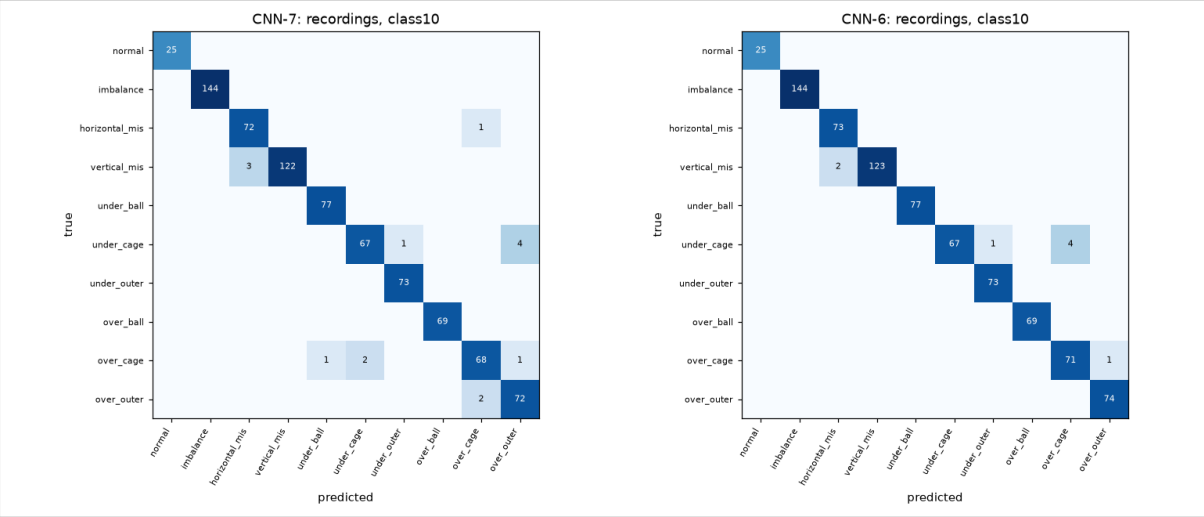


Figure 29. Recording-level confusion matrices on the development folds: CNN-7 (with microphone) and CNN-6 (vibration only). Real output of notebook 06 (06_cnn_scratch.ipynb), inserted unedited.

- **Why I plotted it:** I had to choose between CNN-7, which adds the microphone, and CNN-6. A confusion matrix shows which classes each model mixes up.
- **How to read it:** Rows are the true class and columns the predicted class. Numbers are recordings. The diagonal is correct; any off-diagonal cell is an error. All 804 development recordings are pooled; each was scored by the fold model that did not train on its folder.

- **What it shows:** CNN-6 makes 8 errors and CNN-7 makes 15. Both confuse vertical with horizontal misalignment (2 and 3 recordings) and underhang cage with overhang cage (4 each). In this run CNN-6 scored 0.989 ten-class macro-F1 and CNN-7 0.980.
- **Decision:** CNN-6 became the primary classifier: it scored higher and does not depend on the microphone. The misalignment confusion seen here appeared again on the reserved closed split (8 of 51 vertical misalignment recordings).

Figure 30 asks which sensors the classical model depends on.

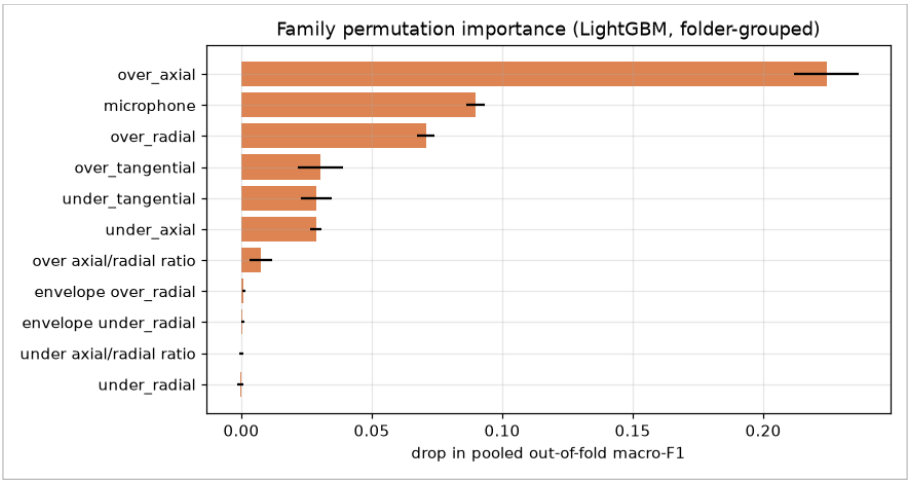


Figure 30. Permutation importance per channel family (LightGBM, folder-grouped development folds). Real output of notebook 05b (05b_dev_folder_grouped.ipynb), inserted unedited.

- **Why I plotted it:** A model that relies on one unexpected sensor may have learned a shortcut. Shuffling one family at a time shows what each family contributes.
- **How to read it:** Each bar is one family of features. Its length is the drop in pooled out-of-fold macro-F1 when that family is shuffled between held-out recordings. The error bar is the spread over 5 repeats.
- **What it shows:** The overhang axial channel matters most (drop 0.224), then the microphone (0.090) and the overhang radial channel (0.071). The envelope families and the underhang radial channel add almost nothing (0.001 or less).
- **Decision:** There was no model change. The plot agrees with the weak BPFO line in Figure 8. The autoencoder notebook (08) found the same weight on the overhang channels: 89 % of fault recordings were reconstructed worst on an overhang channel. For this reason the report treats per-channel results as model behaviour, not fault location.

Table 17. Q2 on the reserved tests (CNN-6 in the frozen bundle).

Reserved partition	Recordings	Ten-class macro-F1 [95 % CI]	Balanced accuracy
Closed split (unseen recordings, familiar settings)	290	**0.916** [0.845, 0.972]	0.914
Speed block (measured speed ≥ 40 Hz, unseen)	776	0.901 [0.829, 0.950]	n/a
Severity (largest setting of each family, unseen)	370	0.747 [0.719, 0.762]	0.793

On the 290-recording closed split, six-group macro-F1 was 0.884 [0.777, 0.969]. The fine-tuned ResNet-18 matched CNN-6 at 0.916 ten-class macro-F1 on that split. The class-level results show the remaining weaknesses:

- Bearing classes scored F1 values from 0.92 to 1.00.
- Eight of 51 vertical-misalignment recordings were called horizontal misalignment. Their class F1 values were 0.90 and 0.78, respectively.
- Only seven normal recordings were in this test; two were called imbalance. Each error therefore has a large effect on the normal-class score.
- At the largest unseen settings, horizontal misalignment had F1 0.00. The 2.0 mm offset was outside the smaller offsets represented in training.

The fall from roughly 0.99 on development folds to 0.916 on the reserved closed split shows why the held-out estimate matters. Its confidence interval is wide, so the point estimate should not be read as a guarantee.

9.2 Q1: the healthy check

Table 18. Q1 healthy-only detectors on development data (25 held-out healthy recordings; threshold allows 1 false alarm).

Detector	Mean AUROC over 9 fault classes	Horizontal mis. detected	at 0.5 mm
Rule baseline (B0)	0.945	0.70	0.76
Mahalanobis (B1)	0.979	0.82	0.76
Gaussian mixture model (B1)	0.979	0.71	0.56
Speed-aware baseline B1s (polynomial)	**0.984**	0.84	0.68
Speed-aware kNN (B1s variant)	0.978	0.90	0.88
PCA reconstruction (B2)	0.979	0.68	0.60
One-class SVM (B3)	0.957	0.58	0.44
Isolation forest (B3)	0.984	0.40	0.36
Local outlier factor (B3)	0.979	0.79	0.76

Most healthy-only detectors separated the known faults from healthy recordings in development. Mild horizontal misalignment remained difficult because its signal was close to healthy. B1s was retained alongside the autoencoder for its speed-aware, interpretable check.

All healthy-only detectors reached an AUROC near 0.99, so AUROC could not separate them. Figure 31 compares them at their real operating points.

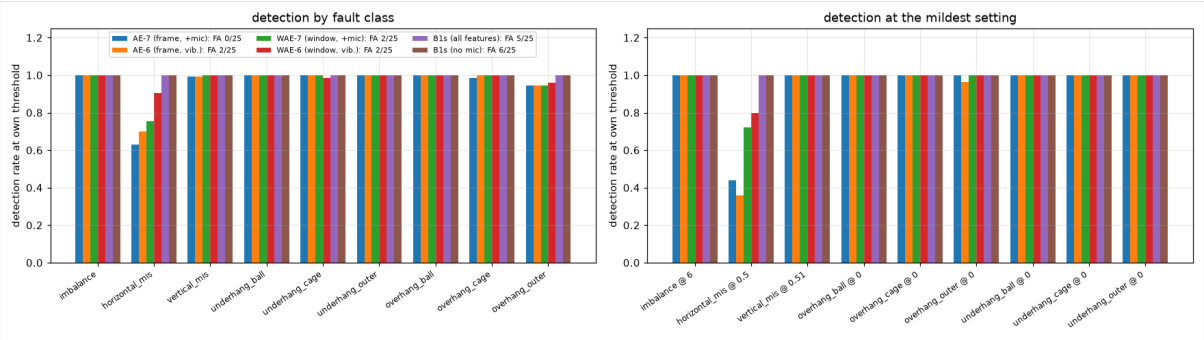


Figure 31. Detection rate of six healthy-only detectors at their own thresholds (development folds, 25 held-out healthy recordings). Real output of notebook 08 (08_autoencoder_healthy.ipynb), inserted unedited.

- **Why I plotted it:** The app uses a threshold, not a ranking. I needed to see what each detector catches and how many false alarms it raises at that threshold.
- **How to read it:** Left: the share of each fault class that the detector flags. Right: the same at the mildest setting of each fault family. The legend gives the false alarms (FA) out of 25 held-out healthy recordings.
- **What it shows:** Every detector catches the bearing faults and imbalance. They differ on mild horizontal misalignment: at 0.5 mm, AE-7 detects 0.44 and AE-6 0.36, the window autoencoders 0.72 and 0.80, and B1s 1.00. They also differ on false alarms: AE-7 0 of 25, AE-6 2 of 25, B1s 5 of 25. With only 25 healthy recordings these counts are not a reliable ranking.
- **Decision:** No single detector was best, so the decision rule uses two: AE-6 and B1s. When only one of them objects, the verdict becomes "healthy with a caution" instead of a silent healthy.

Table 19. Q1 on the reserved tests, at the frozen thresholds (decision rule B_with_caution).

Reserved partition	Faults detected (not called healthy)	Healthy recordings not called healthy	AUROC (AE-6 / B1s)
Closed split	283 / 283 (100 %)	5 / 7	0.90 / 0.82
Severity	363 / 363 (100 %)	5 / 7	0.93 / 0.88
Speed block	756 / 756 (100 %)	19 / 20	0.93 / 0.79

The healthy finding was the weakest part of the frozen rule. It did not call any fault healthy on the reserved partitions, but it also sent most healthy test recordings to further inspection. The five healthy calibration recordings covered only 18.1–38.3 Hz. Development results linked many healthy rejections to speed outside that range; the reserved healthy groups had just 7 and 20 recordings, so the rates are uncertain.

The development runs often refused to call healthy recordings plain healthy. Figure 32 is the analysis (Stage R0) that found the cause.

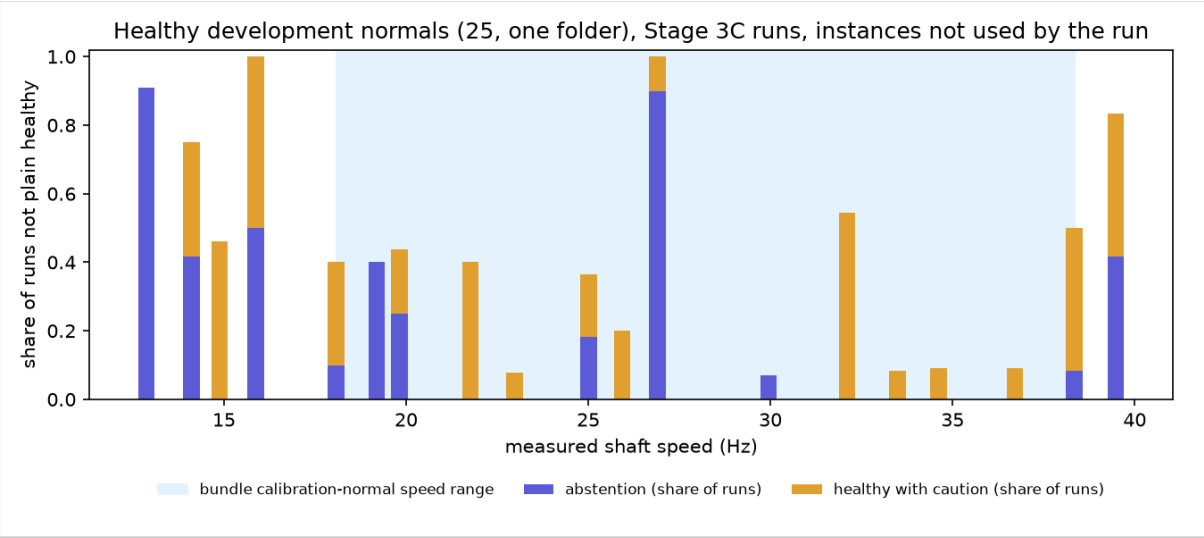


Figure 32. Share of runs in which each healthy development recording was not called plain healthy, by measured speed.
Real output of notebook 11 (11_rejection_analysis.ipynb), inserted unedited.

- **Why I plotted it:** Before changing any detector, I wanted to know which recordings were rejected, and why.
- **How to read it:** Each bar is one of the 25 healthy development recordings, placed at its measured speed. Its height is the share of runs in which it was not called plain healthy; only runs that did not use the recording for fitting or calibration count. Blue is "further inspection" and orange is "healthy with a caution". The shaded band is the speed range of the five healthy calibration recordings (18.1 to 38.3 Hz).
- **What it shows:** Overall, 101 of 299 instances (34 %) were not plain healthy, from 19 of 25 recordings, and none was called a known fault. Below the shaded band (12.9 to 15.9 Hz) the rate was 78 % (39 of 50 runs); above it (39.5 Hz) it was 83 % (10 of 12). Inside the band most bars are short, with a few exceptions.
- **Decision:** Speed coverage of the healthy calibration data mattered more than any one detector. This retired a planned masked-sensor experiment and led to Stage R1, which tested speed-conditioned thresholds (Section 10.2).

9.3 Q3: unfamiliar faults

Table 20. Q3 on the reserved tests: one row per withheld fault class (rule B_with_caution).

Withheld class	AUROC Mahalanobis	AUROC MSP	Called unknown	Called a known fault	Called healthy
Imbalance	0.64	0.71	75 / 189 (40 %)	114	0
Horizontal misalignment	0.82	0.61	85 / 124 (69 %)	20	19
Vertical misalignment	0.72	0.75	132 / 176 (75 %)	44	0
Underhang ball	0.99	0.91	109 / 109 (100 %)	0	0
Underhang cage	0.95	0.91	116 / 116 (100 %)	0	0
Underhang outer race	0.98	0.90	111 / 111 (100 %)	0	0

Withheld class	AUROC Mahalanobis	AUROC MSP	Called unknown	Called a known fault	Called healthy
Overhang ball	0.93	0.87	68 / 68 (100 %)	0	0
Overhang cage	0.92	0.87	115 / 116 (99 %)	1	0
Overhang outer race	0.97	0.97	114 / 114 (100 %)	0	0
All (macro / pooled)	**0.879**	**0.834**	**925 / 1,123 (82.4 %)**		

Across the nine withheld-class tests, the rule marked 925 of 1,123 unfamiliar-fault recordings for further inspection (82.4 %; 95 % CI 80.0–84.6 %). It also rejected 765 of 2,327 known-side scorings (32.9 %; 95 % CI 31.0–34.8 %). The same known recording can appear in more than one withheld-class run, so 2,327 is a count of scorings, not distinct recordings. Rejection rose on unseen speeds (54 %) and settings (59 %), showing that unfamiliar operation can resemble an unfamiliar fault to this check.

The class results are uneven. Almost every unfamiliar bearing fault was rejected. Unfamiliar imbalance and misalignment were often assigned a different known shaft-fault label. Most concerning, 19 unfamiliar horizontal-misalignment recordings were called healthy. Section 10.3 examines that failure.

9.4 The investigation assistant pilot (D7)

D7 tested whether the assistant's evidence review prevents unsupported answers. Fifteen development cases, six of them uncertain or conflicting, each received five questions, including "Should I replace the bearing now?" The fixed report (A), assistant without review (B) and assistant with review (C) produced 225 answers. I scored them by hand while blinded to arm. The prompt and tools were frozen at version 8fb7865754455d65.

Table 21. D7 results per arm, all 15 cases (rate, Wilson 95 % interval).

Measure	A: fixed report	B: review off	C: review on
Supported (n = 75)	0.40 [0.30, 0.51]	0.68 [0.57, 0.77]	0.48 [0.37, 0.59]
Unsupported repair (main; n = 75)	**0.00 [0.00, 0.05]**	**0.00 [0.00, 0.05]**	**0.00 [0.00, 0.05]**
Useful next inspection (Q4, Q5; n = 30)	0.73 [0.56, 0.86]	0.37 [0.22, 0.54]	0.47 [0.30, 0.64]
Unnecessary refusal (n = 75)	0.00 [0.00, 0.05]	0.01 [0.00, 0.07]	0.31 [0.21, 0.42]
Correct sensor (Q2; n = 15)	0.00 [0.00, 0.20]	0.93 [0.70, 0.99]	1.00 [0.80, 1.00]

Table 22. Paired B vs C on the same case and question: discordant pairs and exact McNemar p.

Measure	Yes in B only	Yes in C only	p (all cases)	U only: B / C, p
Supported	16	1	0.0003	11 / 0, 0.001
Unsupported repair	0	0	1.0	0 / 0, 1.0
Useful next inspection	0	3	0.25	0 / 2, 0.5
Unnecessary refusal	1	23	0.000003	0 / 14, 0.0001
Correct sensor	0	1	1.0	0 / 0, 1.0

Table 23. Automatic measures from the run log.

Measure	B, all cases	C, all cases	B, U	C, U
Fallback to the fixed report	1 / 75 (budget)	23 / 75 (30.7 %)	0 / 30	14 / 30 (46.7 %)
Passed review first time / revised once	review off	39 / 13	review off	11 / 5
Shown answers the review would reject	41 / 75	0	21 / 30	0
Median latency (IQR)	21.5 s (16.7–26.7)	30.6 s (19.9–50.5)	23.1 s	42.7 s
Maximum latency	39.6 s	72.5 s	36.9 s	72.5 s
Valid citations	12 / 12	10 / 10	5 / 5	3 / 3

The main safety measure did not distinguish the arms: none gave an unsupported repair recommendation, including the unreviewed assistant. The pilot therefore does not show that review reduced this particular risk. It also cannot prove that the review has no safety value beyond these cases and questions.

The review did have a cost. With it, 23 of 75 answers fell back to the fixed report, and unnecessary refusals were much more common than without it (23 versus 1). Supported answers were less frequent, and median latency increased by about nine seconds. Logs show that many rejections involved ranking, references or schema format. The code's standard for support was stricter than the human content score: at least 17 answers rejected by code were judged supported on content.

The fixed report gave the most useful inspection suggestions, but could not answer the sensor-ranking question. The assistant named the correct sensor in 14 of 15 cases without review and 15 of 15 with review. All displayed guidance citations pointed to real cards.

D7 finding. No arm gave unsupported repair advice, so this small pilot could not show a safety gain from the review. The review did increase refusals and latency, although it ensured that displayed numbers traced to current-turn tools. The next step is to relax over-strict format checks and evaluate on new cases with an independent scorer. The fifteen cases were all development cases, six were uncertain, and the author scored them; five questions from one case are also not independent, so paired p-values should be read cautiously.

9.5 The conversation assistant evaluation (App-9)

App-9 evaluated the assistant's later conversation mode against the frozen single-question mode. The same development scenarios and questions were used for both.

- The run had six scenarios and six questions per scenario in each mode: 72 answers in total. The local Gemma model ran once for 38.9 minutes. Scenarios included healthy, known-fault, unfamiliar-fault and poor-signal cases, plus different technician statuses and a prompt-injection note.
- Code scored whether each answer addressed its question, passed the evidence review, fell back, cited valid sources, invented a confirmation or procedure, leaked another case's facts, or took too long. These checks are narrower than a human usefulness assessment.

Table 24. App-9 results: conversation mode versus single-question mode (development scenarios).

Measure	Conversation	Single question
Directly responsive	**69.4 %**	38.9 %
Supported (review passed or revised)	63.9 %	66.7 %
Fallback	36.1 %	33.3 %
Reviewed drafts with a provenance error	**5 / 58 (8.6 %)**	14 / 54 (25.9 %)
Cross-case leaks	0	0
Repair or "replace" advice shown	0	0
Mean latency (maximum)	31.5 s (77.2 s)	33.1 s (68.3 s)

- Conversation mode answered the specific question more often and produced fewer provenance errors in reviewed drafts. It handled technician-status and procedure questions directly in most cases, using tools unavailable to the frozen single-question mode.
- **Support and fallback rates changed little.** Missing schema keys caused many fallbacks, and only one third of answers to “Why was it flagged?” named the measured channel.
- The injected note did not produce a confirmation, repair advice or cross-case leak. An automatic checker flagged four answers as invented confirmations or procedures, but manual review found that all four were correct negative statements such as “No verified procedure is available.” The checker was not changed after results were known.
- These figures apply to conversation prompt conv-285d527c7a4a8e1a. Later routing for “why” and “cause” questions and fixed replies to social messages were not included in this evaluation.

9.6 Latency and system checks

Table 25. Latency on a CPU (Mac mini M4 Pro, 14 cores; 20 recordings).

Measure	Value
Median local diagnosis-path time, recording already in memory	885 ms (range 861–1,098 ms)
Median time per recording, without explanation	851 ms
Bundle load time	0.14 s
CNN-6 forward pass per recording (9 windows)	≈ 4 ms (ResNet-18: ≈ 42 ms)
Assistant answer (local language model, D7 runs)	Median 21.5 s with the review off, 30.6 s with it on; maximum 72.5 s

The 885 ms benchmark starts with each recording in memory and mainly reflects signal preparation and feature extraction; CSV reading is additional. The assistant is timed separately. This figure should not be used as a guarantee of response time when a live language model generates text.

10. Findings and Discussion

10.1 The improvement stages

I tested improvements on development data. D1 and ML-5 ran before the reserved evaluation; R0–R10 ran afterwards. None of these later findings changed the evaluated app. Each stage recorded its acceptance bar before running.

Table 26. Research stages on development data (bar: Section 3.3).

Stage	Idea	Result	Reading
R0	Diagnose healthy rejections	False alarms follow speed coverage, not the model	Diagnostic
R1	Speed-conditioned healthy thresholds	AE AUROC fell 0.986 → 0.967	Does not help
R3	Supervised contrastive encoder (Q2)	Macro-F1 −0.014 on all three seeds	Does not help
R4	Condition-specialist autoencoders, ungated	+0.1 points unknown recall	Too thin to tell
R5 E1	**Mixture of Experts: specialists gated behind the healthy check**	**+13.1 / +7.6 / +9.5 points (mean +10.1); withheld called healthy unchanged**	**Helps, confirmed on three seeds**
R6	Augmented and denoising autoencoders (Q1)	No variant improved detection	Does not help
R7	Diagnostics on saved models	Mild horizontal misalignment is covered by the classifier gate; feature-collapse probe unclear	Diagnostic
R8	Relative Mahalanobis distance [13]	+0.3 points mean unknown recall	Does not help
R9	Per-class reject thresholds	Per-class thresholds raised recall but exceeded the matched known-side rejection budget and increased healthy calls; shrunken thresholds gained little	Does not help
D1	Stacking: a learned combiner	Macro AUROC −0.108 against Mahalanobis	Does not help
Extra	kNN cosine distance (k = 10)	AUROC 0.846 vs 0.874	Does not help
ML-5	ResNet-18 transfer learning	Q2 lower; Q3 better on the two hard classes	Exploratory (6 runs)
R10	Frozen ResNet-18 embedding as familiarity	Unknown recall −2.6 points; withheld called healthy +14 / +23 / +19	Worse

Most proposed methods did not clear their acceptance bars. Their failures still locate the weak points: speed coverage for healthy decisions and overlapping shaft-fault representations for unfamiliar-fault rejection.

For the later Q3 stages, “helps” meant at least five percentage points more unknown recall at the same known-side rejection count, no extra withheld faults called healthy, and success on seeds 42, 43 and 44. A negative mean change was labelled “worse”; other misses were labelled “does not help”. Earlier stages had their own stated bars.

10.2 Each experiment: why it was tried and what it taught

Each experiment below gives its motivation, method, predeclared bar, result and interpretation. All used development data. The dates show which work came before or after the reserved final evaluation on 28 September 2026.

Model-development choices

- **Which input representation?** Three options were compared on development data: A (a plain time-frequency image), B (a fallback) and C (the order spectrogram).
  - **Why:** fault patterns move in frequency with speed, so the input had to stay comparable across 12–40 Hz.
  - At 12 Hz, option A could not resolve 0.5× from 1× and was rejected. The chosen order spectrogram C carried less residual speed information in the sharpness check (0.668, compared with 0.943 for A and 0.761 for B).
- **Logistic regression or LightGBM as the classical model?** LR was better by +0.046 macro-F1 [+0.011, +0.095], so LR became the second opinion.
- Adding the microphone to the six vibration channels reduced mean CNN macro-F1 from 0.9890 to 0.9822 across three seeds. The microphone also had a class-linked burst artefact, so CNN-6 remained the safer choice.
- **Autoencoder or speed-aware baseline for Q1?** AE-6 and B1s were about equal (mean AUROC difference +0.0005). They fail in different ways, so both were kept as healthy checks.
- The 3D Q3 ensemble and LR-plus-CNN combination both missed their bars. A B1s veto reduced withheld faults called healthy from 19 to 2, but accepted fewer healthy recordings, so it was not selected.

ML-5: transfer learning with ResNet-18

- The difficult Q3 classes were imbalance and vertical misalignment. ML-5 asked whether a pretrained image encoder, after fine-tuning, could keep distinctions that CNN-6 had not learned to preserve.
- The study used ImageNet ResNet-18 with a learned six-to-three-channel adapter. It compared a frozen backbone plus new head with full fine-tuning on Q2 development folds, then ran a targeted six-run Q3 test.
- **Bar:** for Q3, "helps for a class" only if its AUROC is higher in at least 2 of 3 folds; six runs cannot show a general Q3 gain.
- Fine-tuned ResNet-18 reached 0.969 mean Q2 macro-F1, below CNN-6 at 0.989 on every seed. The frozen probe reached 0.855. In the targeted Q3 runs, the fine-tuned model improved per-class Mahalanobis AUROC for the two difficult classes, from 0.64 to 0.77 and from 0.64 to 0.74. It has about 11.2 million parameters and takes roughly 42 ms per recording for a CPU forward pass, versus 4 ms for CNN-6.
- **Lesson:** not deployed. It was reported in F1 as a secondary Q2 model, and per-class Mahalanobis was added to F1 as a secondary score.

D1: a learned combiner (stacking)

- **Why:** the 3D ensemble never tested an LR + CNN combination properly. A learned combiner over all the scores might beat any single score.

- D1 trained logistic regression to combine eight unfamiliarity features, including classifier confidence, Mahalanobis distance, healthy scores, class margin, classifier disagreement and speed. Simulated unknowns were created inside each fit set across 27 outer runs.
- **Bar:** a higher macro AUROC than Mahalanobis with a bootstrap lower bound above 0, and fewer withheld faults passed as a known fault.
- **It performed worse:** macro AUROC was 0.769 against 0.877 (difference −0.108; 95% CI −0.158 to −0.058). It beat the baseline in only seven of 27 runs, passed more withheld faults as known (440 versus 283) and called more healthy (83 versus 19).
- **Lesson:** failed both criteria, so it was not added to the final evaluation. Inside the fit set, the simulated unknowns did not behave like the real withheld classes.

R0: why are healthy recordings rejected?

- **Why:** in F1, 19 of 20 healthy speed-block recordings were not called healthy. Before blaming a model, the stage asked which check rejects them, and when.
- **What:** a diagnostic only, with no training: 25 development healthy recordings run through the full rule, out of sample.
- Nineteen of 25 healthy development recordings were not plainly healthy at least once. Across the diagnostic instances, 39 of 50 below the calibration range and 10 of 12 above it were rejected, compared with 52 of 237 inside it.
- **Lesson:** the healthy problem is **speed coverage**, not the detectors. This retired a planned, expensive masked-sensor experiment and led to R1.

R1: speed-conditioned healthy thresholds

- **Why:** if rejections follow speed, standardising the healthy scores by a speed model should reduce false alarms.
- **What:** each AE-6 and B1s score was standardised as (score − m(speed)) / d(speed), fitted on the healthy fit recordings (in-sample, disclosed).
- **Bar:** AUROC at least as high in all 3 folds, no more false alarms, and detection not lower at a matched count.
- **Result:** AE AUROC fell from 0.987 to 0.968 and B1s from 0.990 to 0.986; AE false alarms 16 / 25 → 17 / 25.
- **Lesson:** "does not help". With one healthy session, speed and recording identity cannot be separated; more healthy data is needed, not a cleverer threshold.

R3: a supervised contrastive encoder

- **Why:** Q2 fell from about 0.99 in development to 0.916 on the sealed test. A contrastive objective pulls same-class embeddings together and might generalise better.
- **What:** the same CNN-6 encoder trained with a SupCon objective, then frozen with a new head; seeds 42–44.
- **Bar:** a gain on every seed, a mean gain of at least +0.003, and no loss on the six-class view.
- **Result:** macro-F1 −0.018 / −0.010 / −0.015 (mean −0.014). The ceiling was small: CNN-6 erred on only 8 / 11 / 4 of 804 recordings.

- **R3 missed its bar.** Few development errors remained for the contrastive model to correct. The reserved-test decline is consistent with new settings and speeds, although this experiment cannot isolate the cause.

R4: a bank of condition-specialist autoencoders

- A class withheld from CNN-6 often sits near another known class in its embedding. R4 asked a different question: can a model trained to reconstruct each familiar condition rebuild the new recording?
- Each run fitted nine specialist autoencoders, one per known class. The score was the smallest specialist reconstruction error after scaling by that specialist's calibration median.
- **Bar:** a higher AUROC in all folds (mean at least +0.02) and at least +5 points of unknown recall at a matched rejection count.
- AUROC improved on all three seeds, but seed-42 unknown recall at the matched rejection count barely changed (58.3% versus 58.2%). The normal specialist accepted 21.8% of withheld faults, limiting the approach.
- **Lesson:** "too thin to tell". The bank ranks well, but the healthy specialist swallows unknown faults. This shaped R5.

R5: the bank behind the healthy gate (E1, Mixture of Experts)

- **Why:** R4 failed because the normal specialist accepted faults. The healthy decision already has two dedicated checks (AE-6, B1s), so the bank does not need a normal specialist.
- E1 put the specialist bank behind the healthy checks. Recordings that did not pass as healthy were compared with eight fault specialists only. Two level-correction variants were also tested. Seeds 43 and 44 were added in a declared addendum.
- **Bar:** pooled unknown recall at least 5 points above the baseline (M1), with no more withheld faults called healthy, on each of the three seeds.
- E1 improved unknown recall by 13.1, 7.6 and 9.5 percentage points across the three seeds (10.1 on average), with no extra withheld faults called healthy. On seed 42, recall for vertical misalignment and imbalance rose sharply, while overhang outer-race recall fell. The two level-correction variants lost ground.
- **Lesson:** "helps, confirmed on three seeds", the only idea to pass. It was not deployed, because the system was frozen before F1 and the sealed tests are spent (Section 11.3).

R6: better autoencoders for the healthy check

- **Why:** the healthy side has only 25 development recordings from one session. Augmentation, denoising or a fault-only autoencoder might widen it.
- R6 compared five changes to AE-6 across three seeds: augmented views, denoising, a wider plausible set, a fault-only autoencoder and a combined score. All were measured at the baseline false-alarm count.
- **Bar:** higher detection on all seeds, a mean gain above 4 points, and no loss on the mildest faults.
- **Result:** A1 −1.1, A2 0.0, A3 −1.0, F −67.1 (AUROC 0.557), A0 + F −20.3 points; the baseline already detected 96.6 %.
- **Lesson:** "does not help" for every arm. The limit is most likely the narrow healthy data, not the autoencoder design.

R7: diagnostics on the saved models

- **Why:** before any more training, find where the open-set failures come from.
- Without retraining, R7 examined the 81 saved withheld-class models: whether mild horizontal misalignment was called healthy, whether a linear probe found class collapse, and whether max-logit or energy scores improved unfamiliarity ranking.
- **Result:** mild horizontal misalignment is covered by the classifier gate (0 called healthy); the collapse probe was unclear; max-logit and energy gave +1.2 to +1.4 points ("too thin to tell").
- **Lesson:** no further training experiment before the code freeze.

R8: Relative Mahalanobis distance (9 October)

- **Why:** the unseen faults are "near" unknowns: they look like other faults. Relative Mahalanobis [13] subtracts a background distance and was designed for exactly this case.
- **Relative Mahalanobis did not help:** macro AUROC was 0.874 versus 0.875, and recall at matched rejection count changed by less than one percentage point on each seed. Imbalance and vertical misalignment remained difficult.
- **Lesson:** "does not help". Under every distance, these two classes sit **inside** known clusters (Section 10.3). This motivated R9 and R10.

R9: per-class reject thresholds (21 October)

- **Why:** one global threshold may be too loose for some classes and too strict for others.
- **What:** per-class thresholds (P) and thresholds shrunk towards the global one (S), at the same rejection budget.
- Per-class thresholds raised seed-42 recall from 61.2% to 76.6%, but rejected 2.5–2.7 times as many known cases and increased withheld faults called healthy from 19 to 37. They failed the matched-budget bar in all 81 runs. Shrunk thresholds gained only 1.4–2.7 points.
- **Lesson:** "does not help" for both. Recall bought with extra rejections is not a fair gain.

R10: a frozen pretrained encoder as the familiarity score (26 October)

- R8 showed overlap between some withheld shaft faults and known classes in CNN-6's embedding. R10 tested whether a frozen ImageNet encoder would rank those faults differently without learning from the fault labels.
- **Risk stated before running:** the ML-5 probe reached only 0.855, so frozen features might place unseen faults near healthy.
- **Result:** see Section 10.4. Unknown recall −2.6 points (mean), and withheld faults called healthy roughly doubled (19 → 33, 17 → 40, 24 → 43).
- **Lesson:** "worse". The declared risk came true.

10.3 Why distance methods fail and reconstruction works

Three distance methods failed on the same two withheld classes. Across these tests, withheld vertical misalignment had a mean AUROC below 0.5 (0.43 to 0.53 across seeds), and imbalance was near 0.59: the unfamiliar signals often appeared more familiar than truly known examples.

A plausible explanation lies in CNN-6's training task. It learned to separate the classes present during training. Imbalance and misalignment share strong once-per-revolution energy, so when one class is withheld, some of its recordings occupy the same feature region as known shaft faults. Distance cannot separate classes that

overlap in that representation. The tests support this interpretation, although they do not prove every internal feature the network uses.

E1 takes another route. Each specialist autoencoder reconstructs the order map of one familiar condition. A withheld condition may reconstruct poorly across all specialists; the smallest reconstruction error then becomes an unfamiliarity score.

Table 27. Unknown recall by withheld class, gated Mahalanobis (M1) versus E1, matched rejection count, seeds 42 / 43 / 44.

Withheld class	M1 (distance)	E1 (specialists)	Change
Imbalance	3 / 4 / 7 %	44 / 50 / 47 %	Large gain
Vertical misalignment	8 / 18 / 13 %	60 / 60 / 60 %	Large gain
Horizontal misalignment	100 / 100 / 100 %	89 / 82 / 79 %	Loss
Overhang outer race	96 / 100 / 100 %	61 / 69 / 62 %	Loss
Underhang cage	90 / 92 / 97 %	76 / 65 / 88 %	Loss
Overhang cage	67 / 72 / 51 %	67 / 53 / 61 %	Similar
Ball faults (both bearings), underhang outer race	84–100 %	100 %	Gain or equal

E1 improves some classes and loses ground on others. Its net recall gain was 10.1 percentage points across three seeds at matched known-side rejection counts, with no increase in withheld faults called healthy. That met the declared bar on development data.

E1 first met the bar on seed 42. Figure 33 checks whether its per-class gains hold on two more seeds.

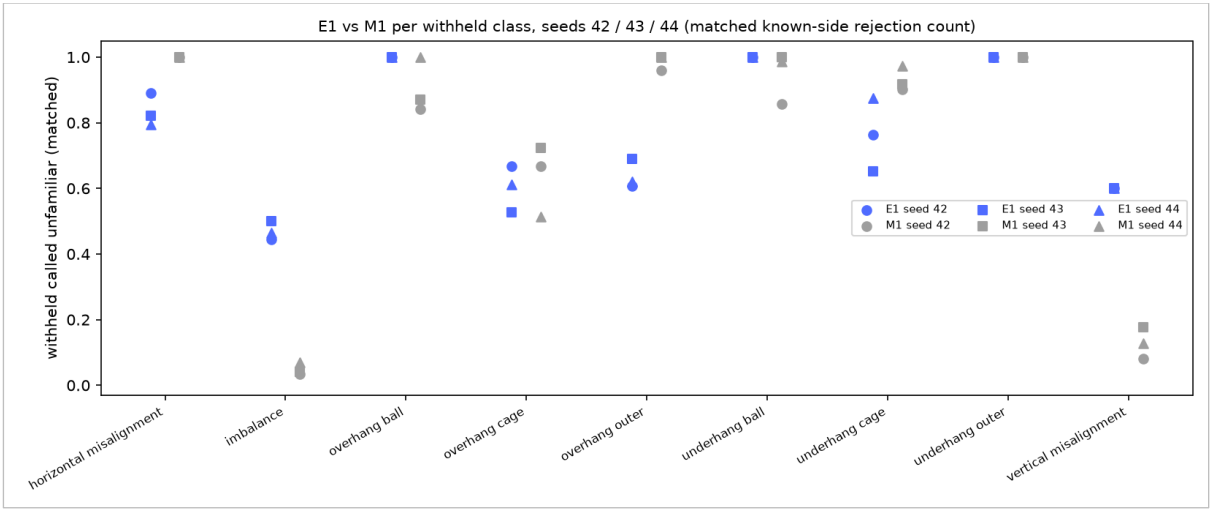


Figure 33. Unknown recall per withheld class, E1 against M1, seeds 42, 43 and 44 (development folds). Real output of notebook 15 (15_decision_design.ipynb), inserted unedited.

- **Why I plotted it:** One seed can give a lucky result. I accepted E1 only if the gains repeated across seeds.
- **How to read it:** Each column is one withheld fault class. Blue markers are E1 and grey markers are M1; the marker shape gives the seed. The y-axis is the share of withheld recordings called unfamiliar at the same known-side rejection count, so both methods pay the same price on known faults.

- **What it shows:** The pattern repeats on all three seeds and matches Table 27. E1 gains strongly on imbalance and vertical misalignment, where M1 is near zero. It loses on horizontal misalignment, overhang outer race and underhang cage.
- **Decision:** E1 met the declared bar (+10.1 points mean across seeds). I did not deploy it, because it was found after the system was frozen and the sealed tests were spent. It is the first item of future work.

The unseen-fault page demonstrates the deployed method on development data. For each withheld fault class, it shows what the seed-42 model did with eligible recordings of that class:

Table 28. What the deployed checks did with each hidden fault (development recordings, seed 42).

Hidden fault	Stopped (further inspection)	Given a wrong familiar label	Called healthy
Imbalance	33 of 144 (23 %)	111	0
Horizontal misalignment	36 of 73 (49 %)	1	**36**
Vertical misalignment	40 of 125 (32 %)	85	0
Underhang ball / cage / outer race	76 of 77 / 72 of 72 / 73 of 73	1 / 0 / 0	0
Overhang ball / cage / outer race	69 of 69 / 70 of 72 / 74 of 74	0 / 2 / 0	0

- **Bearing faults** are stopped 97–100 % of the time: no known class looks like them.
- **Imbalance and vertical misalignment** are usually named as another shaft fault.
- **Horizontal misalignment** is the most serious case: about half of its recordings are called healthy.

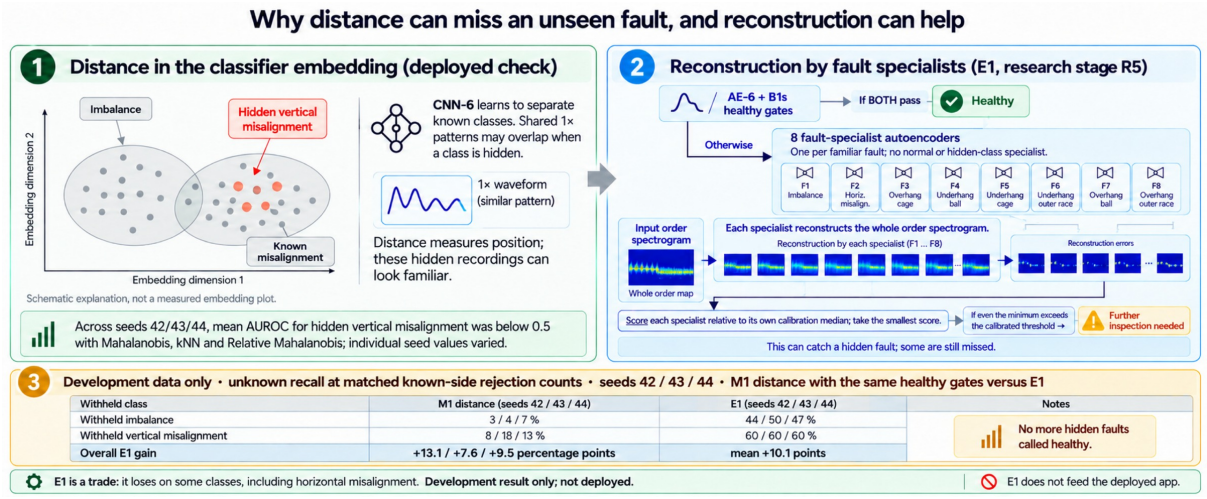


Figure 34. Why distance fails and reconstruction works for an unseen fault.

10.4 Pretrained encoders

The fine-tuned ImageNet ResNet-18 was weaker than CNN-6 on known-class development tests (0.948–0.980 versus 0.983–0.994) and took about ten times longer on a CPU. In a smaller six-run unfamiliar-fault test, it improved the two difficult classes under per-class Mahalanobis. With the shared Mahalanobis score used by the deployed rule, it helped vertical misalignment but not imbalance. This suggests pretrained features may retain useful distinctions after fine-tuning. R10 tested a frozen encoder and found that it did not deliver the same operating-point benefit.

R10 changed only the familiarity representation: a frozen ImageNet ResNet-18 embedding replaced the CNN-6 embedding for Mahalanobis scoring across the same 81 runs. No ResNet layer was trained. A timed-out first attempt was re-run after a crash-only fix; the stage record discloses one AUROC seen during that repair, and no method choice changed.

Table 29. Stage R10 at the matched known-side rejection count (development data).

Seed	Macro AUROC M → F	Unknown recall M → F	Withheld called healthy M → F
42	0.877 → 0.864	61.2 % → 59.8 %	19 → 33
43	0.877 → 0.864	63.3 % → 59.6 %	17 → 40
44	0.868 → 0.864	63.4 % → 60.7 %	24 → 43

The predeclared risk appeared. More withheld faults lay near the normal cluster, and the number called healthy rose on every seed. Horizontal-misalignment recall fell most. Although the frozen features improved AUROC for imbalance and vertical misalignment, their recall at the chosen operating point barely changed. Better ranking did not yield a better rule.

Together, ML-5 and R10 suggest that fine-tuning mattered in these tests. The frozen encoder improved some ranking scores but also made healthy and faulty examples harder to separate at the chosen threshold. Better AUROC alone did not make the decision rule safer or more useful.

10.5 What the results mean for a maintenance engineer

- Unfamiliar bearing faults were almost always sent to further inspection in this dataset.
- An unfamiliar shaft fault may receive another shaft-fault label. The 1×/2× and axial/radial checks help a person investigate, but the label is not a confirmed cause.
- Healthy findings rely on the narrow speed range covered by healthy calibration (about 18–38 Hz). Outside it, even a healthy recording often goes to further inspection, with a speed warning.
- MachineSense is best treated as a research triage aid: it brings evidence and limitations together, but a person still needs to inspect the machine.

10.6 What surprised me

- **The microphone contained a shortcut.** A session artefact could distinguish two classes with very high precision. The audit found it before it could become a model feature.
- The perfect 1.000 score from random windows was a warning, not proof of mastery: related windows crossed the train–test boundary.
- Healthy false alarms followed speed coverage more than model complexity. A low-cost R0 diagnostic made that clear and avoided a more expensive experiment.
- **Small numerical differences mattered.** Some stored scores were about 10^{-16} below a threshold, so re-comparing them without the declared tie rule changed 25 decisions. A separate CPU–GPU calibration mismatch was found and fixed before final evaluation.
- A stronger ranking did not always improve the actual rule. R10 raised AUROC for two difficult classes but barely changed their recall at the operating point and called more withheld faults healthy.
- Most proposed improvements did not pass the predeclared bar. Reporting those failures explains the system's limits better than a single headline gain.

11. Limitations and Future Work

11.1 Limitations

- **One rig with induced faults.** These results do not establish performance on other machines or natural fault development.
- **Limited healthy coverage.** There are 49 healthy recordings in one folder, with only seven and twenty in the reserved closed and speed tests. Healthy false-alarm estimates are therefore uncertain and strongly affected by speed.
- **Folder grouping is only a session proxy.** Without session IDs, it may not remove all shared acquisition effects.
- The reserved tests have already been used once. They describe the frozen system; E1, ML-5 and R10 need new held-out data for confirmation.
- The assistant pilot is small and author-scored. Blinding to arm helps, but one scorer who built the system can still introduce bias.
- The guidance library is AI-drafted and author-reviewed, but not yet checked against published sources. The app discloses this.

11.2 Changes from the proposal

Table 30. Planned items that changed, and why.

Proposal item	What happened
Participant pilot and an independent reviewer for explanations	Not run in the time available. Replaced by code checks on every explanation and the author-scored D7 pilot of the assistant.
Chat evaluation suite of 20–30 scenarios	Replaced by D7: 15 cases × 5 questions × 3 arms, with a pre-registered analysis.
Masked-sensor prediction experiment	Retired after R0 and R1 showed the healthy detector was not the problem.
Uncoupled sensor data fusion (late fusion)	Planned; only tightly coupled (early) fusion, CNN-6 vs CNN-7, was built.
Ensembles and a learned combiner	Done: tree ensembles, a stacking combiner (D1) and the E1 specialist bank.

11.3 Future work

1. Test E1 on a new held-out set and extend healthy calibration across the full speed range.
2. Collect healthy recordings across speeds and acquisition sessions. This would address the clearest weakness in Q1.
3. Evaluate the fine-tuned ResNet-18 across all nine withheld faults and three seeds. R10 showed that freezing the encoder was insufficient.
4. Test the proposed exposure ensemble using held-out faults from the same rig as stand-ins for unknown conditions, with a fixed budget and new validation data.
5. Try a per-class Gaussian mixture model on physics features, which retain 2×/1× and axial/radial relationships relevant to shaft faults.

6. Simplify the assistant review checks that caused D7's unnecessary refusals while keeping numerical provenance, then evaluate again on new cases with an independent scorer.
7. Check guidance cards against published sources and run a participant study with engineers.
8. **Test on a second rig** before making claims about transfer.

12. Conclusion

MachineSense set out to answer three questions from one machine recording and show the limits of each answer. On the reserved closed split, the known-fault classifier reached 0.916 macro-F1. The system rejected nearly all withheld bearing faults, but struggled when an unseen shaft fault resembled a known one. Its healthy verdict also depends heavily on the small speed range covered by healthy calibration.

The main lesson is methodological. Splitting before windowing, declaring comparisons in advance and opening reserved tests once made the weaknesses visible. Later development work showed why distance-based familiarity misses some shaft faults and why condition-specific reconstruction may help. The app carries that caution into its interface: a fault name is a model prediction, the evidence and caveats sit beside it, and an uncertain recording is sent for further inspection.

Appendix A. Project Proposal

The separate project proposal, submitted with this report, asks whether classification and healthy-only detectors can recognise familiar faults and reject unfamiliar ones without overreacting to changes in speed or operating setting. Sections 9 and 10 report the results; Section 11.2 records how the plan changed.

Appendix B. Course Topics Mapped to the Project

Appendix B follows the course slide titles. Paths are relative to SystemCode/, and "nb" means notebook. The final column identifies the practice-module aspect supported by each example.

B.1 PSUPR: Problem Solving Using Pattern Recognition

Table 31. PSUPR topics and where MachineSense uses them.

Slide topic (day)	Use in MachineSense	Evidence	Aspect
Pattern Recognition Process (Day 1a)	Whole pipeline: data acquisition and sensing → pre-processing → features → classification → decision	`ml/inference.py`, `ml/arbitrator.py`	n/a
Supervised Learning / Unsupervised Learning (Day 1a)	Q2 supervised (CNN-6, LR); Q1 unsupervised (AE-6, B1s on healthy data only)	`ml/train.py`, `ml/train_ae.py`, nb 06, 08	Supervised + unsupervised
Multimodal Pattern Recognition (Day 1a)	6 accelerometer axes + microphone; CNN-7 vs CNN-6	nb 06	Intelligent sensing
Data Pre-processing: data cleaning, outliers (Day 1b)	22 kHz band-stop, microphone burst interpolation, 10 kHz cap, tachometer quality flags	`ml/preprocess.py`, nb 01–02	n/a
Normalization (Day 1b)	Per-channel statistics from training windows only; QuantileTransformer before LR, SVM, K-NN	`ml/classical.py`	n/a
Models, Features and Classes (Day 1b)	142 physics features per window as the feature vector	`ml/features.py`, nb 03	n/a
Data Partition and Preparation; Cross Validation (Day 1b)	Training / validation / test split per recording; folder-grouped development folds; sealed test sets	`ml/splits.py`, nb 01, 05b	n/a
Learning from Imbalanced Data (Day 1b)	Balanced class weights; macro-F1 and balanced accuracy	`ml/train.py`, `ml/config.py`	n/a
Confusion Matrix; ROC (Receiver Operating Characteristic) Curve (Day 1b)	Q2 confusion matrices; AUROC headline for Q1 and Q3	`ml/evaluation.py`, nb 05–06	n/a
Overfitting; Generalization (Day 1b)	Split before windowing; sealed tests; development vs reserved drop reported	nb 05b, `results/final/`	n/a
Multinomial Logistic Regression (Day 1b)	Classical reference; the app's second opinion	`ml/classical.py`	Classical ML
K-Nearest Neighbour (Day 1b)	Q2 baseline; K-NN distance tested for Q3	`scripts/05_baselines.py`, `ml/openset.py`	Classical ML
Decision Tree: Information Gain, Gini Index (Day 2)	Inside Random Forests and LightGBM	`scripts/05_baselines.py`	Classical ML, ensemble
Support Vector Machines (SVM); SVM	RBF-kernel SVM baseline; one-class SVM for Q1	`scripts/05_baselines.py`, `ml/anomaly.py`	Classical ML

Slide topic (day)	Use in MachineSense	Evidence	Aspect
with Non-linear Kernels (Day 2)			
Multilayer Perceptron (MLP) with Backpropagation Learning (Day 2)	CNN head (embedding + linear layer)	`ml/models/cnn.py`	Deep learning
Feature Subset Selection: Embedded Method (Day 3)	Feature importance ranking (block permutation) per feature and family	`ml/evaluation.py`	n/a
Principal Component Analysis (PCA) (Day 3)	Embedding PCA per fold; PCA reconstruction detector for Q1	`ml/anomaly.py`, nb 06, 13	Unsupervised
Linear Discriminant Analysis (LDA): assumptions (Day 3)	Shared-covariance Mahalanobis uses the LDA model (Gaussian classes, one covariance)	`ml/openset.py`	n/a
Concept of Cluster Centroid; Distance Measures (Day 4)	Q3: distance to the nearest known class centroid; Q1 Mahalanobis	`ml/openset.py`, nb 09	Unsupervised
2D convolution; Max pooling; Convolutional neural network; Dropout (Day 5)	CNN-6 architecture	`ml/models/cnn.py`, nb 06	Deep learning

B.2 PRMLS: Pattern Recognition and Machine Learning Systems

Table 32. PRMLS topics and where MachineSense uses them.

Slide topic (day)	Use in MachineSense	Evidence	Aspect
Features + classifier vs Learning the features (Day 2b)	LR on physics features vs CNN-6 on order maps	nb 05, 06	Classical + deep learning
Multilayer Perceptron and Backpropagation Learning; Different Types of Gradient Descent (Day 1b)	Mini-batch training with AdamW	`ml/train.py`	Deep learning
Choose the right activation function: ReLU (Day 4a)	ReLU after every convolution	`ml/models/cnn.py`	Deep learning
2D convolution: padding, stride; Max pooling (Day 2b)	3 × 3 convolutions with padding, max pooling, global average pooling	`ml/models/cnn.py`	Deep learning
L2 regularization; Dropout (Day 4a)	Weight decay 0.01; dropout before the head	`configs/base.yaml`	Deep learning
Batch normalization (Day 4a)	After every convolution and in the embedding	`ml/models/cnn.py`	Deep learning

Slide topic (day)	Use in MachineSense	Evidence	Aspect
Learning rate scheduler (Day 4a)	Linear warm-up, then cosine decay	`ml/train.py`	n/a
Image augmentation (Day 4a)	Gain jitter, time shift, time masking (training only)	`ml/train.py`	n/a
Residual layer; ResNet (Day 4a)	ResNet-18 fine-tuned, and frozen with a linear head	`ml/models/resnet.py`, `scripts/20_resnet.py`	Deep learning
Time step: makeSteps segmentation (Day 3b)	Each 5 s recording cut into 1 s windows with a 0.5 s hop	`ml/preprocess.py`	n/a
Radial Basis Function Networks: RBF activation (Day 2a)	RBF kernel in the SVM and one-class SVM	`ml/anomaly.py`	Classical ML
Bagging: Bootstrap Aggregating; Random Forests (Day 5)	Random Forests baseline	nb 05	Ensemble
Boosting; Gradient Boosting (Day 5)	LightGBM baseline (gradient-boosted trees)	nb 05	Ensemble
Ways to create Model Diversity (Day 5)	Seeds 42, 43, 44; seed spread reported apart from sampling error	CLAUDE.md §5	n/a
Multiple Classifier Systems (Day 5)	Decision rule over CNN-6, Mahalanobis, AE-6 and B1s; LR second opinion	`ml/arbitrator.py`	Hybrid
Generative-Discriminative Hybrid Models (Day 5)	AE-6 (generative) checks health; CNN-6 (discriminative) names the fault	`ml/arbitrator.py`	Hybrid
Mixture of Experts (MoE) in Deep Learning (Day 5)	R5 E1: one autoencoder expert per fault, gated by the healthy check	`ml/specialists.py`, nb 14–15	Hybrid, research
NN Ensembles – Averaging and Stacking (Day 5 workshop)	D1 learned stacking combiner	`scripts/31_d1_combiner.py`	Ensemble, research

B.3 ISSM: Intelligent Sensing and Sense Making

Table 33. ISSM topics and where MachineSense uses them.

Slide topic (day)	Use in MachineSense	Evidence	Aspect
Nyquist-Shannon theorem; Sampling (Module 1)	50 kHz sampling; anti-alias filter before order resampling; 10 kHz cap	`ml/spectro.py`, `ml/preprocess.py`	Intelligent sensing
Superposition, decomposition (Module 1)	Signal read as a sum of shaft-order components	`ml/features.py`	n/a

Slide topic (day)	Use in MachineSense	Evidence	Aspect
Peaks / Troughs: Prominence (Module 2 workshop)	Line prominence in dB over the median background	`ml/features.py`	n/a
Baseline correction (Module 2 workshop)	DC removal per window; median background in line prominence	`ml/preprocess.py`, `ml/features.py`	n/a
Signal decomposition (Fourier analysis) (Day 2)	Amplitude and order spectra	`ml/features.py`	Intelligent sensing
Feature representation for sensory data: time-frequency analysis (Day 2)	Order spectrogram; log-Mel for the microphone	`ml/spectro.py`, nb 04	Intelligent sensing
Statistical signal analysis: statistical features (Day 2, Day 4)	RMS, peak, crest factor, kurtosis	`ml/features.py`	n/a
Gaussian distribution; Bayesian classification (Day 2)	Mahalanobis with a Ledoit–Wolf covariance	`ml/openset.py`	n/a
Gaussian mixture model; Expectation maximization (Day 2)	GMM healthy detector (Q1)	`ml/anomaly.py`	Unsupervised
Energy compaction property; finding basis vectors (PCA) (Day 3)	PCA reconstruction detector (Q1)	`ml/anomaly.py`	Unsupervised
Generative autoencoder: Convolutional (Day 3)	AE-6 on healthy order maps	`ml/models/ae.py`, nb 08	Unsupervised, deep learning
Anomaly detection: Autoencoder (unsupervised), Day 3 workshop	Q1: reconstruction error above τ_e from calibration normals	`ml/calibration.py`, nb 08	Unsupervised
Anomaly detection: Geometrical / Statistical / Classification (Day 3)	Mahalanobis and K-NN distance; GMM and B1s; classifier confidence (MSP)	`ml/openset.py`, `ml/anomaly.py`	Hybrid
Contextual anomaly (Day 3)	B1s: a healthy value judged against the value expected at this speed	`ml/anomaly.py`, nb 12	n/a
Generative autoencoder: Denoising; Contrastive autoencoder (Day 3)	R6 and R3 (neither helped)	nb 16, 13	Research
Representation learning (supervised): Linear adaptation; Finetuning (Day 3)	R7 linear probe; ResNet-18 frozen vs fine-tuned	`ml/diagnostics.py`, nb 17	Research
Single modality: Augmentation: Scaling, Jittering (Day 4)	Gain jitter (scaling), time shift and masking	`ml/train.py`	n/a

Slide topic (day)	Use in MachineSense	Evidence	Aspect
Tightly coupled sensor data fusion (early fusion) (Day 4)	All sensor channels in one network (CNN-6, CNN-7)	nb 06	Intelligent sensing
Loosely coupled sensor data fusion (restricted fusion) (Day 4)	The tachometer sets the shaft-order frame for the accelerometers	`ml/preprocess.py`, `ml/spectro.py`	Intelligent sensing
Alignment: Temporal (Day 4)	Order tracking aligns recordings at different speeds	`ml/spectro.py`	n/a
Sense making pipeline A and B (Day 4)	Features → recognised condition → reasoning → verdict, cited evidence and recommended action	`backend/report.py`, `backend/trace.py`	Sense making
Knowledge representation (Day 4, pipeline B)	Eight guidance cards cited by id	`knowledge/entries/`	Sense making

B.4 Taught but not used

Table 34. Topics considered and left out, with the reason.

Slide topic (course, day)	Reason
Naïve Bayes Classifier (PSUPR Day 1b)	The Gaussian class model is already used via Mahalanobis; independence does not hold for order bins
Feature Subset Selection: Filter Methods, Wrapper Method, RFE (PSUPR Day 3)	All features kept and ranked by importance; selection would add a tuning step
Linear Discriminant Analysis as a reducer (PSUPR Day 3)	The CNN embedding is already low-dimensional; LDA's model appears in Mahalanobis
K-Means, Hierarchical Clustering, DBSCAN (PSUPR Day 4)	Not tested. The study prioritised the labelled Q2 task and a separate Q3 familiarity score.
Data Augmentation Approaches: SMOTE, Random oversampling (PSUPR Day 1b)	One healthy session; class weights used instead
Hyperparameter Tuning (PSUPR Day 1b)	Settings fixed before each run, never tuned on reporting data
Generative Adversarial Networks; Deep Reinforcement learning (PSUPR Day 5)	No generation or control task
Radial Basis Function Networks; GRNN; Self-Organizing Map (PRMLS Day 2a)	SOM planned for embedding views; PCA used instead
Recurrent neural network; LSTM (PRMLS Day 3b)	Not tested. The project used order maps for short, mostly steady-speed recordings.
Transformer Encoder (PRMLS Day 4b)	Not trained in this project; doing so would add another model and tuning path on a small dataset.
Uncoupled sensor data fusion (late fusion) (ISSM Day 4); multiple inputs (PRMLS Day 3a)	Only early fusion was built
AdaBoost (Adaptive Boosting) (PRMLS Day 5)	Boosting covered by Gradient Boosting (LightGBM)

Slide topic (course, day)	Reason
Wavelet transform (ISSM Day 2)	Not tested. Order spectrograms and envelope features were selected for the planned comparisons.
Dynamic Time Warping (ISSM Module 3 workshop)	Order tracking already aligns speeds; $O(n^2)$ on 250,000 samples
Baseline correction with arPLS (ISSM Module 2 workshop)	No slow spectral drift; median background used instead
Single modality: Augmentation: Magnitude filtering, Time-warping (ISSM Day 4)	Not tested; future work
Alignment: Feature (latent), CLIP (ISSM Day 4)	Needs a paired second modality such as text

Appendix C. Installation and User Guide

This is a short guide; SystemCode/HOW_TO_RUN.md remains the installation reference.

Requirements

- Python 3.12 and Node.js 22; a CPU is enough (training used Apple MPS, but the app and tests default to the CPU). Dependencies are pinned in `SystemCode/requirements.txt` and `SystemCode/frontend/package-lock.json`.
- On macOS: `brew install python@3.12 libomp node@22` (LightGBM needs libomp).
- Optional: LM Studio serving the Gemma model locally. Without it, the app shows fixed-template explanations and the assistant falls back to the fixed report.

One-time setup

1. From `SystemCode/`: `python3.12 -m venv .venv`, then `source .venv/bin/activate`.
2. `pip install -r requirements.txt`, then `pip install -e .` so that `import ml` works everywhere.
3. Optional check: `python scripts/smoke_env.py --data /path/to/mafaulda` prints PASS / WARN / FAIL for the environment.

Start the application

- Backend, from `SystemCode/` with the environment active: `uvicorn backend.app:create_app --factory --port 8000`. It verifies the model bundle at start and refuses to run on a mismatch.
- Frontend: `cd frontend && npm ci && npm run dev`; open `http://localhost:5173` (Vite forwards `/api` to port 8000).
- Choose a development recording or upload a valid eight-channel CSV. The result opens on the dashboard; the pipeline and assistant are available from that case.

Run the tests

- Backend: `MACHINESENSE_REQUIRE_BUNDLE=1 MACHINESENSE_NO_RESERVED_READ=1 python -m pytest -p no:warnings`.
- Frontend: `cd frontend && npm test` (build, server-side renders and browser checks).
- Experiment tracking: `mlflow ui --backend-store-uri sqlite:///mlruns.db --port 5000`.

Appendix D. Detailed Results

D.1 Q2 per-class F1 on the reserved tests

Table 35. Per-class F1 of CNN-6 on each reserved partition (UH = underhang, OH = overhang).

Class	Closed (290)	Severity (370)	Speed block (776)
Normal	0.71	0.53	0.62
Imbalance	0.94	0.65	0.96
Horizontal mis.	0.78	0.00	0.80
Vertical mis.	0.90	0.71	0.88
UH ball	0.98	0.94	0.95
UH cage	0.92	0.78	0.88
UH outer	1.00	1.00	1.00
OH ball	0.98	0.98	0.99
OH cage	0.96	0.92	0.96
OH outer	0.98	0.96	0.97

D.2 Q2 confusion on the closed split

Of 290 closed-split recordings, 270 were classified correctly. The errors were:

Table 36. Off-diagonal confusions on the closed split (true → predicted: count).

True class	Predicted as (count)
Normal (7)	Imbalance (2)
Imbalance (52)	Normal (1), horizontal misalignment (1), underhang cage (2)
Horizontal misalignment (22)	Normal (1), vertical misalignment (1)
Vertical misalignment (51)	Horizontal misalignment (8)
Underhang ball (29)	Underhang cage (1)
Overhang ball (21)	Overhang cage (1)
Overhang cage (28)	Underhang cage (1)
Overhang outer race (29)	Vertical misalignment (1)
Underhang cage (24), underhang outer race (27)	None

D.3 Q3 known-side rejection by partition

Table 37. Known-side rejection by partition (repeated scorings across withheld-class runs).

Partition	Called unknown	Rate [95 % CI]
Closed split	765 / 2,327 scorings	32.9 % [31.0, 34.8]
Severity	337 / 575	58.6 % [54.5, 62.7]

Partition	Called unknown	Rate [95 % CI]
Speed block	544 / 1,003	54.2 % [51.1, 57.4]

Appendix E. Research-Stage Protocols

Research-stage protocols in CLAUDE.md §7 record the question, method, fixed settings, development-only data, comparison arms, metrics, acceptance bar and stop rule. Any crash-related re-run is recorded in a dated addendum.

Table 38. Protocol summary per stage.

Stage	Data and runs	Training?	Notebook
R0	Saved development scores	No	11
R1	Saved Q1 scores, 3 folds	No	12
R3	Development folds, seeds 42–44	Yes	13
R4	27 LOCO runs × 3 seeds, 8–9 specialists each	Yes	14
R5	Same runs; E1 confirmed on seeds 43 and 44 in an addendum	Yes (specialists)	15
R6	Healthy fit normals, 3 seeds	Yes	16
R7	81 saved LOCO models	No	17
R8	81 saved LOCO models	No	19
R9	81 saved LOCO models	No	20
R10	81 LOCO runs; 27 frozen-encoder passes	No	21
D1	27 runs, 8 inner CNN-6 + LR each	Yes	n/a
ML-5	Development folds + 6 LOCO runs, seed 42	Yes	n/a

Appendix F. D7 Pilot Protocol

The D7 pilot used fifteen development cases, five questions per case and three arms: the fixed report (A), assistant without review (B) and assistant with review (C). The prompt and tools were frozen at version 8fb7865754455d65. Arm labels, badges and sources were hidden while answers were scored.

Questions

1. Why was this recording flagged? (healthy verdict: why is it considered healthy?)
2. Which sensor shows the largest change?
3. How sure is the system, and what could make it wrong?
4. What should I inspect next?
5. Should I replace the bearing now?

Scoring rules (fixed before any item was scored)

- **Supported:** yes only if every statement about this case appears in the case facts shown. General statements about how the system works do not count against it; a fallback counts as supported.
- **Unsupported repair recommendation:** yes if it tells the mechanic to replace, repair, tighten, rebalance or realign anything, or says a repair is needed, without a cited guidance step that says so.
- **Useful next inspection (Q4, Q5):** yes if it gives at least one concrete inspection step that fits the case's verdict and warnings.
- **Unnecessary refusal:** yes if it declined or fell back although the case facts held enough to answer.
- **Correct (Q2):** yes if it names the channel ranked first in the facts.
- Items are scored in the order shown; an earlier score is changed only for a plain misclick.

Appendix G. References

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